Eroding Democracy, Eroding Prosperity*

Vanessa Boese-Schlosser[1], Rodolphe Desbordes[2], Markus Eberhardt[3], and Mario Larch[4]

[1]WZB Berlin Social Science Center (Germany)
[2]SKEMA Business School (Paris, France); Université Côte d'Azur (Nice, France)
[3]University of Nottingham (U.K.); CEPR (London, U.K.)
[4]University of Bayreuth (Germany); CEPII (Paris, France); CESifo and ifo Institute (Munich, Germany); GEP (Nottingham, U.K.); WIFO (Vienna, Austria)

This draft: May 15, 2026

Abstract

Recent research has demonstrated that U.S. states are bellwethers of national institutional decline, serving as 'laboratories of autocratisation' through voter repression and gerrymandering. We provide the first empirical evidence on the economic consequences of sub-national democratic erosion in the United States. We find that autocratisation episodes during 2000-2023 lead to lower state per capita income and an increase in income inequality and deprivation. Innovation efforts (business R&D expenditure) and outputs (patenting) contract sharply, undermining the endogenous growth engine of the state economy. International exports are unaffected, suggesting that foreign accountability operates through national-level institutions rather than sub-national ones.

Keywords: autocratisation, difference-in-differences, interactive fixed effects, heterogeneity, economic prosperity, bilateral trade flows

JEL codes: P16, F13, F14, C23

*This work was supported by the Centre for Inclusive Trade Policy (CITP) Innovation Fund, ESRC (Boese-Schlosser and Eberhardt) [grant number ES/W002434/1]. We thank Andreas Fuchs, Aaron Günther, Sarah Langlotz, James Snyder Jr., Martina Uccioli and seminar participants at the Nottingham School of Economics brownbag seminar, the First WZB Interdisciplinary Conference on 'The Future of Democracy?', the American Political Economy Exchange (APEX)-Research Network in Political Economy (RNPE) Partnership Meeting, the CITP online seminar, the University of Göttingen and the annual NICEP conference for helpful comments and suggestions. The usual disclaimers apply. Correspondence: Markus Eberhardt, School of Economics, Sir Clive Granger Building, University of Nottingham, University Park, Nottingham NG7 2RD, U.K. Email: markus.eberhardt@nottingham.ac.uk.

1 Introduction

"Congratulations America! The Polity Project has downgraded its score for the USA from +8. . . to 0."
Monty G. Marshall, Center for Systemic Peace, February 12, 2025 (posted on LinkedIn)

"Americans trust in President Trump's America First economic agenda that continues to prove the so-called 'experts' wrong. President Trump has reduced America's reliance on foreign products, boosted investment in the US, and created thousands of jobs — delivering on his promise to Make America Wealthy Again."
News Release (excerpt), Press Secretary Karoline Leavitt, The White House, July 30, 2025

Democracy is in retreat around the world. Over the past decade, the number of countries reverting from democracy to autocracy has outpaced the number of democratising nations, and the level of democracy experienced by the average citizen of the world in 2023 has reverted to that of 1985, before the collapse of the Eastern Bloc and the ensuing 'Third Wave of Democratisation' (Nord et al., 2024). Many democratic countries are also said to have experienced persistent deterioration of their democratic institutions, a process referred to as autocratisation, democratic erosion, or democratic backsliding (Edgell et al., 2020).[1] While so far during the twenty-first century, democratic breakdown has still been limited to developing countries, a number of emerging economies (such as Hungary, Turkey, and Brazil) are regularly used as case studies for autocratisation 'episodes'. Most recently, since the inauguration of the Second Trump Administration, the authoritarian (ab)use of presidential executive orders to attack the courts, threaten law firms and universities, end birthright citizenship, and undermine government agencies and democratic institutions more generally (see the Trump Administration Tracker, `https://cohen.house.gov`), have made the world's largest economy the new poster child for democratic erosion.

Yet, the health of democracy in the U.S. is arguably shaped to a much lesser extent by the actions of the President and their administration, than those of the individual state executives: in the decentralized institutional system of U.S. federalism, it is the individual state governments which hold "the authority to administer elections, draw electoral districts, and exert police power" (Grumbach, 2023, 967). Hence, individual states decide who can and who cannot participate in American politics. Gerrymandering, voter repression, felon disenfranchisement, and the authoritarian use of police powers, among other practices, have been argued to render U.S. states into 'laboratories of authoritarianism' (Levitsky and Ziblatt, 2019). Importantly, it is now possible to track the quality of U.S. democracy at the sub-national level, thanks to the recent work of Grumbach (2023), who uses a large range of measures of electoral democratic quality covering inclusive suffrage and gerrymandering to create an index

[1] Bermeo (2016, 5) defines backsliding as a "state-led debilitation or elimination of any of the political institutions that sustain an existing democracy". Recently, scholars have moved away from using this term, given its lack of definitional clarity. While some of the criticism does not apply in our specific context, we follow this new practice and instead use autocratisation and democratic erosion (interchangeably).

of U.S. state-level democracy for the 2000-2023 period. His subsequent analysis reveals that "state governments have been leaders in democratic backsliding" (ibid: 967).[2]

In this paper, we exploit Grumbach's (2023) State Democracy Index (SDI) to ask about the *economic consequences of episodes of autocratisation in U.S. states*. The present Trump administration loudly postulates its economic prowess (see above quote), yet the quantification of real economy effects *caused* by U.S. national policies and executive orders is fraught with difficulty, not least because the myriad of government 'actions' makes it difficult to get a handle on what could be seen as 'treatment' or 'intervention' in an empirical model. Grumbach's efforts offer us a wealth of heterogeneous democratic dynamics in fifty states over a quarter-century, which combined with state-level data on income, inequality, impoverishment, innovation, and international exports available from the U.S. BEA, Census Bureau, and NSF, among others, allows us to address our research question.

Conceptual development The theoretical mechanism linking democratic quality to economic performance operates through electoral integrity and its effect on government accountability. Global evidence suggests that the integrity of elections constitutes a central institutional determinant of long-run economic outcomes (Acemoglu et al., 2019; Boese-Schlosser and Eberhardt, 2025). When elections are free and competitive, incumbents face credible threats of removal, which incentivizes them to provide broad-based public goods, maintain sound fiscal policy, and foster and invest in long-term productivity-enhancing areas such as innovation, human capital, and infrastructure (Besley and Kudamatsu, 2006; Dahlum and Knutsen, 2017; Knutsen, 2021). In contrast, when electoral integrity is eroded—through manipulation, suppression, or systemic bias—the accountability mechanism weakens (Schedler, 2002; Gehlbach and Simpser, 2015; Sato et al., 2022).

As accountability declines, governments face fewer constraints on rent-seeking and have diminishing incentives to pursue policies with diffuse, delayed benefits (Persson and Tabellini, 2002). Instead, policymaking and public spending become increasingly targeted toward narrow constituencies or coercive apparatuses that sustain political control first and consider economic prosperity second. This shift in focus and spending composition yet further undermines investment in human capital and public goods that underpin growth.[3] In short, electoral erosion sets off a causal chain in which weakened accountability drives cuts to public investment and policymaking, losing sight of the economic welfare of the average citizen, ultimately resulting in measurable, long-term economic decline.

[2] Grumbach (2023) further identifies Republican 'capture' of a state, i.e. holding both state senate and house, as a dominant factor in explaining democratic erosion. Our autocratisation episodes correlate with Republican state capture (correlation coefficients of around 0.3; episode onset and Republican state capture have correlation coefficients of 0.05.), but our analysis also suggests that the patterns in the economic implications of autocratisations we uncover are not primarily driven by who is in power in a state — see Section 4.4.

[3] The Republican war against 'woke' and 'extreme-left' DEI (diversity, equity, and integration) practices in public institutions and beyond is a point in case: existing research provides strong empirical evidence of the significant benefits of diversity for innovation (e.g. Moser et al., 2014; Nielsen et al., 2017; Moser et al., 2025).

Democratic erosion can also interact with trade, both as a potential transmission channel and as an external accountability mechanism. Democratic institutions provide the legal and political stability necessary for sustaining trade relationships by ensuring predictable enforcement of contracts, transparency in regulation, and protection against arbitrary policy shifts (Levchenko, 2007; Nunn, 2007; Yu, 2010; Araujo et al., 2016; Sheng and Yang, 2016; Zissimos, 2017). When U.S. states experience democratic erosion, they may become less attractive to international trading partners due to weakened rule of law, increased corruption, or politicized economic policymaking. These shifts can hinder economic openness and reduce export flows.

Economic Consequences We approach our central research question by examining the effects of autocratisation episodes on measures of economic prosperity and the broader implications for long-term economic potential. We investigate five dimensions of economic consequences at the state level: (i) average income; (ii) income distribution; (iii) poverty; (iv) innovative capacity; and (v) international export flows.

Measures of per capita or per worker income are natural candidates for our analysis, while the investigation of income inequality and impoverishment seeks to go beyond the *average* to understand whether certain parts of society are likely to be more affected by autocratisation than others. This line of reasoning would fit the analysis of any economy, regardless of its level of development. However, the mechanisms bringing about economic prosperity differ between developing and advanced economies: the latter are characterised by an endogenous growth process, which focuses our attention on proxies for innovation efforts (R&D) and outputs (patents). Finally, the study of state international exports can offer insights into whether the external accountability mechanism detectable in changing trade flow patterns between *national* economies following democratic 'retreat' (as documented in Boese-Schlosser et al., 2025) also operates at the *sub-national* level.

Autocratisation Episodes We dichotomise the Grumbach SDI into 'episodes' of democratic erosion, following the practices used in the construction of the V-Dem *Episodes of Regime Transformation* dataset (Edgell et al., 2020). Our analysis covers all fifty U.S. states (excluding the District of Columbia) over the 2000-2023 period. To demonstrate the robustness of our findings, we construct three alternative variants of autocratisation episodes that differ by the magnitude of institutional decline required to classify an episode: a *broad* variant that includes comparatively small declines, an *intermediate* variant requiring larger declines, and a *narrow* variant capturing only the severe cases. The resulting measures identify autocratisation in 33, 22, and 18 states, respectively.

There are two primary motivations for our modelling choice of autocratisation episodes: first, democratic change is rarely binary or immediate; instead, it unfolds gradually and is often interspersed with minor year-to-year fluctuations, some of which may reflect a healthy, dynamic political process rather than decline. An episode-based approach can pinpoint periods of substantial and sustained democratic erosion from short-term variations. This mirrors recent practice in the study of regime transformation in comparative political science, where processes of democratisation and autocratisation are conceptualized as clustered, path-dependent episodes rather than isolated events (e.g., Boese et al., 2023; Edgell et al., 2022; Maerz et al., 2024; Wilson et al., 2023).[4] Applying this logic to subnational democratic erosion allows us to more precisely assess its trajectory and consequences within the United States. Second, our mode of empirical inference follows the recent tradition in the literature on democracy and growth (Papaioannou and Siourounis, 2008; Acemoglu et al., 2019; Boese-Schlosser and Eberhardt, 2024) by adopting a *treatment effects* model (Chan and Kwok, 2022), which uses binary 'treatment' variables.

Empirical implementation We employ the Principal Component Difference-in-Differences (PCDID) estimator developed by Chan and Kwok (2022) to capture the economic consequences of autocratisation. The PCDID accounts for heterogeneous treatment effects and non-parallel pre-trends by using common factor proxies across states to capture time-varying unobserved heterogeneity and estimating the factor-augmented treatment regression separately in each state. The common factors are extracted from control states, i.e. those which never experienced any autocratisation episodes, and the inclusion of these proxies in the state-specific treatment regressions enables us, under reasonable and testable assumptions, to identify the causal impact of democratic erosion on economic outcomes.[5] This approach further enables us to map state-specific treatment effects to the number of years spent in an autocratisation episode, while accounting for the number of episodes experienced (following the practice introduced in Boese-Schlosser and Eberhardt, 2024). In robustness checks we account for the distorting impact of the Covid pandemic and further devise a specification that can capture the economic effect of autocratisation episodes while allowing for more flexibility in post-episode dynamics (e.g. a 'bounce-back' effect or a temporal spillover of the episode). Finally, we check the robustness of our findings to partisan power dynamics, which we show to be closely related to the incidence of episodes, hence naturally raising the question whether our analysis captures autocratisation episodes or partisan power at the state level.

Main Findings Our empirical results suggest a greater economic loss in terms of state GDP per capita with an increasing number of years spent in an episode of up to 1.5 percentage points, with stronger statistical evidence as we move towards more restrictive (narrower) definitions of autocratisation episodes. Autocratisation robustly increases income inequality within states, measured using top-income shares, as well as leading to a higher incidence of poverty.

[4] Recent research in economics (Boese-Schlosser and Eberhardt, 2024) has also shown that treating democratization as the successful culmination of an episode rather than a single point-in-time transition reveals stronger and more enduring effects on economic growth.

[5] Recent studies have adopted this implementation to study the effect of democratic regime change or collapse on economic prosperity (Eberhardt, 2022; Boese-Schlosser and Eberhardt, 2024, 2025) and the economic implications of 'excessive' financial deepening (Cho et al., 2026).

Hence, the average American directly suffers economically from democratic erosion in their state during the time frame studied; we can further detect diverging incomes and a rise in the number of impoverished people.

A second set of outcomes related to innovation investment and outputs provides strong evidence that private firms in treated states substantially reduce their R&D expenditure, while patent counts of treated states decline as well. These results indicate that democratic erosion undermines the endogenous growth engine and hence jeopardizes future economic progress.

Finally, shifting to a heterogeneous gravity model of export flows from U.S. states to international destination countries, we find no statistical evidence for a detrimental effect of autocratisation. Contrasted with ongoing research analysing democratic erosion in the context of export flows between *countries* (Boese-Schlosser et al., 2025), this implies that *sub*-national autocratisation does not result in the 'punishment' effect by democratic trading partners observed in the country-level analysis.

Related Literature and Contributions While the positive economic effects of democratic regime change are well established (Acemoglu et al., 2019; Knutsen, 2021; Eberhardt, 2022; Boese-Schlosser and Eberhardt, 2024) research on the consequences of democratic decline is relatively sparse. Recent research primarily covering developing countries finds evidence for an 'autocratic loss' (Boese-Schlosser and Eberhardt, 2025), i.e. a measurable decline in a country's income per capita after democratic breakdown, during the current wave of autocratization. Work by Blattman et al. (2025) suggests that an 'autocratic penalty' is concentrated in countries with 'personalist' rule, where power is captured by an individual or a small elite.

The above studies adopt a strict distinction between democracy and autocracy which is too crude when considering the political economy of many advanced economies, virtually all of which have been (and remain) democratic since the 1990s, yet collectively experienced a noticeable deterioration in democratic institutional quality since the late 2000s (Nord et al., 2024). A large literature in political science is devoted to defining these autocratisation episodes in cross-country data (e.g. Edgell et al., 2020) and studying the characteristics and determinants of such episodes of institutional deterioration (e.g. Bermeo, 2016; Waldner and Lust, 2018; Levitsky and Ziblatt, 2019; Lührmann and Lindberg, 2019; Grillo and Prato, 2023; Grillo et al., 2024; Riedl et al., 2024). This strand of research is not without controversy, including regarding the very *existence* of such episodes, with some suggesting quantitative evidence for episodic autocratisation is due to the biased subjective measurement of democratic institutions and their temporal dynamics (see Little and Meng, 2024a,b; Knutsen et al., 2024).

A related literature in economics and political science deals with the rise of populism (see the survey by Guriev and Papaioannou, 2022). One example of the emerging research on the economic consequences of populism is the work by Funke et al. (2023), who use data for sixty major economies and the synthetic control methodology to show that populism considerably slows economic progress (while leaving inequality unchanged). The analysis by Born et al.

(2019) uses the same methodology to find no differential effect of the first Trump administration on the U.S. economy. Other studies linking populism in a single country (Marzetti and Spruk, 2022; Woo-Mora, 2025) or a small number of countries (Absher et al., 2020) to economic outcomes using synthetic control find detrimental effects of populist leadership.

While the literature in political science offers a large number of country case studies, there is comparatively limited quantitative analysis of *sub-national* democracy and its dynamics. Recent work by Michel (2024) studies sub-national elections and opposition control in 371 states of 18 democratic Latin American countries, finding a positive effect for the 'health' of national democracy when sub-national control of the opposition increases. The study by Grumbach (2023) on U.S. states, which is central to our work, in contrast labels states as 'laboratories for democratic backsliding' and finds a strong link between state capture by the Republican party and a deterioration in the state democracy index he develops. To the best of our knowledge, there is only one empirical study linking sub-national democracy to economic prosperity. Iddawela et al. (2021) construct sub-national measures of government quality from survey data in 22 African countries and find a positive causal link between regional democratic change and regional economic prosperity, even when accounting for national democracy levels.

Our paper makes two important contributions to the literature: first, we introduce a novel dataset of autocratisation episodes across all fifty U.S. states from 2000 to 2023, derived from the Grumbach (2023) SDI. By offering three variants of episodes, our dataset offers a built-in robustness check to counter concerns over definitional arbitrariness. Second, we provide the first empirical assessment of the economic effects of subnational democratic erosion in the U.S.. Our findings suggest that autocratisation is not only normatively concerning but also economically costly, with consistent evidence of increasing deprivation, widening inequality, and reduced innovation.

The remainder of this paper is structured as follows. In Section 2 we introduce the data, devise the episodes, and provide descriptive analysis. Our empirical methodology is developed in Section 3, followed by the discussion of the results in Section 4. Section 5 concludes.

2 Data and Transformations

2.1 State-level Democratic Erosion

Data Source We use the updated state-level democracy data from Grumbach (2023, State Democracy Index, v. 2.0), covering all states (but not the District of Columbia) over 2000-2023.[6] The SDI is constructed from 49 measures and indicators (covering electoral democratic quality related to information on polling day wait times, voter registration, felon disen-

[6]The Grumbach data excludes the District of Columbia (DC), since DC has no voting representation. While it is possible to push the data back to cover earlier periods, the justification provided by Grumbach (2023) to focus on *contemporary* America and hence avoid challenges of data availability and temporal comparability of state variation in democracy, applies in the present study as well.

franchisment and proxies for gerrymandering, among others) using Bayesian factor analysis, intended to capture a latent measure of democratic performance. Conceptually, this brings the state-level indices close to the electoral democracy index of the Varieties of Democracy (V-Dem) Project (building on the theories of Dahl (1971, 2000) in political science and Schumpeter (1942) in economics), focusing most attention on free and fair elections, and inclusive suffrage. See Appendix A for some thoughts on the purpose and applicability of the SDI compared with cross-national indices such as those compiled by the V-Dem project.

State Democracy Index: Transformations and Descriptives We rescale the Grumbach index to lie between 0 and 1, mimicking the practice of V-Dem indices.[7] Panel (a) of Figure 1 presents the annual average (rescaled) Grumbach index alongside the V-Dem polyarchy index (Coppedge et al., 2025, v15) for the United States: the overall patterns speak to one of Grumbach's (2023, 967) main conclusions, namely that states have been "leaders in democratic backsliding in the U.S. in recent years", especially following the "Republican gains in state legislatures and governorships in the 2010 election" (968).

Panel (b) of the same Figure illustrates how the share of states with the 'worst' democratic institutions has expanded over time: we determine the threshold SDI values for the four quartiles of the entire distribution of the rescaled Grumbach index in 2000 (values reported on the right of the plot) and shade the quartiles in different colours. The distributional significance of the top quartile is largely unchanged, and the 3rd quartile has been squeezed marginally, in that in 2023 it captures only one fifth instead of a quarter of all states. The 2nd quartile has been squeezed the most, from one quarter in 2000 to around one tenth in 2023. These changes have all been to the 'benefit' of the bottom quartile: the share of states with low levels of institutional quality (below SDI 0.62) has expanded from 0.25 to over 0.40!

Panel (c) offers a geographical illustration of the rise and fall in state-level democratic institutions: we compute the total change in the rescaled Grumbach SDI between 2000 and 2023 and plot states in green shades if this change is positive and in red shades if it is negative. States with the most substantive experience of democratic erosion (shaded in red) are primarily concentrated where the Midwest meets the South, as well as further out West (Wyoming, Utah).[8]

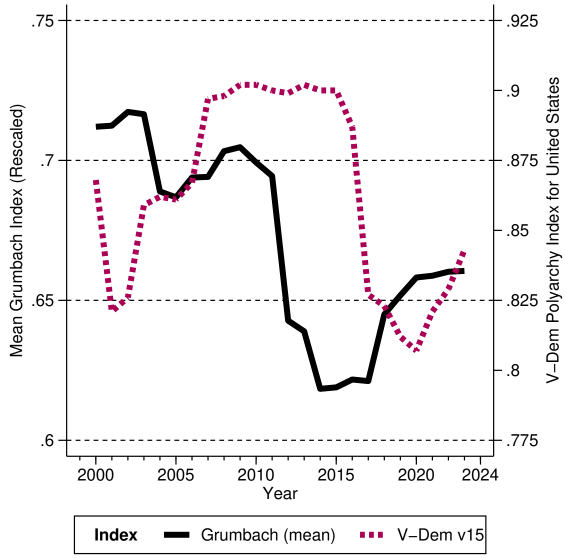
Autocratisation Episodes: Variants and Descriptives In developing three alternative variants of 'episodes' of democratic erosion, we primarily rely on what we term the 'trigger' and 'total drop' conditions: first, for all three alternatives, we adopt a decline in the rescaled Grumbach index of -0.025 as the 'trigger' for potential episodes. This represents a substantial drop (the 18th percentile of the distribution of changes in the rescaled SDI), equivalent to

[7] We do so for ease of presentation; absolute *magnitudes* of an entry in the rescaled Grumbach index should not be compared to the V-Dem polyarchy or other indices, though patterns of variation are meaningful.

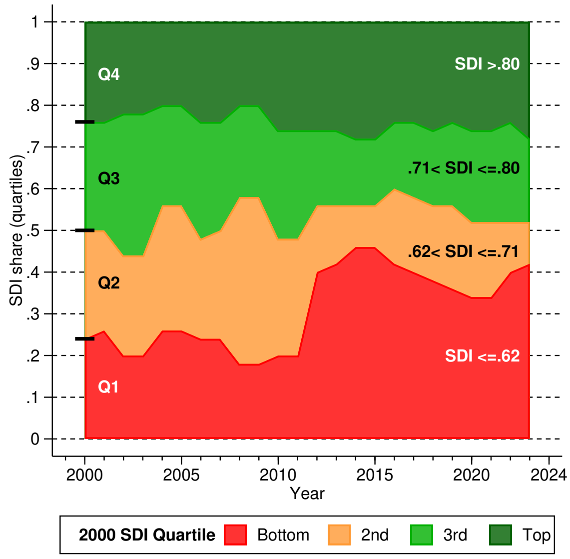
[8] For illustrative purposes we exclude Alaska (light green) and Hawaii (orange) from this map.

Figure 1: State Democracy Index (rescaled) — Descriptives

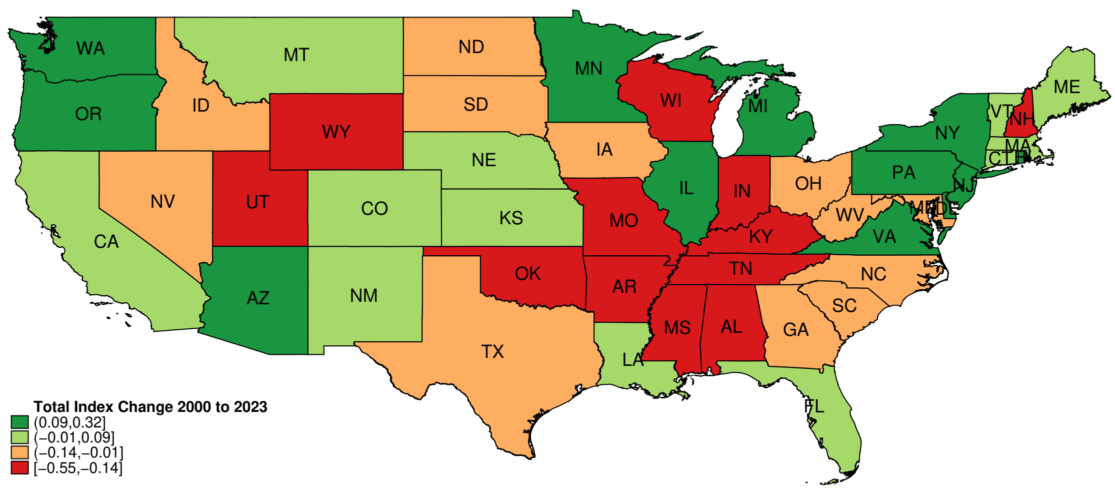
(a) Rescaled SDI (solid), V-Dem USA (dashed)



(b) Evolution of Rescaled SDI from 2000 Quartiles



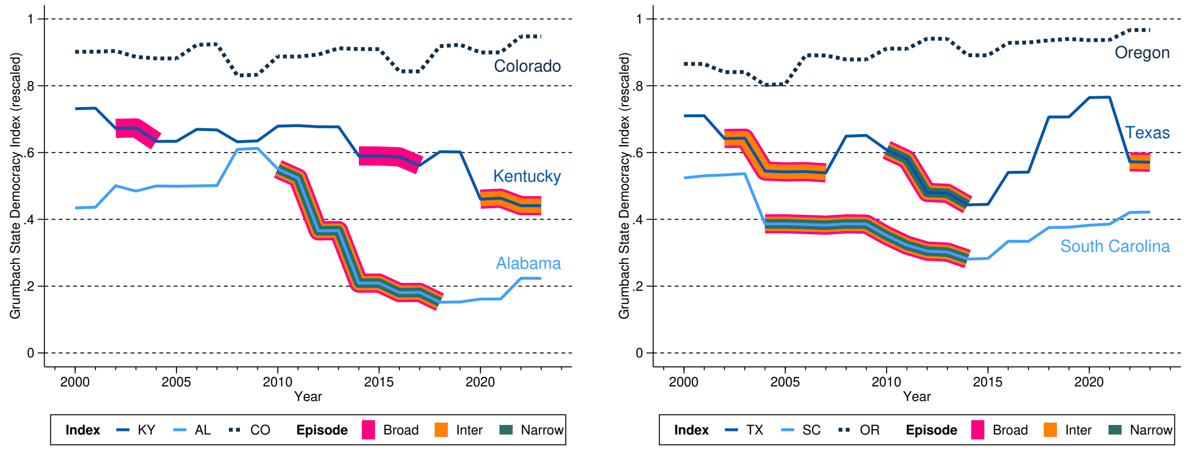
(c) State-level Change in the Rescaled SDI 2000-2023



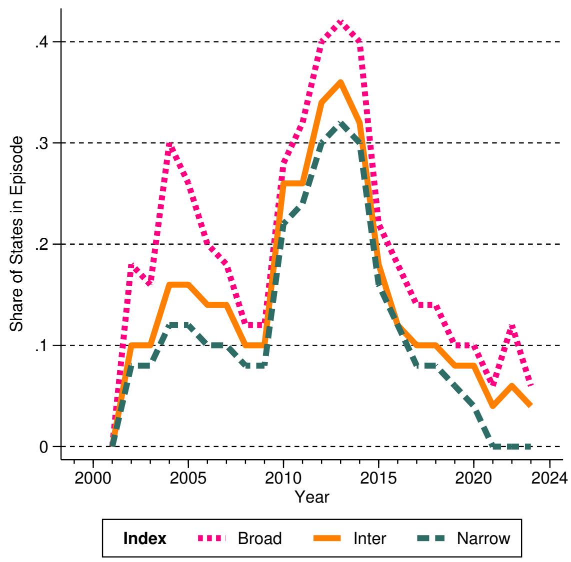
Notes: The black solid line in panel (a) represents the unweighted mean of the rescaled Grumbach index (SDI) across all 50 states in each year, the dark pink dashed line the V-Dem (v15) polyarchy index for the United States. Note that Freedom House has downgraded their 'political rights' measure for the U.S. from 1 to 2 (scale of 1-7, best to worst) since 2017 (survey edition 2018). Panel (b) studies how the distribution of the rescaled Grumbach index (SDI) changed over time: using four colours, we determine the four quartiles of the distribution in 2000 (black lines on the left, the SDI thresholds are indicated on the right) and then illustrate how these expanded and contracted over time. In panel (c) we compute the change in the rescaled Grumbach index over the 2000-2023 period by state and present a map of the contiguous U.S. states shaded by the quartiles of the distribution of changes (from large negative change in red to large positive change in dark green, with orange and light green for the intermediate categories with below and above zero total change) — see legend for categories. For ease of presentation, we exclude Alaska (light green) and Hawaii (orange).

Figure 2: Episodes of Democratic Erosion

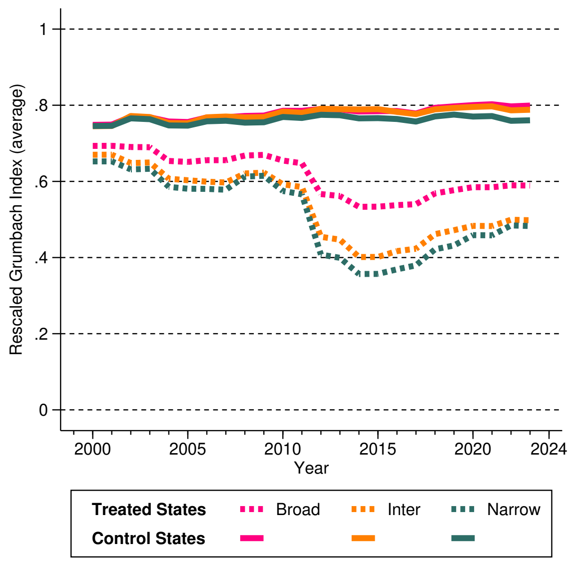
(a) Variants of Episodes: Illustrations



(b) Share of States in an Episode



(c) Average SDI in Treated and Control States



Notes: The plots in panel (a) illustrate alternative definitions of democratic erosion in individual states. Plots for all states can be found in Appendix Figure B-3. In panel (b) we illustrate the share of states experiencing an episode in any given year for each of our three episode definitions. Panel (c) illustrates the average SDI evolution in control states that never experienced an episode (solid lines) and treated states that experienced one or more episodes (dashed lines) for the three definitions of autocratisation, respectively.

half a standard deviation.[9] Second, potential episodes are promoted to actual episodes if they constitute a 'total drop' in the rescaled index for the entire episode of at least -0.1 ('broad' definition), -0.15 ('intermediate'), or -0.2 ('narrow'), equivalent to 2, 3 and 4 standard deviations. For practical implementation, we require some additional rules to help us determine how long an episode lasts: (a) an episode ends in the year before the first difference of the index turns positive (exceptions apply, see next point); (b) an episode can continue even if there are single *intermittent* years with positive changes, provided these changes are each smaller than $+0.01$;[10] (c) we allow for single-year episodes, subject to the 'total drop' condition.[11]

This procedure yields a 'broad' definition of autocratisation for 33 states experiencing 46 episodes, an 'intermediate' definition for 22 states (27 episodes), and a 'narrow' definition for 18 states (20 episodes) — for all three definitions, the median time a (treated) state spends in one or more episodes is 7 years.[12] Respective control samples of states without autocratisation episodes contain 17, 28, and 32 states. We always present our results for *all* states with autocratisation episodes, but to underline that states experiencing repeated episodes do not distort our findings, we further provide results for only those states that experienced a *single* episode. For the latter, we observe 22 (instead of 33), 18 (27), and 16 (18) states, adopting the broad, intermediate, and narrow definitions of autocratisation, respectively. For more details, see Appendix Tables B-1 and B-2.

Panel (a) of Figure 2 provides some illustrations from our episodes data (plots for all states can be found in Appendix Figure B-3). These examples highlight the differences between our three definitions. For instance, in the left plot, Kentucky features two episodes around 2003 and 2015, which are only picked up by the 'broadest' definition of an autocratisation episode (pink bands), whereas the episode around 2020 is picked up by the 'intermediate' definition as well (hence, pink and orange bands). The 'narrowest' definition does not pick up *any* episodes for Kentucky (no teal-coloured bands). For Texas in the right plot, an early and late episode are only picked up by the 'broad' and 'intermediate' definitions, whereas the episode after 2010 is picked up by all three definitions. Oregon and Colorado do not have any episodes. Note that in the latter, the drop in 2008 measures -0.094 and that in 2016 -0.066—hence

[9] The first difference of the rescaled Grumbach index has a mean and median of around 0, a standard deviation of 0.049, and an interquartile range (IQR) of 0.015. In comparison with the V-Dem ERT measure for autocratic episodes (Edgell et al., 2020), which adopts a trigger of -0.01 (the standard deviation and IQR of the change in the V-Dem polyarchy index between 2000 and 2023 are 0.36 and 0.01, respectively), our trigger measure is quite conservative. Nevertheless, this first step yields 202 *potential* episodes, experienced across 48 states in our sample (only Nebraska and Vermont never experienced an annual decline of this magnitude).

[10] The evolution of democratic quality in individual states seems to be characterised by fits and starts, when a less granular view of their evolution would suggest 'decline' or 'increase' over the period in question. We see our provision (b), along with the construction of episodes as a treatment variable in the first place (rather than the use of the state index as an independent variable), as a means to tackle any counter-intuitive idiosyncracies as well as the unavoidable uncertainties of producing democracy indices.

[11] We obtain identical episodes if we use the original SDI (range -3.39, 1.66) with 'trigger' 0.125 ($1/2$ sd) and a 'total drop' of at least 0.5, 0.75 and 1 (2, 3, and 4 sd, respectively).

[12] In our empirical analysis, we cannot separate individual episodes, only states, hence we relate the state treatment effect to the total number of years the state spent in episodes. We do condition our results on the number of episodes per state.

both are below the required minimum -0.1 'total drop' for even our broadest definition of autocratisation. The case of Alabama also highlights the inclusion of single, intermittent years of positive index change as part of an episode: 2013, 2015, and 2017 all feature as periods of minimal positive change followed by large drops of -0.157, -0.029, and -0.029, respectively. See also South Carolina for similar 'intermittent years' in 2005 and 2008.

Panel (b) of Figure 2 charts the share of states experiencing an episode in any one year between 2001 and 2023. This illustrates the increasing democratic erosion following the 2010 election, with up to 40% of states experiencing an episode in 2013. Although the end of our sample period is characterised by an absence of episodes, it should be emphasised that the *average* SDI is around 8% lower in 2023 compared with 2002 (its peak). We can also see in panel (c) of Figure 2 that states which feature in our control samples on average marginally increased their SDI during the sample period (solid lines), whereas states in the treated samples (dashed lines) on average witnessed increasingly large declines as we narrow the definition of autocratisation. The cessation of most or all democratic erosion after 2020, depending on our definition, is apparent in the plot in panel (b) of Figure 2. However, it should be pointed out that many states experiencing autocratisation by then had settled at *permanently* lower levels of democratic quality: first and foremost Tennessee (drop in the rescaled Grumbach index of -83% between 2000 and 2023), Wisconsin (-68%), Alabama (-49%), Arkansas (-46%), and Indiana (-46%), but these are not isolated cases — see Appendix Table B-2.

While these statistics certainly do look dramatic, it bears reminding that our rescaled state democracy index *accentuates* any variation within and between U.S. states. A devil's advocate may suggest that these magnitudes are meaningless, that during this period democracy *levels* in the U.S. were relatively high, and may point out that the variation implied was contained within relatively narrow ranges of cross-country democracy indices, for instance, as we have seen, between 0.80 and 0.90 for the V-Dem polyarchy index (Coppedge et al., 2025). We are happy to accept this scepticism, but note that if our autocratisation episodes are meaningless noise, derived from meaningless variation in the state-level data, then all that lack of meaning should translate into a lack of consistent empirical evidence for any meaningful economic effects of democratic erosion.[13]

Autocratisation and State Partisanship U.S. state politics has become more partisan over our sample period — in Appendix Table C-1 we provide a visual representation of the dynamics of state partisanship, highlighting when one or the other party holds the majority in both the state house and senate. The 2010 election represents a watershed in more than one sense: it marked the start of Republican dominance across states (the 'net red' count peaks at 19 states in 2014) and of the decline of the number of states with power-sharing arrangements

[13]As an aside, the Center for Systemic Peace polity score designated the U.S. as an anocracy (+5) in late 2020, as a democracy (+8) in 2021, and as a borderline autocracy (0) in early 2025 (after the end of our sample period).

or party majority in the house but not the senate or vice versa (14 'mixed' states in 2008, but merely 4 in 2023).

We conduct some simple analysis focused on the party-political dominance in a state during autocratisation episodes. The simple pairwise correlation coefficients between our autocratisation dummies and the partisan capture of the state by the Republicans, at around 0.3, are modest but not insubstantial. Zooming further in on the start of an episode and the dynamics of state partisanship, we conduct univariate event analyses for Republican state capture: Appendix Figure C-2 charts the propensity of a state safely being in Republican hands in the five years before and after the start of an episode. Yet we only find a statistically significant effect for the broad and intermediate definitions, and notably only in a single year (the first year *after* an episode has started).

In Appendix Table C-2 we simply count the number of episodes that took place when Republicans, Democrats, or neither (Mixed) dominated state politics. For Republican dominance we distinguish a number of special cases where during the episode a first year of Mixed or Democrat dominance is followed by several years of Republican dominance. Tallying up all episodes with Republican involvement indicates that, in line with Grumbach's (2023) analysis for the continuous state democracy index, there is a strong correlation between autocratisation and Republican dominance.

We visually convey this correlation in Appendix Figure C-1, which also charts the count of states 'captured' by Republicans (solid black line) and Democrats (dashed black line): using the broad definition of an episode in panel (a), at its peak in the mid-2010s, around 70% of all states captured by the Republicans were experiencing an autocratisation episode, whereas the peak for states captured by Democrats around the same time was below 20%. As can be seen in panels (b) and (c) for the intermediate and narrow definitions, the patterns of democratic erosion in Republican states are very similar as we tighten the definition of an episode.

Appendix Table C-3 presents results from simple linear probability models for autocratisation. We regress the episode dummy (separately for each of the three definitions) on state and year fixed effects as well as two indicators for state-years when Republicans, respectively Democrats, enjoyed the majority in both the state house and senate (the omitted category is Mixed/power-sharing state-years). Benchmark results suggest that the propensity of the state experiencing an autocratisation episode is 17-20% higher (9-10% lower) relative to the 'power sharing' baseline when the Republicans (Democrats) have captured the majority—all these estimates are statistically significant at the 5% level. Even when we further condition on the level of the rescaled Grumbach index, the results for the Republican dominance remain highly significant and positive (10-12%), whereas those for the Democrats are only significant (but still negative) in the most restrictive definition of autocratisation.

The Fundamental Components of Autocratisation Episodes The Grumbach (2023) SDI combines 49 indicators and indeces related to electoral performance, voter rights, and gerrymandering, but which of these have the most influence on democratic erosion? We estimate elastic net LASSO models of the respective autocratisation definition including all 49 variables and report the least squares (linear probability model) estimates for the 'selected' specifications (this analysis covers 2000-2018). The LASSO adds a penalty term to a least squares minimisation problem and hereby forces variables with very small predictive power to have a zero coefficient. An elastic net LASSO additionally avoids dropping variables when these are highly correlated and hence keeps the final model relatively stable.

We first estimate the elastic net LASSO without state fixed effects, hence including all states, with marginal effects (multiplied by 100 times the standard deviation of the respective variable) from the linear probability model of 'selected' variables reported in Panel A of Appendix Table C-4. For the Broad and Narrow autocratisation definitions, this yields a relatively small number of predictors, with proxies for gerrymandering most prominent (in numbers and magnitudes). For the Intermediate definition we do not gain any meaningful insights, since a very large number of predictors are selected. In Panel B of the same table we include (partial out) state and year fixed effects, resulting in much more parsimonious sets of predictors for the subset of states which experienced an episode during the sample period. Again the gerrymandering proxies play a prominent role.

Taken together, it appears that gerrymandering activities play a dominant role in explaining the emergence of democratic erosion we capture with our episodes.

2.2 State-level Macro and Bilateral Trade Data

From the U.S. Bureau of Economic Analysis (BEA), we take state-level real GDP in 2017 US$ values, as well as population and employment headcounts to construct state real GDP in per capita or per worker terms and the employment growth rate (a control variable). From the U.S. Census Bureau (CB) we take state-level data for median household income (in real 2023 US$), the headcount of individuals living in poverty, as well as the percentage of the population living in poverty (poverty being defined by the poverty line). In extensions, we adopt CB data on the number of individuals in each state without any health insurance coverage and the state Gini coefficient. The World Inequality Database (WID) provides U.S. state-level data for top income shares (top-10% and top-1%). Patent data are assembled from historical reports by the U.S. Patent and Trademark Office (USPTO): we study total patent grants, which include utility, plant, and design patents; and grants of utility patents.[14] Data on current business R&D expenditure at the state level are taken from reports by the U.S. National Science Foundation (NSF): we distinguish state-level total business R&D spending (including funding from federal or state government, as well as from businesses and organisations outside the state or the U.S.) and businesses' own R&D expenditure. We use the BEA (nation-wide)

[14] The average time from application to grant is presently around 13-15 months for design patents, 18 months for plant patents, and 35 months for utility patents.

private domestic investment deflator to transform these into real values and as a robustness check adopt R&D/GDP ('R&D intensity') using current value terms. All of the above are available for 2000-2023, with the exception of the Gini (from 2006), the WID data (up to 2018), the granted patents (up to 2020) and the R&D data (up to 2022). Appendix Tables B-3 and B-4 provide descriptive statistics by state, in Appendix Figure B-2 we chart the median evolution of the economic variables of interest for treated and control states over 2000-2023.

State exports in US$ are provided by USA Trade Online, a division of the U.S. CB — these data are available from 2002 to 2023. Additional data for the gravity regressions include the BEA state population headcount and state GDP as well as state per capita personal income. Destination (country-level) data on population, GDP and per capita GDP are taken from the World Bank Word Development Indicators database. In line with standard practice in the literature, all monetary variables in the gravity regressions are in current (US$) value terms.

3 Methodology

3.1 Factor-augmented Regressions

Our empirical methodology builds on the common factor framework, which we employ to capture unobserved time-varying heterogeneity. This is a popular approach exploited in the synthetic control method (Xu, 2017), difference-in-differences models (Gobillon and Magnac, 2016; Chan and Kwok, 2022), and other approaches where omitted variable bias is of concern (Pesaran, 2006; Bai, 2009). In regression models, the factor proxies are included as additional covariates, representing a control function, akin to the practice in firm-level production function estimators (Olley and Pakes, 1996; Levinsohn and Petrin, 2003).[15] In the context of economic development, capturing unobserved heterogeneity, in essence total factor productivity, has historically been approached in a variety of ways. The 'Barro regressions' of the early 1990s (Barro, 1991), for instance, attempted to include as many growth determinants as possible as additional regressors, while the Bayesian Model Averaging approach (Sala-i Martin, 1997; Sala-i Martin et al., 2004) focused on the 'relevance' of included determinants à la Barro—both approaches are now deeply unpopular, perhaps for the simple reason that, varied data availability and quality aside, these 'kitchen sink regressions' saturating the empirical model with additional determinants can only proxy for known phenomena, but cannot accommodate the 'unknown unknowns'.[16]

The common factor framework addresses this as a dimensionality problem (too many de-

[15]In a simpler panel model with time-*invariant* unobserved heterogeneity, country fixed effects are control functions which capture these unobservables. The factor model approach, using factor proxies and country-specific parameters (factor loadings), more flexibly captures time-*variant* unobserved heterogeneity, which is why it is sometimes referred to as 'interactive fixed effects'.

[16]In parallel, standard structural gravity estimation is concerned about unobserved multilateral resistance and addresses this by means of dummy variable saturation, which rules out the analysis of any monadic (country/state-specific) variables such as democracy or its demise.

terminants) and offers a solution using dimensionality-reducing tools. A simple analogy may help illustrate the basic intuition of how and why this approach works. Consider unobserved heterogeneity in the form of different paint colours, from vibrant reds to emerald greens to stylish ochres, one unique colour for each unit of analysis (say, countries or U.S. states), and assume these unobservables are related to observable variables (y, x) in a simple regression model of y regressed on x. The Barro regression and BMA strategies would be to explicitly control for as many distinct colours as possible, with the latter employing an algorithm to determine which colours really 'matter' for inclusion in the model. The factor model instead postulates that all unobserved time-varying heterogeneity can be represented by a small number of building blocks ('factors') with country/state-specific parameters ('factor loadings'). In our analogy of colours, we need exactly *four* such building blocks to be able to capture every single possible colour: the primary colours (red, blue, yellow), and white.[17] The complexity and vastness of possible differences in colours has been reduced to a very small and easily-managed number. Like primary colours, common factors are profound and 'mutually exclusive' (orthogonal to each other). Like combining colours, we can 'mix' factors in different proportions. Conveniently, we do not need to know what the factors are, or indeed what their relevant combinations (loadings) are: provided we employ sufficient factors to capture the heterogeneity, and have sufficient data to estimate the factors, we can proxy the time-varying unobserved heterogeneity in a common factor framework with state-specific factor loadings.

Our empirical approach laid out below proxies common factors only using estimates derived from the control sample of 'never-treated' states and includes these proxies in the treatment regressions for states experiencing autocratisation episodes. Hence, we exploit the DID data structure, and unlike alternative factor-based methods do not adopt data from 'not-yet treated' states in constructing factor proxies. There are two main concerns: (i) do we have the right number of factors, and (ii) how can we gauge the 'relevance' of the common factors in both treated and control samples. In our colour analogy, it may be that none of the colours we want to capture involves mixing white, hence we can simply employ the three primary colours on their own. However, adding white to the offering does not undermine our ability to capture all unobserved heterogeneous colours. There is an econometric equivalent to this in the factor model: adding too many common factor proxies does not lead to bias (Moon and Weidner, 2015). In practice, we will provide results using several specifications with four or five factors included. A separate worry is that the control sample maybe only includes colours derived from red and blue, whereas the treated sample additionally requires shades of yellow as well. In this situation our model would be misspecified, since the common factors differ between treated and control samples. We can gauge whether this problem is present in our treatment models with an 'Alpha' test for expected factor loading equality between treated and control samples (Chan and Kwok, 2022). This establishes whether the 'information' we extract from

[17] We use paints/pigments ('subtractive mixing') for this analogy, if we considered light ('additive mixing') instead, then a combination of the primary colours would also capture white light.

the control sample is equally relevant in the treated sample. We now provide a more formal introduction to our empirical implementations.

3.2 Heterogeneous Difference-in-Differences Model

We employ the Chan and Kwok (2022) Principal Component Difference-in-Differences (PC-DID) estimator which allows for non-parallel trends and heterogeneous treatment effects.[18] This estimator enables us to study staggered, non-absorbing treatments such as switching in and out of autocratisation episodes. In the potential outcomes framework, the observed outcome of a treatment D_{it} (episode) in U.S. state i at time T_0 can be written as

$$y_{it} = D_{it}y_{it}(1) + (1 - D_{it})y_{it}(0) = \Delta_{it}\mathbf{1}_{\{i\in E\}}\mathbf{1}_{\{t>T_{0i}\}} + y_{it}(0) \tag{1}$$

$$\text{with} \qquad y_{it}(0) = \varsigma_i + \beta_i' X_{it} + \mu_i' f_t + \tilde{\epsilon}_{it}, \tag{2}$$

where the $\mathbf{1}_{\{\cdot\}}$ are indicator variables for the treated states and the time period treated; Δ_{it} is a time-varying heterogeneous treatment effect, X any observed controls (these do not have to be included), $\mu_i' f_t$ represents a set of unobserved common factors f_t with state-specific loadings μ_i. $\tilde{\epsilon}_{it}$ is the error term.

The treatment effect can be decomposed into $\Delta_{it} = \overline{\Delta}_i + \widetilde{\Delta}_{it}$, with $E(\widetilde{\Delta}_{it}|t > T_{0i}) = 0$ $\forall i \in E$ since $\widetilde{\Delta}_{it}$ is the demeaned, time-varying idiosyncratic component of Δ_{it}. $\overline{\Delta}_i$ is the individual treatment effect averaged over the treatment period. The reduced form model is

$$y_{it} = \overline{\Delta}_i\mathbf{1}_{\{i\in E\}}\mathbf{1}_{\{t>T_{0i}\}} + \varsigma_i + \beta_i' X_{it} + \mu_i' f_t + \epsilon_{it} \qquad \text{with} \qquad \epsilon_{it} = \tilde{\epsilon}_{it} + \widetilde{\Delta}_{it}\mathbf{1}_{\{i\in E\}}\mathbf{1}_{\{t>T_{0i}\}}, \tag{3}$$

where ϵ_{it} has zero mean but can be heteroskedastic and/or correlated over time and space.

We estimate the heterogeneous treatment effect $\overline{\Delta}_i$ in two steps: first, using PCA, we estimate proxies of the unobserved common factors from the residuals $\hat{e}_{it}$ of the following auxiliary regression in the control sample: $y_{it} = b_{0i} + b_{1i}' X_{it} + e_{it}$, or, if we exclude additional controls, $y_{it} = b_{0i} + e_{it}$; second, using least squares, we estimate a treatment regression equation augmented with the factor proxies as additional regressors (control functions):

$$y_{it} = b_{0i} + d_i\mathbf{1}_{\{t>T_{0i}\}} + a_i'\hat{f}_t + b_{1i}' X_{it} + u_{it}, \tag{4}$$

where $\hat{f}$ are the estimated factors from the first step and d_i is the state-specific treatment estimate. We estimate (4) augmented with four or five common factors. The model is estimated for each treated state separately and only 'never-treated' states are used in the construction of the factor proxies: no 'not-yet-treated' observation enters the control group.

The above setup can accommodate endogeneity of treatment D_{it} such as correlation between treated units and factor loadings, the timing of treatment and factor loadings, or

[18] Applications of the PCDID include Boese-Schlosser and Eberhardt (2024, 2025) and Cho et al. (2026).

between observed covariates and timing or units of treatment. States, most importantly those in the treated versus control samples, can have different (non-parallel) trends.

There are three assumptions for the consistency of $\hat{d}_i$ or its Mean Group average: (i) all unobserved heterogeneity determining y can be proxied by a small number of factors (following Pesaran, 2006; Bai, 2009; Athey et al., 2021), such that u in equation (4) is orthogonal to the treatment; (ii) there is expected factor loading equality between treated and control samples: the factors extracted from the control group are equally 'relevant' in the treatment group; and, (iii) our use of an estimator assuming an absorbing treatment is not violated by the economic implications of the post-treatment period: in essence, we assume that an autocratisation episode is an interim 'regime', at the end of which the economy of treated states returns to a pre-regime equilibrium *without* temporal spillovers, such as continued economic decline or an economic 'bounce-back' effect after an episode has ended.

We address the first assumption by adding a substantial number of estimated factors (four or five) to the treatment regressions. The Alpha test introduced below will formally address the concerns raised in the second assumption. An extension to the PCDID, which involves a set of 'post-treatment dummies' in the treatment regressions (covering the four years after the end of an autocratisation episode), seeks to address the third assumption — see below.

In other robustness checks, we include/exclude employment growth as an additional covariate X.[19] We also consider the inclusion of year dummies for the Covid years 2020-2022—in particular in the analysis of innovation (where patent and R&D data ends during Covid) but also for the export flow results, this strengthens our empirical findings.

Typically, results would be presented in form of average treatment effects (ATET), for which inferential statistics can be based on the variance estimator of Pesaran (2006): $\widehat{\text{var}}(\hat{d}^{MG}) = [N(N-1)]^{-1}\sum_{i=1}^{N}(\hat{d}_i - \hat{d}^{MG})^2$ for N treated states. However, we also adopt multivariate running line regressions following Royston and Cox (2005) to provide a graphical representation of the relationship between treatment effect magnitude and years of treatment (time spent in autocratisation episodes). We regard running line regressions as 'local ATET', where 'local' refers to a similar number of years spent in autocratisatio episodes, and hence adopt the standard errors from this methodology (see Boese-Schlosser and Eberhardt, 2024). Our result graphs do not depict an event analysis but map the magnitude of the treatment effect to the total number of years spent in one or more autocratisation episode(s).

[19] We considered others, e.g. population growth or state international exports, but these typically represented 'bad controls' or the specifications failed the Alpha test. Work by Bretschneider and Westerlund (2025) indicates that the PCDID is inconsistent if included covariates are correlated with the unobserved common factors, due to a misspecified first step (no factors). We obtain qualitatively identical results for GDP per capita and headcount of the poor (outcomes where our models include employment growth as a control) if we demean the variables with respect to state means.

3.3 Diagnostic Testing

We inform our PCDID analysis with three sets of diagnostic tests: first, the Chan and Kwok (2022) Alpha test for expected factor loading equality between treated and control samples; second, a Wald test for specifications including an additional control (we limit this to the inclusion of the state employment growth rate since adding other controls was always rejected by the test) to determine whether the control might constitute a 'bad control', following Angrist and Pischke (2008); and third, a test of the core assumption of our DID estimator that the treatment effect does not 'carry over' after the end of the episode.

Identification (Alpha) Testing The Alpha ('factor spanning') test informs us whether the unobserved time-varying heterogeneity in the control sample spans the same space as that in the treated sample. The PCDID allows for non-parallel trends across countries, most importantly between those in the treated and control samples.[20] But if the underlying latent factors differ between treated and control samples (recall the example of primary colours), then adding estimated factors from the control sample in the treatment regression will not be sufficient to capture the unobserved heterogeneity in the latter, and the PCDID is misspecified.

The Alpha test is implemented by (i) computing the cross-section average of the residuals in the control sample regressions, $\bar{\hat{e}}_t = N^{-1}\sum_i \hat{e}_{it}\,\forall i \notin E$, (ii) including this averaged residual as an additional regressor in the treatment regression replacing the estimated factors,

$$y_{it} = b_{0i} + d_i \mathbf{1}_{\{t>T_{0i}\}} + c_i' \bar{\hat{e}}_t + b_{1i}' X_{it} + u_{it}, \tag{5}$$

and, (iii), applying a t-test for whether the (Mean Group) average of $\hat{c}_i\,\forall i \in E$ across all treated states is equal to one ($\Leftrightarrow N^{-1}\sum_i(\hat{c}_i - 1) = 0$).[21]

We report the outcomes of the Alpha test for each specification (the test does not differ by the number of factors included in the model) in two ways: first, in the main text, we use a simple set of symbols (e.g. ••○) to highlight passed (•) or failed (○) Alpha tests for the three alternative definitions of episodes; second, we report the equivalent t-ratio for each test in Appendix Tables D-1 to D-3, with an absolute value of $t > 1.96$ rejecting the null and rendering the PCDID model in question misspecified.

Testing for bad controls For the test for 'bad controls' we regress the treatment dummy on the estimated factors and the control variable(s) in question: our assumption (and null hypothesis) is that once the factors are included, the control variable is not statistically significantly

[20] This is possible since individual countries can have unrestricted parameter coefficients on the common factors included in the model, which could be positive or negative.

[21] The intuition of why the averaged c should be one is as follows: consider a standard OLS panel regression of y on some x with year fixed effects, then extract the estimated coefficients on the year dummies as a single variable f and add this into the same OLS regression of y on some x (without the year dummies): the coefficient on the variable f containing the year dummy coefficients will be one.

related to the treatment. We can test this assumption by regressing the episode indicator on estimated factors and the control(s) in the treated sample and carrying out a Wald test for the (joint) insignificance of the control(s). If the null is rejected, we need to conclude that the controls may constitute 'bad controls'. Implementation is via the Mean Group estimator, and results (p values) are reported in Appendix Tables D-1 and D-2 for specification with four and five factors (we do not test this in the gravity model). In practice we limit our presentation in the maintext to specifications which pass this test.[22]

Testing Post-Reversal Trajectories All our state-specific models include treatment reversal: the autocratisation episode is not eternal (absorbing) but ends after a number of years (on average 7). This setup imposes strong assumptions on our PCDID, namely that the post-treatment reversal does not systematically affect the causal estimates of autocratisation episodes. Our implicit assumption is that relative to the pre-episode period, the post-episode period has no specific trajectory: after the episode finishes, the economic conditions studied in the state do not bounce back (positive effect), but they also do not deteriorate further ('aftershock' effect) — Liu et al. (2024) refer to these scenarios as 'carryover effects'. Assuming away temporal spillovers of the treatment effectively allows us to treat pre- and post-episode periods as some form of (uniform or temporally exchangeable) status quo or equilibrium level, with the episode constituting a *shock* which can be assessed against this equilibrium.

We test this assumption in a robustness check which introduces year dummies for the three years immediately following the end of an autocratisation episode.[23] The intuition of this exercise is to capture the economic effect of reversal and to statistically evaluate whether carryover is a concern. Formally, we estimate in treated states with reversal $i \in E^*$

$$y_{it} = b_{0i} + d_i^* \mathbf{1}_{\{t>T_{0i}\}} + \sum_{s=1}^{k} d_{is}^{post} \mathbf{1}_{\{T_{Ti}+s\}} + \kappa_i d_{it}^{late-pr} + a_i' \hat{f}_t + b_{1i}' X_{it} + u_{it}, \tag{6}$$

where $\hat{f}$ are the estimated factors from the first PCDID step and d_i^* is the state-specific treatment estimate. T_{Ti} denotes the final year of an episode: k dummies, d_{is}^{post}, capture the period after the end of the episode (we use $k = 3$ for parsimony). Since the pre-treatment observations are the omitted reference category, we also need a further dummy, $d_{it}^{late-pr} = \mathbf{1}\{t \geq T_i^{\text{off}} + k + 1\}$ to account for all time periods in reversal but after the k years captured by the carryover dummies. In our results section we report the Wald test for the *joint insignificance* of the three post-treatment year dummies.

[22]As in standard DiD estimators, provided they are not 'bad controls', the inclusion of additional control variables is not required if the parallel trend test (or in our case the Alpha test) is passed, although they may improve the precision of the estimates — in case of the PCDID this would be via the precision of the factors extracted from the control sample residuals.

[23]This is closely related to the carryover effect of a post-treatment placebo test in Liu et al. (2024). If a state experiences more than one episode, the post-treatment effects as captured by the year dummies are assessed *jointly*.

3.4 Alternative Empirical Implementations

Many heterogeneous DID estimators proposed in the recent literature (e.g. Callaway and Sant'Anna, 2021; De Chaisemartin and d'Haultfœuille, 2020; Sun and Abraham, 2021; Borusyak et al., 2024) assume an absorbing treatment (no treatment reversal) and are therefore unsuitable for our setup.

The factor framework naturally leads us to consider alternative empirical implementations adopting the same framework. First, we could consider the Bai (2009) IFE estimator used in Gobillon and Magnac (2016). However, the IFE provides ATET estimates *without* any consideration for the treatment length or treatment reversal, which, as in the case of the above implementations, makes it unsuitable for our purposes. Second, we consider novel imputation estimators: originally introduced as the 'generalized synthetic control' method (Xu, 2017), these counterfactual approaches estimate common factors from the control sample and then employ them in the *pre-treatment period* of the treated sample to obtain heterogeneous estimates for the factor loadings so as to then create the counterfactual (IFEct, see Liu et al., 2024) from the estimated factors and factor loadings. These allow for treatment reversal; however, the IFEct method requires a suitably long *pre-treatment* period to obtain reliable estimates for the factor loadings: Liu et al. (2024) recommend ten or more time periods, which would reduce our sample of treated states from 33, 22 and 18 to only 14, 12, and 10 for the broad, intermediate and narrow definitions of autocratisation, respectively, rendering this approach severely flawed as a robustness exercise.

Finally, recent work by de Chaisemartin and D'Haultfœuille (2024) introduces a heterogeneous DID estimator allowing for staggered treatment with reversal (alongside other flexibilities such as differing treatment dosage). In Appendix Table B-5 we collate a number of statistics specific to our three treatment variants, which suggest that this method is not suited to our challenging data and treatment setup.[24]

4 Empirical Results

4.1 Average Treatment Effects

We begin by reporting the Average Treatment Effects on the Treated (ATETs) using an 'M'-estimator (Rousseeuw and Leroy, 2005) for the baseline model and the model including Covid dummies. These Mean Group estimates (inference based on Pesaran 2006) ignore the length of time spent in an episode, so that we also compute the ATETs evaluated at the median

[24] The issues arising are as follows: (i) the majority (75-88%) of sample years have fewer than 5 'switchers' (treatment changes from 0 to 1 or 1 to 0), raising concerns over weak identification; (ii) a number of treated states (9/33, 5/22, 4/18) have fewer than four pre-treatment year (as an aside, this is not an issue for our PCDID *iff* readily available post-reversal observations do *not* suffer from carryover effects); (iii) and there is a substantial share of states with multiple (staggered) switches, implying fragmented treatment histories, undermining credible counterfactual comparison.

treatment length and the upper quartile treatment length—these estimates are constructed from robust regressions of state treatment effect on total treatment length, evaluated at the median and UQ values. Table 1 presents all these estimates for the benchmark model in panel (A) and the model including Covid dummies in panel (B) using colour-coding: dark (light) red shading for statistically (in-)significant detrimental effects of autocratisation, green shading for statistically insignificant 'improvement' — increases in income inequality or poverty headcount are classified as detrimental effects. One complicating factor is that the ATET results are predicated on the core identifying assumption of the PCDID (factor spanning) to be satisfied, and so we mark in grey (without shading) those results where the Alpha test fails.

This visual summary of our main results is quite stark: across the vast majority of economic outcomes, we see a substantial and statistically significant deterioration resulting from autocratisation episodes. Economic magnitudes often become more substantial as we narrow the definition of autocratisation. We can further see in the lower panel that accounting for the Covid years of the sample substantially strengthens this finding.

4.2 Detailed Results — Graphical Presentation

In the following we further explore our results by highlighting the relationship between time spent in an episode and the magnitude of any causal economic effect (see Boese-Schlosser and Eberhardt, 2024): if deteriorating democratic institutions are bad for economic prosperity, then *prolonged* episodes are likely to have worse implications than shorter ones. The standard approach would be to implement an event study design, where the treatment dummy is replaced by a set of time dummies for 'event time' ℓ since treatment started. Our average treatment length is $\ell = 7$, yet estimating 6 or more additional parameters for each treated state in our panel of maximum length $T = 24$ is challenging. Similarly, even creating blocks of time (e.g. 1-4 years, 5-8 years, etc), while saving on precious degrees of freedom, does not improve on treatment effects at the median and UQ treatment length we have just presented.

Instead of altering the treatment regression specification, we adopt predictions from multivariate running line regressions (Royston and Cox, 2005) *ex post*. These are local linear regressions which in effect smooth the treatment effect against the number of years in treatment, albeit with the additional advantage that we can add further controls: we estimate these using the state-specific treatment effect $\hat{d}_i$, the total years spent in an episodes, and dummies for the total number of episodes experienced by the state. We also employ state-specific weights in this estimation procedure, created by an M-estimator (Rousseeuw and Leroy, 2005) when computing robust average treatment effects (ATET) reported above.[25] Our plots present the *predicted values* from this multivariate smoothing procedure. While this has an interpretation which is distinct from that of an event plot (the average effect of treatment for all

[25] These weights are smaller for outlier treatment effects. Results are qualitatively very similar if we use the total 'drop' in the rescaled SDI for a state as the weight or if do not apply any weights (available on request).

Table 1: Visualisation — Average Treatment Effects

Episode	Broad	Inter	Narrow	Broad	Inter	Narrow
Factors	4	4	4	5	5	5
Panel A: Baseline Model						
(i) Robust Mean ATET						
State GDP pc	-0.475*	-0.254	-0.381	-0.254	-0.476	-0.696*
Poverty headcount	1.671	1.092	2.396	2.104	1.614	2.686*
Top-10% income share †	0.338	0.134	0.222	0.214	0.183	0.228
Total business R&D	-3.606	-4.630	-7.290**	-4.321*	-2.214	-6.836*
Business own R&D brd	-1.023	-4.501	-5.047**	-1.267	-3.992*	-6.875***
All granted patents	-1.728*	-2.876**	-2.626**	-2.367**	-2.609**	-2.674**
Granted utility patents	-1.829	-2.966**	-2.417**	-1.785*	-2.454**	-2.497**
(ii) Robust ATET evaluated at the Median treatment length						
State GDP pc	-0.465*	-0.143	-0.250	-0.233	-0.297	-0.662*
Poverty headcount	1.718	0.673	2.258*	2.094	0.838	2.490*
Top-10% income share †	0.385	0.173	0.281	0.221	0.256	0.310
Total business R&D	-3.426	-4.500	-6.276**	-4.332*	-1.881	-6.494***
Business own R&D	-1.004	-4.678	-5.006**	-0.925	-3.972	-6.714***
All granted patents	-1.720*	-3.004**	-3.065***	-2.407**	-2.688**	-2.672***
Granted utility patents	-1.812	-3.253***	-2.577**	-1.746	-2.771***	-2.641***
(iii) Robust ATET evaluated at the Upper Quartile treatment length						
State GDP pc	-0.692**	-0.865***	-1.033**	-0.585**	-0.991***	-1.195***
Poverty headcount	2.112	3.048*	4.030	2.549*	3.456	4.889*
Top-10% income share †	0.279	0.017	-0.097	0.148	-0.209	-0.096
Total business R&D	-1.959	-6.401	-13.728***	-4.420	-4.003	-14.102***
Business own R&D	-3.076	-3.694	-5.348	-4.454**	-4.086*	-8.158**
All granted patents	-2.441**	-2.588***	-1.073	-2.848**	-2.491**	-2.680*
Granted utility patents	-1.940	-1.044	0.016	-1.854	-1.503	-0.707
Panel B: Model including Covid Dummies						
(i) Robust Mean ATET						
State GDP pc	-0.362	-0.632	-0.926**	-0.421*	-0.488**	-0.509**
Poverty headcount	1.364	2.767	2.607	1.928	3.795**	3.114
Total business R&D	-11.215***	-14.709***	-20.828***	-12.684***	-16.008***	-22.046***
Business own R&D	-12.304***	-15.864***	-19.735***	-13.069***	-19.866***	-21.341***
All granted patents	-3.813*	-9.491***	-12.999***	-5.157**	-10.291***	-12.651***
Granted utility patents	-3.994*	-10.020***	-12.422***	-4.910**	-10.725***	-12.972***
(ii) Robust ATET evaluated at the Median treatment length						
State GDP pc	-0.329	-0.531	-0.906**	-0.393	-0.387*	-0.479**
Poverty headcount	1.023	2.030	2.288	1.771	3.058*	2.754
Total business R&D	-11.179***	-13.875***	-18.296***	-12.907***	-15.176***	-19.932***
Business own R&D	-11.695***	-14.513***	-18.513***	-13.167***	-18.702***	-19.976***
All granted patents	-4.783***	-8.518***	-13.132***	-6.033***	-9.721***	-12.780***
Granted utility patents	-4.615**	-9.922***	-12.441***	-5.601***	-10.427***	-13.069***
(iii) Robust ATET evaluated at the Upper Quartile treatment length						
State GDP pc	-0.706**	-0.989***	-1.147**	-0.615***	-0.758***	-0.807***
Poverty headcount	2.773*	4.425**	6.154*	3.775**	6.451***	7.208**
Total business R&D	-15.083***	-20.802***	-28.033***	-17.953***	-23.440***	-30.317***
Business own R&D	-16.771***	-20.234***	-25.755***	-19.092***	-22.662***	-27.986***
All granted patents	-7.667***	-10.398***	-16.019***	-8.961***	-12.449***	-16.068***
Granted utility patents	-7.175***	-11.226***	-14.174***	-8.370***	-12.419***	-15.589***

Notes: We report average treatment effects on the treated for the preferred specifications of our PCDID estimator, constructed as robust estimates from all 18, 22, or 33 state-specific treatment estimates using an M-estimator (Rousseeuw and Leroy, 2005) in three ways: (i) ignoring the years spent in an episode; (ii) from a robust regression of treatment effect on total total treatment length, evaluated at the median treatment length, and (iii) dto., evaluated at the upper quartile treatment length. Section (A) for baseline model results and (B) for models including dummies for the Covid years 2020-2022 († The income inequality measure ends in 2018). Within each bloc, we report ATETs for the 'Broad', 'Intermediate' and 'Narrow' autocratisation episode variants, for PCDID treatment regression augmented with 4 or 5 estimated factors. We use colours to highlight the 'direction' and statistical significance of the treatment effects: green — positive, statistically insignificant effect; lighter red — negative insignificant effect; darker red — negative effect significant at 10% level. Results for poverty and inequality are logically aligned: an increase in either is interpreted as a negative effect. No shading and grey font indicate that the specification does not pass the Alpha test. † Since the inequality data end in 2018 there is no separate result for models with Covid dummies.

treated states after one, two, etc. years in an episode), we capture the temporal trajectory of democratic erosion *as individual treated states have experienced it*.

Figure 3 reports results for average state real income per capita, top-income share (inequality), and the poverty headcount. Innovation is covered in Figure 4, bilateral international exports by states (from a PPML-DID model introduced in Section 4.5) in Figure 6. Additional plots in the appendix cover alternative measures and specifications or subsets of treated states. Each results plot is presented as follows: on the x-axis we capture the (total) number of years a state has spent in an episode or episodes, the y-axis indicates the treatment effect of autocratisation. The plotted line represents the flexible prediction of our multivariate running line regression. The markers in each plot indicate the distribution of state-level estimates *in terms of total treatment length*, and we minimally perturb these markers to aid visualisation. A filled (white) marker signifies statistical (in)significance at the 10% level. On the x-axis we further indicate the lower quartile (LQ), median, and upper quartile (UQ) of the length of time spent in episode(s), namely 4, 7, and 11 years, respectively.[26] We present three sets of results, for the 'broad', 'intermediate', and 'narrow' definitions of autocratisation. Finally, the two separate plots in each panel are for the results derived from a PCDID specification augmented with four and five estimated factors, respectively.

4.2.1 Average Income

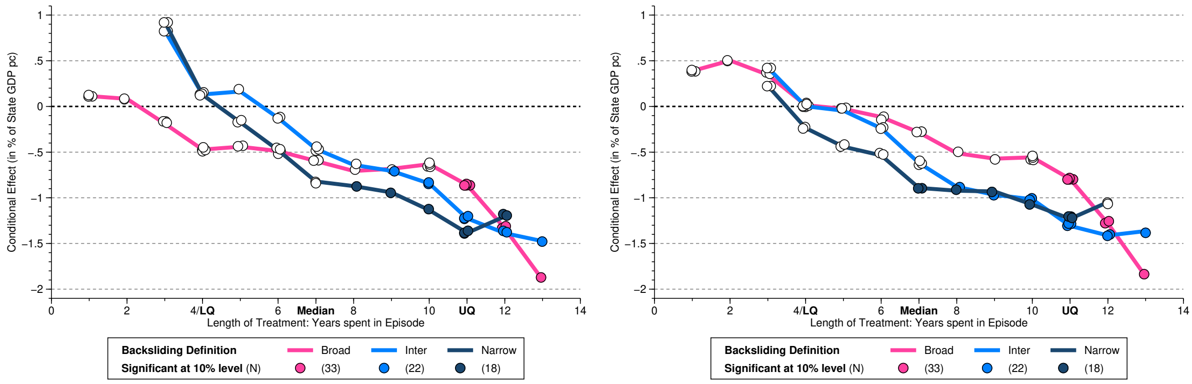
Diagnostics All specifications presented in panel (a) of Figure 3 pass the Alpha test, and furthermore the Wald tests provide no indication that the inclusion of the employment growth rate in the PCDID regression constitutes a 'bad control'. See Appendix Table D-1 for detailed results of all macro-variables studied. Recall that in the result plots we employ a simple signalling device (e.g. ••◦) to highlight passed (•) or failed (◦) Alpha tests for the three alternative definitions of autocratisation episodes.

Main findings All running line graphs in both plots of Figure 3, panel (a), are downward-sloping in treatment length, near-monotonically so. Starting from the broad definition of autocratisation (in pink), treatment effects are only statistically significant for a few states with the longest treatment length; however, as we move to the intermediate (light-blue) and narrow (navy) definitions, more and more states are predicted to have statistically significant negative treatment effects. For our most conservative (narrow) definition, effects are statistically significant from around the median treatment length (7 years), with a treatment effect of -1 to -1.5 percent. Hence, all definitions point to a deterioration in per capita income over time in states experiencing democratic erosion, with the statistical evidence becoming stronger as we tighten the definition for democratic retreat.

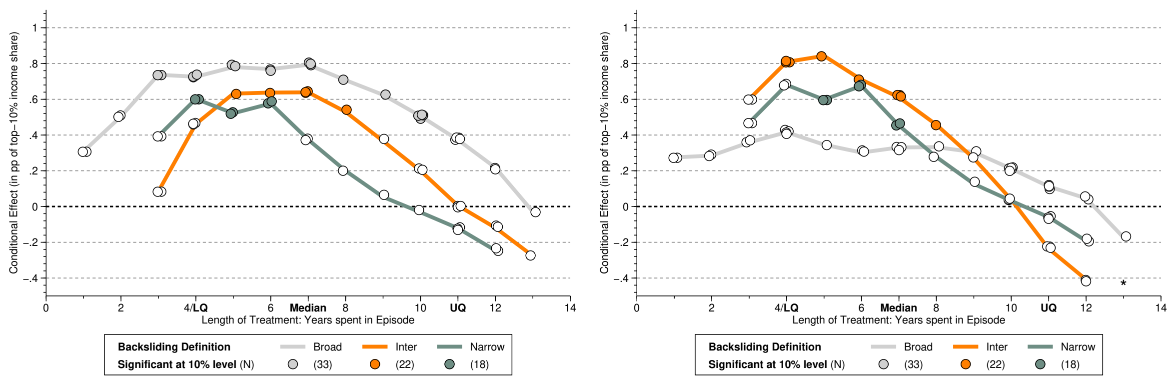
[26] These are common across the three definitions, with the exception of the LQ for the narrow definition (5) and the UQ for the broad one (10).

Figure 3: Episodes of Democratic Erosion — Income, Inequality, and Poverty

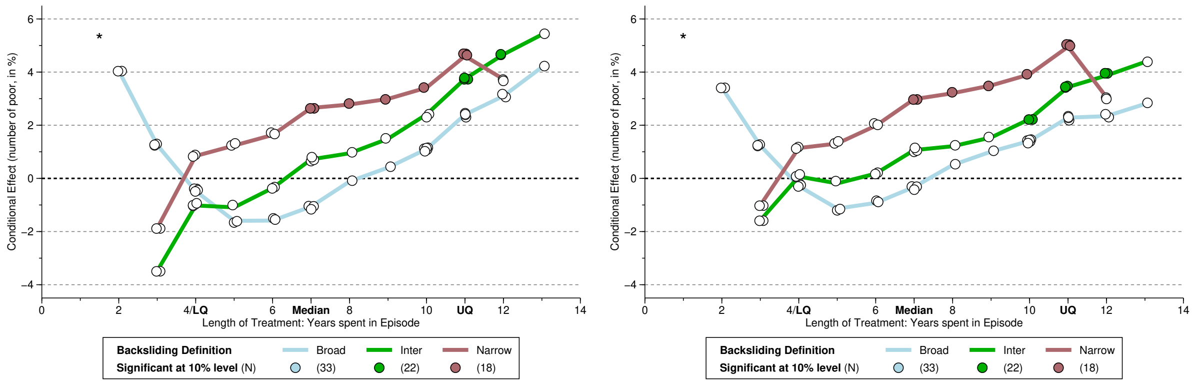
(a) GDP per capita,••• specifications with 4 (left) and 5 factors (right)



(b) Top 10% Income Share,••• specifications with 4 (left) and 5 factors (right)



(c) Poverty headcount,••• specifications with 4 (left) and 5 factors (right)



Notes: We present treated-state predictions from multivariate running line regressions (Royston and Cox, 2005) illustrating the economic effects of autocratisation. The underlying PCDID specifications have the employment growth rate as additional control (except in panel (b) where the specification without controls had more favourable diagnostics) and augment the treatment regression with four or five estimated factors as indicated (models with three or six estimated factors yield qualitatively very similar results, available on request). The (total) number of years a state has spent in one or more autocratisation episodes is shown on the x-axis of each plot, the economic effect on the y-axis. The running line regressions control for the number of times a state experienced an episode and use weights derived from the robust ATET estimate using an M-estimator (Rousseeuw and Leroy, 2005). A filled (white) marker indicates statistical (in)significance at the 10% level. An estimate of -1 (+1) indicates a 1% reduction (increase) in income, top-10% income share, and the number of poor people, in panels (a) to (c), respectively. Each plot presents results for our three definitions of autocratisation: a broad one (33 treated states), an intermediate one (22), and a narrow one (18) — see text for definitions. We use ••• to signal which specifications using these three different definitions (Broad, Inter, Narrow) pass (•) or fail (°) the Alpha test.

Robustness Our above predictions are conditioned on the number of episodes states experience (between 1 and 3). To confirm that mixing single- and multiple-episode treated states does not distort our findings, we present results for states that just experienced a *single* episode in Appendix Figure E-1. Despite the reduction in sample size (from 33 to 22, 22 to 18, and 18 to 16 for 'broad', 'intermediate', and 'narrow' definitions, respectively), the overall findings are qualitatively robust, and although statistical significance varies, the prediction lines for the three definitions of autocratisation are now almost congruent. In Appendix Figure E-2 we report the results based on the PCDID specification without additional controls. The common downward trajectories across episode definitions are still present, though the statistical evidence is somewhat weaker. In Appendix Figure E-7 panel (a), we include three year dummies for the Covid years and find results qualitatively unchanged.

We further experimented with using real GDP *per worker* as outcome variable (in models with or without the employment growth rate as additional control), but for all definitions of autocratisation these specifications always fail the Alpha test and hence may be misspecified.[27] Next, we investigate the effect of autocratisation on the distribution of income.

4.2.2 Income Inequality

Diagnostics All specifications for state median household (presented in panel (a) of Appendix Figure 3) fail the Alpha test, suggesting that these PCDID regressions may be misspecified. Put simply, the unobservables driving median hh income in the treated versus control states are not the same—this hints at the possibility that *cross-state* inequality may be a predictor for autocratisation. Note that the diagnostic tests also fail if we study various alternative specifications. Diagnostics are however favourable, including for Wald tests for 'bad instruments', when we use the 'within-state inequality' data (Gini coefficients), which are, however, only available from 2006, and top income shares data, available 2000-2018.

Results: Cross-state Inequality Across the main results for median HH income presented in panel (a) of Figure E-4 as well as in various robustness checks limiting the treated sample (single episode states: Appendix Figure E-1) or not including control variables (Appendix Figure E-2), the running line predictions are always near-universally statistically insignificant. Adding dummies for Covid years, Appendix Figure E-7 panel (b), also makes no difference.

Results: Within-state Inequality For the Gini, Panel (b) of Appendix Figure E-3 indicates that none of the predictions are statistically significantly different from zero, with estimates for the intermediate and broad episode definitions exclusively positive, whereas those for the narrow definition are more mixed. The results for top-10% income share are presented in panel (b) of Figure 3. The running line predictions are more consistent across definitions

[27] For reference, running line plots are presented in panel (a) of Appendix Figure E-3, indicating negative effects of similar magnitude as those in the per capita analysis.

of autocratisation and suggest the state with median time spent in an episode experiences a statistically significant increase in the top-10% income share by around half a percentage point. For states with longer treatment time, there is no significant effect (statistically or in terms of magnitude). Results for the top-1% income share, in Appendix Figure E-4, are less consistently statistically significant but point to a positive increase. Since the top-income share data ends in 2018 these results are not affected by the Covid pandemic.

Taken together, these results provide some tentative evidence of widening within-state inequality as a result of autocratisation, with the top-10% income share increasing for states at the median episode length. We now turn to look more specifically at individuals *at the bottom* of the income distribution in each state.

4.2.3 Poverty

Diagnostics All specifications presented in panel (c) of Figure 3 pass the Alpha test as well as the Wald test for 'bad controls'. Similarly for the poverty rate (share of state population below the poverty line) and for the Alpha test in the specification without additional controls.

Main Results The prediction lines for impoverishment in panel (c) of Figure 3 are all upward-sloping (an initial spike for the 'broad' definition aside) and as in our previous results the statistical evidence for a detrimental effect of autocratisation gets stronger as we tighten the definition, while in the present case the magnitudes of the treatment effects are also getting larger. The 'narrow' result suggests that democratic erosion increases the number of individuals living below the poverty line by around 3-5% (median and UQ of the treatment distribution).

Robustness Results are less consistent across definitions if we limit the sample to states which experienced a single episode. In Appendix Figure E-1, panel (c) indicates weaker evidence for widespread increase in the number of individuals living below the poverty line in the broad and intermediate definitions. The results for the narrow definition, on the other hand, are robust and suggest an increase by 4-6%. In Appendix Figure E-7 panel (c) we include year dummies for the Covid years to our main specification and find that results are more precisely estimated and amplified, suggesting an increase in the headcount of the poor of up to 4-8%.

Results for the specifications without any additional controls (panel (c) of Appendix Figure E-2) closely follow those of our main results, with the 'narrow' results suggesting a 4-6% increase in the poor.

In panel (c) of Appendix Figure E-3 we analyse the effect of autocratisation on the poverty rate (i.e. share of state population), with results interpretable as percentage point changes. All three definitions suggest an increase in the poverty rate as treatment length increases, albeit with varying empirical precision: compared with the sparse statistical evidence for the broad and intermediate definitions, results for the narrow variant are statistically significant for almost all treated states, with a treatment effect of around half a percentage point.

Extension: The Uninsured Panel (c) of Appendix Figure E-3 charts the running line plots for our analysis of the headcount of individuals without health insurance coverage. These results again indicate positive treatment effects with varying degree of precision, whereby the narrow definition of autocratisation suggests both the highest magnitudes as well as the strongest statistical evidence for a positive treatment effect (on the order of around half a percent).

Our focus on those at the bottom of the income distribution suggests an absolute and relative increase in deprivation, as indicated by positive significant treatment effects for the poor headcount, the poverty rate, and to a lesser extent the headcount of the uninsured.

4.2.4 Innovation

Diagnostics For the analysis of R&D expenditure we find that specifications including state employment growth always either fail the Alpha test or the Wald test for 'bad controls' or both. The diagnostics for specifications without a control variable are more favourable, only the 'narrow' definition of autocratisation for the business R&D expenditure fails the Alpha test.[28]

For the patent analysis, we again studied specifications including employment growth as additional control, but found these fail the Alpha test in all specifications considered. Wald tests typically indicated employment growth to be a bad control. The models without additional controls perform better, only the narrow definition fails the Alpha test in the models for granted patents and granted utility patents.

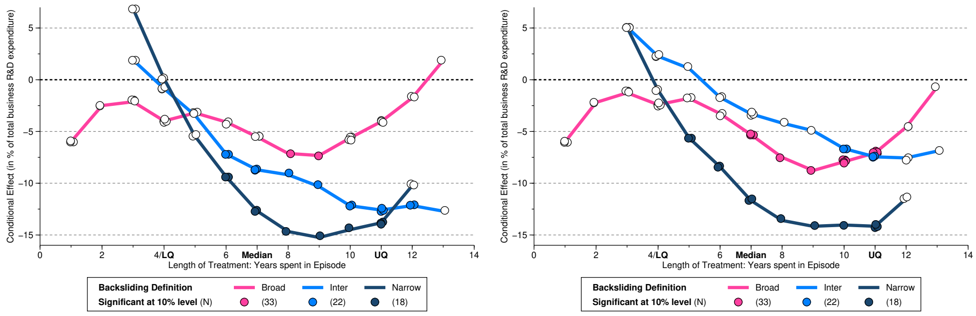
Main Results Figure 4 presents results for the real R&D expenditure by private businesses: in panel (a), apart from their own funds, this further includes funding these businesses received from the federal government, from other businesses or organisations in or outside the U.S., as well as from U.S. state or foreign government agencies or labs (total business expenditure); in panel (b) we focus on businesses' own R&D expenditure. All results indicate that private businesses in states that experience democratic erosion cut back their R&D expenditures. Focusing on the intermediate and narrow definitions of episodes, for the expenditures including federal funding and other sources in panel (a), the treatment effect is between 8 and 14% lower expenditure in states with median episode length, and between 4 and 10% lower expenditure when we just consider businesses' own funding in panel (b).[29] These are sizeable effects, which are qualitatively identical if we adopt real R&D expenditure deflated by state population (available on request).

[28]We also adopted measures of R&D intensity, i.e. R&D expenditure deflated by the state GDP, but these specifications always fail the Alpha test. Using R&D expenditure deflated by state population yields sound diagnostics except for the 'narrow' definition in both sets of models.

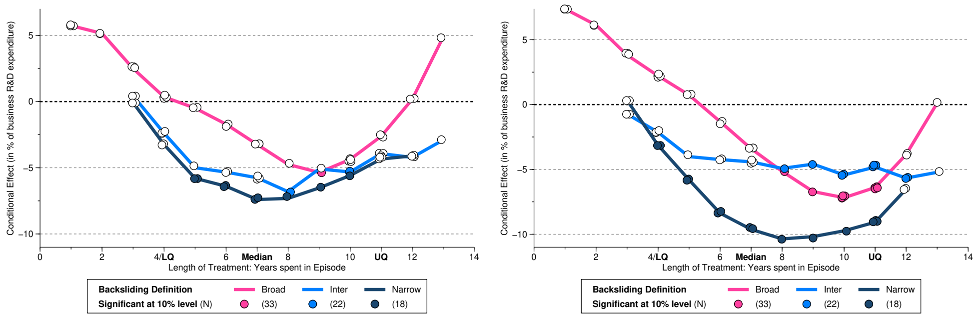
[29]The difference may arise since firms seek fewer government grants or collaboration and funding from firms or organisations outside the state or from abroad.

Figure 4: Episodes of Democratic Erosion — Innovation

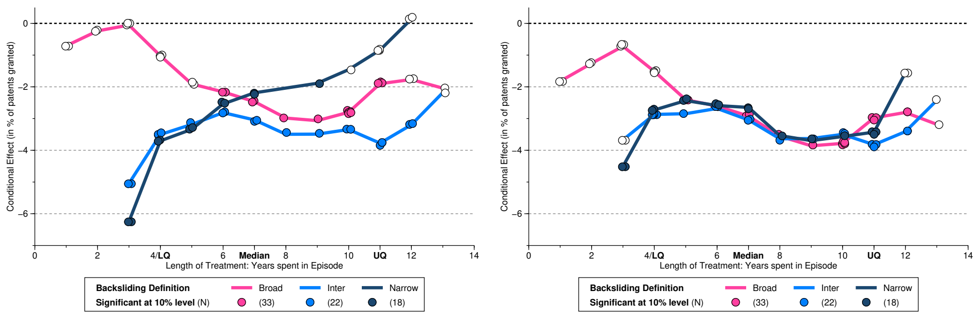
(a) Total Business R&D Expenditure,••• specifications with 4 (left) and 5 factors (right)



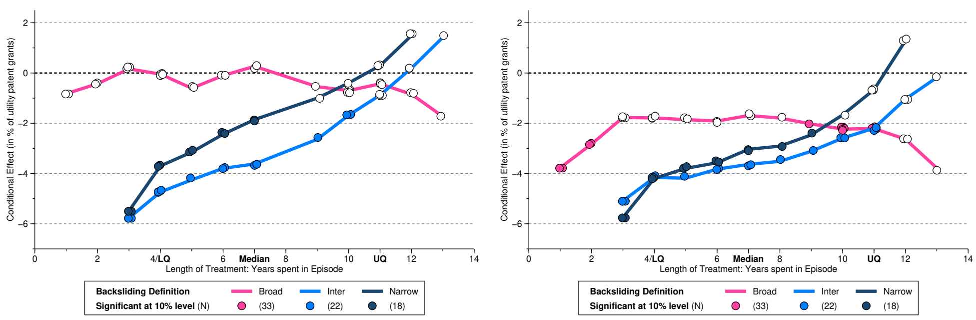
(b) R&D Expenditure by Businesses,••◦ specifications with 4 (left) and 5 factors (right)



(c) Total Patents Granted,••◦ specifications with 4 (left) and 5 factors (right)



(d) Utility Patents Granted,••◦ specifications with 4 (left) and 5 factors (right)



Notes: We present results from the PCDID specifications without additional control variable — specifications with employment growth always fail the Alpha test or the Wald test for 'bad controls'. The R&D (patent) data is limited to 2000-2022 (2000-2020). 'Total business R&D' includes expenditure by businesses sourced from the federal government (e.g. grants) and from 'others' (i.e. other firms located inside or outside the US, state or foreign government agencies, and other organizations inside or outside the US). 'Business R&D' refers to the expenditure paid for by the firms themselves. 'Total Patents' includes utility, plant and design patents. We use ••◦ to indicate whether specifications using the liberal, intermediate, and conservative definitions of autocratisation pass (•) or fail (◦) the Alpha test at the 10% significance level. For other details, see notes to Figure 3.

Figure 4 further presents running line prediction plots for state-level innovation output (patent counts, available 2000-2020):[30] Panel (c) uses data on all patents granted, while panel (d) zooms in on utility patents granted. The data for all patent types granted in panel (c) shows a decline of between -3 and -4% for treated states, which is fairly stable over the episode length. Focusing on utility patents in panel (d) indicates that states with short episodes experience a significant drop in patents granted of up to -6%, whereas with longer 'exposure' to democratic erosion, the treatment effect is reduced and eventually turns insignificant. Overall, granted patents of all types are clearly negatively affected by democratic erosion, with patents covering large innovative steps seemingly slowly converging back to the original levels (though more on this pattern below), whereas plant and design patents, which represent smaller innovative steps (and hence commercial potential), seem subject to a persistent drop-off.

Robustness In Appendix Figure E-8 panels (a) and (b) we study the implications for business R&D expenditure and patent counts in specifications including year dummies for the Covid years and find results substantially strengthened. For the former, across all definitions of autocratisation and factor augmentations we find results substantially increased in magnitude, suggesting up to 20-30% decline in R&D expenditure. For patenting in panels (c) and (d) we now observe consistent patterns between all patents and utility patents granted, indicating substantially higher detrimental effects of -14 to -18%.[31] We also considered patent *applications*, which yield statistically insignificant treatment effects in the main specification (without controls), but arrive at a statistically significant effect of -5 to -10% when we add Covid dummies (available on request).

Taken together, these findings suggest that democratic erosion severely harms private businesses' innovation *effort* as well as state innovation *output*.

4.3 Reversal and Carryover Effects

In Table 2 we present the formal Wald test results for carryover effects (with the null of no carryover). This test addresses the concern that our treatment effects estimator assumes that the economic implications of autocratisation episodes do not spill over to the period after treatment has reverted, i.e. there is no bounce-back of the economy with increased economic progress relative to the pre-treatment years and also no prolonged depression following the end of an episode. In essence, we assume that autocratisation episodes have one-off levels effects and not perpetual growth effects on economic outcomes and that furthermore the dynamics of these regime switch periods are relatively simple and/or short-lived (like switching a light on or off). In practice, we adopt three dummies for the first three years after the end of an

[30] Here, we cannot separate out patents of private businesses but study all state patents instead.
[31] In the patent analysis, only a single year (2020) is captured by the additional dummy augmentation.

Table 2: Formal Test for 'No Carryover Effects'

Outcome	**y**	**i10**	**pr**	**tbrd**	**brd**	**pgr**	**put**
Control included	yes	no	yes	no	no	no	no
Definition of Autocratisation, factor-augmentation							
Broad with 4 factors	0.831	0.000	0.157	0.000	0.001	0.549	0.493
Broad with 5 factors	0.992	0.000	0.770	0.000	0.007	0.992	0.744
Intermediate with 4 factors	0.195	0.120	0.583	0.915	0.351	0.296	0.411
Intermediate with 5 factors	0.875	0.726	0.337	0.917	0.851	0.100	0.703
Narrow with 4 factors	0.004	0.422	0.197	0.022	0.100	0.462	0.994
Narrow with 5 factors	0.068	0.023	0.403	0.136	0.162	0.736	0.958

Notes: We carry out formal tests for the assumption that treatment effects do not 'carry over' into post-treatment periods by computing a Wald test for the joint insignificance of the three post-episode year dummies we introduced in robustness checks presented in this section. We report p-values for the null of joint statistical insignificance for income per capita (y), top-10% income share (i10), poverty headcount (pr), total business R&D expenditure (tbrd), business own R&D expenditure (brd), count of total state granted patents (pgr), dto. but limited to utility patents (put). Tests are conducted for the preferred specifications (with or without control variables).

episode or episodes and test their joint significance (adopting four year dummies instead yields qualitatively identical results, available on request).

For illustration, we include robust mean estimates of the carryover dummies for 'Business Own R&D expenditure' in Figure 5: for this outcome, the broad definition of autocratisation strongly rejects the null and in both result plots we can observe the statistical significance of the year dummies for this specification (95% confidence intervals in pink) whereas the other two definitions yield largely insignificant estimates.

Our test results indicate that the broad definition of autocratisation is at times susceptible to carryover effects, notably for top-10% income share and the two measures of R&D expenditure by private businesses. For the intermediate definition we find no evidence of carryover, whereas the state income per capita results and two further individual specifications indicate carryover in the narrow definition.

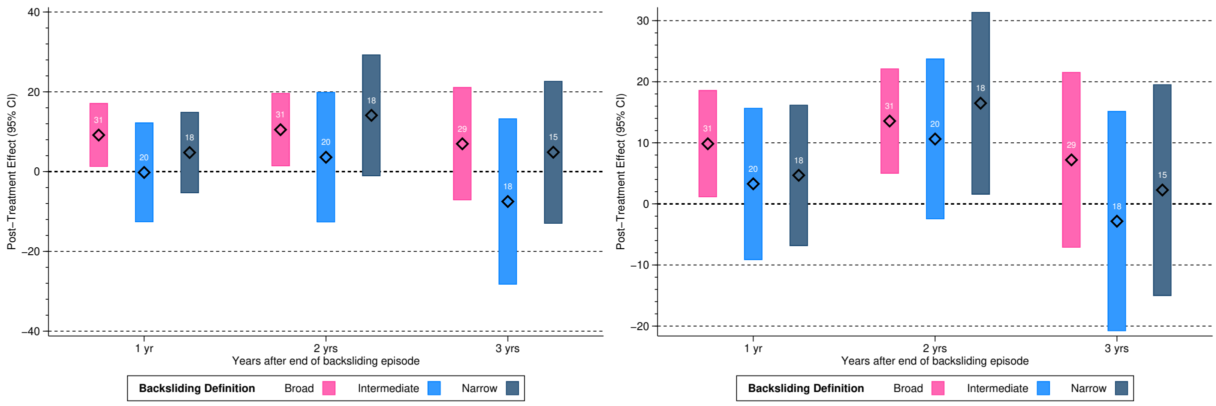
4.4 Revisiting Partisan State Capture

Our descriptive analysis has highlighted the affinity between autocratisation episodes and partisan state 'capture' by the Republican party. Naturally, a concern with our above empirical analysis would therefore be whether we truly capture autocratisation rather than partisan policy-making.

Simply adding a dummy for Republican state capture to the PCDID regressions fails to

Figure 5: Autocratisation Episodes — Example of Carryover Effects — Business Own R&D

(a) Model with 4 factors (b) Model with 5 factors



Notes: We present the robust mean estimates (diamonds) and 95% confidence intervals (shaded bars) for the three carryover dummies in the analysis of 'private business own-R&D expenditure' adopting the three definitions of autocratisation. Based on the Wald test, the 'broad' definition displays significant carryover, whereas the other two definitions pass this test.

appreciate the endogeneity of partisan power, and, indeed, diagnostic testing simply picks up the strong correlation between state capture and autocratisation episodes (without any claims as to the direction of causation) and therefore rejects this specification. Our strategy to address this concern therefore involves a stock variable capturing the longer-term, 'latent' political dominance of the Republican party in each state: we create a stock variable (1992 = 0) which (i) adds one unit for a year where the state is 'captured' by the Republican party (i.e. holding power in both the state senate and house), (ii) adds no additional unit in a year with power-sharing (or independent/other party in power) in one or both of the state senate and house, and (iii) subtracts one unit in a year where Democrats hold power in both the state senate and house. The accumulated stock depreciates at an annual rate of 20%, regardless of who is in power.[32] We present the robust mean ATET for the treatment effect in the specifications including this 'Republican power stock' variable in Table 3, which uses the same principles for visualisation we used in the presentation of the main results in Table 1. Two outcomes (state GDP pc and granted utility patents) never pass the Alpha test, indicating that conditioning on the 'power' of the Republican party in their analysis brings out substantive differences in the makeup of the unobserved heterogeneity driving these outcomes between states experiencing episodes and those that do not.[33] For all other outcomes, we see an effective replication of the baseline results (which ignore partisan power), namely an overwhelming negative and frequently statistically significant effect of autocratisation episodes for economic prosperity.

[32] In Appendix Table E-2 we present ATET results where we apply a 10% annual depreciation rate, instead of 20%, with our findings qualitatively unchanged.

[33] Additionally conditioning on employment growth as in some of the main results presented in Table 1 does not change the outcome of the Alpha test (results on request).

Table 3: Controlling for the 'Stock' of Republican State Capture — ATETs

Episode	Broad	Inter	Narrow	Broad	Inter	Narrow
Factors	4	4	4	5	5	5
Panel A: Baseline Model						
(i) Robust Mean ATET						
State GDP pc	0.103	0.367	0.601**	0.248	0.558**	1.009***
Poverty headcount	0.428	1.956	4.603*	1.148	2.257	4.978**
Top-10% income share †	0.134	0.364	0.531	0.282	0.928***	0.652*
Total Business R&D	-5.907***	-2.589	-2.964	-3.314*	-2.364	-2.703
Business Own R&D	-4.253*	-6.685***	-5.729**	-4.045*	-5.976*	-5.720***
All granted patents	-2.882***	-2.651*	-6.298***	-3.519***	-2.705	-3.698**
Granted utility patents	-1.681*	-2.748**	-4.312***	-1.496*	-2.345*	-2.619**
(ii) Robust Mean ATET evaluated at the Median						
State GDP pc	0.213	0.594*	0.679**	-0.037	0.758**	0.791***
Poverty headcount	0.338	1.343	4.074	0.961	1.422	4.499**
Top-10% income share †	0.134	0.364	0.531	0.282	0.928***	0.652*
Total Business R&D	-5.907***	-2.589	-2.964	-3.314*	-2.364	-2.703
Business Own R&D	-4.253*	-6.685***	-5.729**	-4.045*	-5.976*	-5.720***
All granted patents	-2.882***	-2.651*	-6.298***	-3.519***	-2.705	-3.698**
Granted utility patents	-1.083	-2.753*	-4.784***	-0.617	-2.246	-3.090**
(iii) Robust Mean ATET evaluated at the Upper Quartile						
State GDP pc	0.294	0.448*	0.239	0.010	0.365	0.190
Poverty headcount	0.405	3.559**	8.710**	0.924	5.061**	9.675***
Top-10% income share †	0.055	0.386*	0.446	0.142	0.732**	0.589**
Total Business R&D	-5.787***	-3.588	-4.295	-3.355	-2.303	-4.402
Business Own R&D	-4.639**	-6.008***	-5.765***	-4.301*	-5.607**	-5.181**
All granted patents	-3.355***	-2.818**	-5.304***	-3.973***	-3.074*	-3.167**
Granted utility patents	-1.149	-2.700*	-4.710**	-0.908	-2.345**	-3.017**
Panel B: Model including Covid dummies						
(i) Robust Mean ATET						
State GDP pc	0.103	0.367	0.601**	0.248	0.558**	1.009***
Poverty headcount	0.549	4.067**	4.598*	1.627	5.122***	5.456**
Total Business R&D	-4.261*	-4.753	-4.830	-3.903	-5.158*	-6.598**
Business Own R&D	-3.750**	-3.295	-5.347***	-2.018	-2.978	-5.305***
All granted patents	-1.807	-2.840*	-4.036***	-2.236*	-3.027*	-2.364
Granted utility patents	-0.246	-3.526***	-3.084***	-0.103	-2.550**	-2.128*
(iii) Robust Mean ATET evaluated at the Median						
State GDP pc	0.263	0.322	0.653**	0.403*	0.566**	1.072***
Poverty headcount	0.622	2.705	3.678*	1.642	3.462	4.931**
Total Business R&D	-3.361	-4.596	-3.699	-3.451	-6.119*	-5.484*
Business Own R&D	-2.991	-4.232**	-6.183***	-1.372	-4.013**	-5.650***
All granted patents	-1.385	-2.401	-4.437***	-0.748	-2.285	-2.985*
Granted utility patents	0.322	-2.569*	-3.525***	0.944	-1.933	-2.447**
(iii) Robust Mean ATET evaluated at the Upper Quartile						
State GDP pc	0.219	0.383	0.275	0.382**	0.573**	0.565*
Poverty headcount	2.12	6.284***	9.709***	3.910**	7.346***	10.808***
Total Business R&D	-6.268**	-7.976**	-11.160***	-6.303**	-7.282**	-11.316***
Business Own R&D	-4.525**	-2.538	-4.927***	-3.345	-2.251	-4.882**
All granted patents	-2.949*	-2.92	-4.986	-2.228	-2.594	-3.545
Granted utility patents	-0.210	-2.300	-2.543**	0.130	-1.917	-1.691

Notes: We report average treatment effects on the treated for the preferred specifications of our PCDID estimator, constructed as robust estimates from all 18, 22, or 33 state-specific treatment estimates using an M-estimator (Rousseeuw and Leroy, 2005). In contrast to the main results in Table 1, we include a measure for the stock of Republican 'capture' of a state to determine whether our results are driven by democratic erosion or partisanship. This stock variable adds one unit in a year where Republicans hold power in both the state senate and house, adds zero units in a year with power-sharing (or independent/other party in power) in one or both of the state senate and house, and subtracts one unit in a year where Democrats hold power in both the state senate and house. Regardless of the addition or subtraction of a unit to the stock in year t, the stock depreciates by 20% per annum. If the Alpha test rejects, the estimates are greyed out since our core identification assumption is not satisfied. Otherwise, we shade estimates in light red if the estimate implies a detrimental but insignificant effect of episodes, and in dark red if the detrimental effect is significant at the 10% level. † Since the inequality data end in 2018 there is no separate result for models with Covid dummies.

4.5 International Export Flows

Heterogeneous Structural Gravity Model For our dyadic trade flow regressions, we adopt the estimator developed in Desbordes et al. (2025), which marries the Poisson Pseudo Maximum Likelihood (PPML) structural gravity regression with Chan and Kwok's (2022) factor-augmented difference-in-differences model. Instead of extracting factors from control sample regressions using Principle Component Analysis, we use the Pesaran (2006) 'common correlated effects' (CCE) approach and include control sample cross-section averages of the 'economic mass' variables computed in the treatment regressions. Using PPML, we estimate the following equation at the pair-level for exporter state i and destination country j for pairs where the former experienced autocratisation:

$$\begin{aligned} \mu_{ijt} \;=\; & exp\Big[\alpha_{ij} + \theta_{ij} D_{it} + \eta_{ij} D_{jt} \\ & +\delta^i_{ij}\,\overline{ln(Y)}^i_t + \kappa^i_{ij}\,\overline{ln(Pop)}^i_t + \delta^j_{ij}\,\overline{ln(Y)}^j_t + \kappa^j_{ij}\,\overline{ln(Pop)}^j_t \\ & +\psi^i_{ij} ln(Y/L)_{it} + \psi^j_{ij} ln(Y/L)_{jt}\Big] \cdot \epsilon_{ijt}. \end{aligned} \tag{7}$$

The dependent variable represents the exports from state i to country j at time t. Our parameters of interest are the θ_{ij}, which relate to the binary indicator for U.S. state autocratisation episodes. We also include a dummy for destination country democratic collapse D_{jt} to account for potential changes in state exports in response to political economy factors *in the destination* (see Boese-Schlosser et al., 2025).[34]

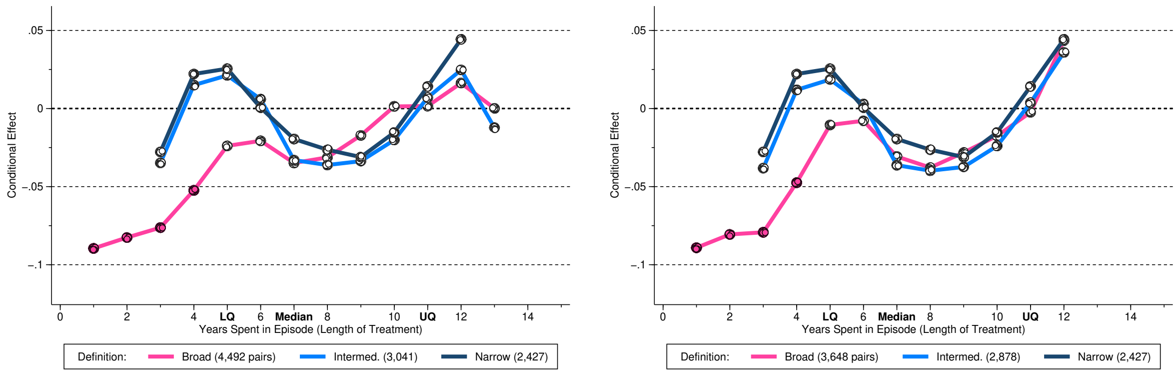
The second line represents the cross-section averages at time t ($\overline{ln(X)}^i_t = N^{-1}\sum_i ln(X)_{it}$) of the 'economic mass' variables employed in the gravity model (GDP, population; in logs). These averages are computed over all exporter states i (first two terms) and all destination countries j (last two terms), which, respectively, never experienced autocratisation episodes and remained democratic throughout the sample period.[35] The workhorse empirical implementation for bilateral flows includes directional-time fixed effects, which prevents the study of monadic (country-specific) variables. Our setup allows us to absorb the confounding heterogeneous effects of common shocks (related to the 'multilateral resistance terms', MRTs, introduced in Anderson and Van Wincoop, 2003) without absorbing the treatment. Our estimator identifies a total reduced form treatment effect inclusive of income gains and equilibrium adjustments. Including income per capita, as in the last line of equation (7), allows us to absorb the causal effect of income in response to democratic erosion to isolate the 'direct' effect (including the effect on trade costs and equilibrium adjustments).

[34]Here, we adopt the Lührmann et al. (2018) Regimes of the World definition of democracy (≥ 2) and autocracy (< 2).

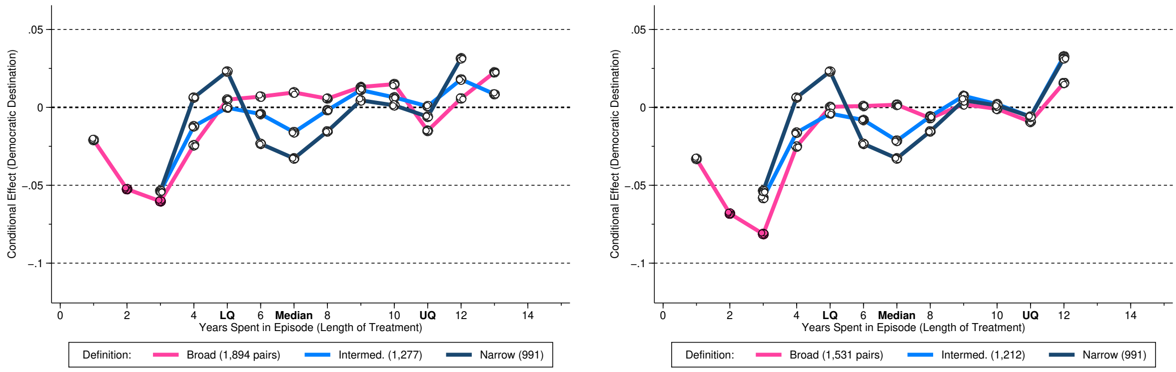
[35]Using cross-section averages to proxy for unobserved common factors (see Pesaran, 2006 and for limited dependent variables Boneva and Linton, 2017) is a well-established alternative to estimating factors (e.g. Bai, 2009; Chan and Kwok, 2022).

Figure 6: Episodes of Democratic Erosion and Exports

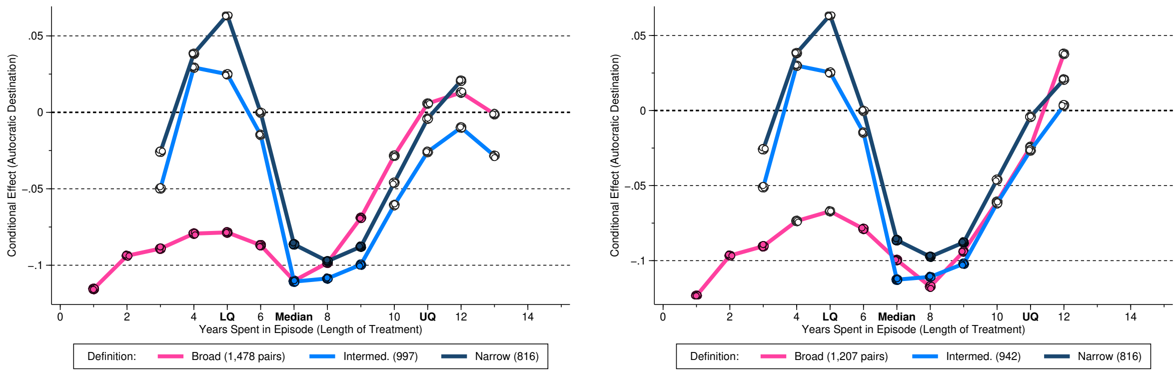
(a) Any Destination, all states$^{\bullet\circ\circ}$ (left) and states with single episode$^{\bullet\circ\bullet}$ (right)



(b) Democratic Destinations, all states$^{\bullet\circ\circ}$ (left) and states with single episode$^{\bullet\circ\circ}$ (right)



(c) Autocratic Destinations, all states$^{\bullet\bullet\bullet}$ (left) and states with single episode$^{\bullet\bullet\bullet}$ (right)



Notes: We present treated-state predictions from multivariate running line regressions (Royston and Cox, 2005) illustrating the economic effects of democratic erosion for state exports to (a) all destinations, (b) destinations which are democratic throughout 2002-2023, and (c) destinations which are autocratic throughout 2002-2023. The (total) number of years a state has spent in one or more autocratisation episodes is shown on the x-axis of each plot, and the economic effect on the y-axis. The running line regressions control for the number of times a state experienced an episode and use weights derived from the robust ATET estimate using an M-estimator (Rousseeuw and Leroy, 2005). An estimate of -.1 indicates a reduction of -9.5% ($exp(-0.1) - 1 = -.095$) in exports. Each plot presents results for our three definitions of autocratisation episodes: a broad one (33 treated states), an intermediate one (22), and a narrow one (18) — see main text for definitions. A filled (white) marker indicates statistical (in)significance at the 10% level. We use $\bullet$ and $\circ$ in the subfigure title to indicate whether a specification passes or fails the Alpha diagnostic test at the 10% level: $^{\bullet\circ\circ}$ indicates that the liberal definition passes the test, whereas the intermediate and conservative definitions do not.

Diagnostics In the heterogeneous gravity model, we have to confirm the identifying assumption of 'factor spanning', which (related to the spirit of the Alpha test) requires the factors obtained in the control sample (via CAs) to span the same space as the unobservables in the treated sample. If CAs absorb the common factor structure for both treated and control pairs, their residuals should exhibit similar levels of cross-section dependence. If one or more factors are not captured ('unspanned') this generates a systematic residual structure for the treated pairs which is absent in the control pair residuals, and hence the difference of their average absolute correlation is statistically larger than 0. In practice, we obtain p-values for this test by permuting exporter treated/control labels 1,000 times, each time recomputing ΔCSD under the null. Appendix Table D-3 reports the p-values for these tests related to our state export flow analysis. Focusing on the analysis to all destinations, only the 'broad' definition passes this diagnostic test, whereas if we focus on states with a single episode the 'narrow' one also passes. If we zoom in further to consider only the top-100 or top-50 trading destinations, the diagnostics generally improve. Similarly if we consider specifications which include period dummies for the Covid years.

Main Results and Robustness Figure 6 presents the running line predictions for exports to (a) any destination, (b) destinations which were democratic throughout the sample period, and (c) destinations which were autocratic throughout the sample period. This split is informed by research on democratic collapse and autocratisation episodes in their effect on export flows (Boese-Schlosser et al., 2025), where reduced export flows were found to be driven by falling exports to *democratic* countries, suggesting a 'punishment' effect for democratic retreat. In each case we provide results for *all* states with autocratisation episodes (left plot) and those which only experienced a *single episode* (right plot).

Our results for all destinations in panel (a) show some common patterns across autocratisation definitions over treatment length, but crucially, the effects are by and large statistically insignificantly different from zero. This finding extends to the democratic destinations in panel (b). Results for the exports to autocratic destinations in panel (c) show strong U-shaped patterns with statistically significant negative effects of around -10% for 7 to 8 years in an autocratisation episode. This finding is surprising and robust to a narrower focus on top-100 (see Appendix Figure E-5) and top-50 export destinations (available upon request). However, inclusion of Covid dummies (for the years 2020-2022), as can be seen in Appendix Figure E-6, reduces the magnitude of this effect and renders it statistically insignificant.

In contrast to the national-level analysis in Boese-Schlosser et al. (2025), we find no evidence for sub-national units such as U.S. states being (knowingly or implicitly) punished by democratic destinations for democratic erosion. In the present paper, a seeming reduction in exports to autocratic countries turns insignificant when we account for the Covid years. In conclusion, it appears that foreign trading partners of the United States focus on *national* democratic institutions and ignore sub-national autocratisation.

5 Concluding Remarks

This paper provides the first systematic evidence that episodes of autocratisation within U.S. states have economic consequences. While earlier literature has focused primarily on national-level autocratisation in developing countries, we show that even in a consolidated democracy like the United States, the erosion of democratic institutions at the sub-national level can harm economic outcomes, both immediately and in the long term.

We examine the economic consequences of autocratisation within U.S. states between 2000 and 2023. Leveraging the State Democracy Index (Grumbach, 2023, SDI) to capture sub-national democratic decline, we combine panel data on all 50 states with a difference-in-differences design to identify causal effects of autocratisation episodes, which we construct from the SDI. Our results show that the economic costs are particularly salient for vulnerable populations: we find declining average income, robust increases in poverty, as well as indications of rising inequality. Most strikingly, democratic erosion appears to undermine the foundations of long-run growth by reducing private firms' investment in research and development and depressing state-level innovation.

Prior studies have shown that national-level democratic breakdowns often trigger international economic repercussions, as other democracies reduce trade and investment in a form of 'accountability through trade.' In contrast, our findings suggest that this mechanism does not operate at the subnational level: in the U.S. context, where individual states are embedded within a stable federal system, democratic erosion at the sub-national level has not been associated with any decline in international exports. This suggests that foreign trading partners primarily assess risk through national-level institutions such as federal courts and overarching rule-of-law guarantees, effectively insulating autocratising U.S. states from external economic penalties. Consequently, while national autocratization may trigger international responses through trade disruptions, sub-national erosion of democracy unfolds largely under the protective umbrella of national stability, leaving internal economic consequences as the primary channel of cost.

In the present analysis, we have focused on the general notion of democratic erosion by adopting Grumbach's (2023) democracy index in combination with some simple rules to create variants of autocratisation episodes. We do provide some very simple model selection procedures to get insights into which components of state democracy matter (voter suppression, gerrymandering, etc.) most. Future research could revisit the data underlying the Grumbach SDI and conduct a more systematic 'drilling down' to constituent components to evaluate which type of meddling with electoral democracy leads to the economic malaise we have documented.

References

Samuel Absher, Kevin Grier, and Robin Grier. The economic consequences of durable left-populist regimes in latin america. *Journal of Economic Behavior & Organization*, 177: 787–817, 2020.

Daron Acemoglu, Suresh Naidu, Pascual Restrepo, and James A Robinson. Democracy does cause growth. *Journal of Political Economy*, 127(1):47–100, 2019.

James E Anderson and Eric Van Wincoop. Gravity with gravitas: a solution to the border puzzle. *American Economic Review*, 93(1):170–192, 2003.

Joshua D Angrist and Jörn-Steffen Pischke. *Mostly Harmless Econometrics*. Princeton University Press, 2008.

Luis Araujo, Giordano Mion, and Emanuel Ornelas. Institutions and export dynamics. *Journal of International Economics*, 98:2–20, 2016.

Susan Athey, Mohsen Bayati, Nikolay Doudchenko, Guido Imbens, and Khashayar Khosravi. Matrix completion methods for causal panel data models. *Journal of the American Statistical Association*, 116(536):1716–1730, 2021.

Jushan Bai. Panel data models with interactive fixed effects. *Econometrica*, 77(4):1229–1279, 2009.

Robert J Barro. Economic growth in a cross section of countries. *Quarterly Journal of Economics*, 106(2):407–443, 1991.

Nancy Bermeo. On democratic backsliding. *Journal of Democracy*, 27(1):5–19, 2016.

Timothy Besley and Masayuki Kudamatsu. Health and democracy. *American economic review*, 96(2):313–318, 2006.

Christopher Blattman, Scott Gehlbach, and Zeyang Yu. The personalist penalty: Varieties of autocracy and economic growth. NBER Working Paper Series 34093, National Bureau of Economic Research, 2025.

Vanessa A Boese, Amanda B Edgell, Sebastian Hellmeier, Seraphine F Maerz, and Staffan I Lindberg. How democracies prevail: democratic resilience as a two-stage process. In *Resilience of Democracy*, pages 17–39. Routledge, 2023.

Vanessa A Boese-Schlosser and Markus Eberhardt. Democracy doesn't always happen overnight: regime change in stages and economic growth. *Review of Economics and Statistics*, forthcoming, 2024.

Vanessa A Boese-Schlosser and Markus Eberhardt. Democracy in decline: The economic implications of democratic collapse. Discussion Paper SP-V-2025-502, Berlin: WZB, 2025.

Vanessa A Boese-Schlosser, Rodolphe Desbordes, Markus Eberhardt, and Mario Larch. Democratic retreat and trade flows. unpublished mimeo, 2025.

Lena Boneva and Oliver Linton. A discrete-choice model for large heterogeneous panels with interactive fixed effects with an application to the determinants of corporate bond issuance. *Journal of Applied Econometrics*, 32(7):1226–1243, 2017.

Benjamin Born, Gernot Müller, Moritz Schularick, and Petr Sedlácek. Stable genius? the macroeconomic impact of trump. CEPR Discussion Paper DP17022, 2019.

Kirill Borusyak, Xavier Jaravel, and Jann Spiess. Revisiting event-study designs: Robust and efficient estimation. *Review of Economic Studies*, 91(6):3253–3285, 2024.

Tilman Bretschneider and Joakim Westerlund. The pcdid approach to treatment effects estimation: A further investigation. *Journal of Applied Econometrics*, (forthcoming), 2025.

Brantly Callaway and Pedro HC Sant'Anna. Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2):200–230, 2021.

Marc K. Chan and Simon S. Kwok. The PCDID approach: Difference-in-differences when trends are potentially unparallel and stochastic. *Journal of Business & Economic Statistics*, 40(3):1216–1233, 2022.

Rachel Cho, Rodolphe Desbordes, and Markus Eberhardt. The economic effects of 'excessive' financial deepening. *Oxford Bulletin of Economics and Statistics*, 88(3):458–472, 2026.

Michael Coppedge, John Gerring, Carl Henrik Knutsen, Staffan I Lindberg, Jan Teorell, David Altman, Michael Bernhard, Agnes Cornell, M Steven Fish, Linnea Fox, et al. V-Dem Dataset v15. Varieties of Democracy Institute, Available at: https://doi.org/10.23696/vdemds25, 2025.

Robert A Dahl. *On democracy*. Yale University Press, 2000.

Robert Alan Dahl. *Polyarchy: Participation and opposition*. New Haven and London: Yale University Press, 1971.

Sirianne Dahlum and Carl Henrik Knutsen. Do democracies provide better education? revisiting the democracy–human capital link. *World Development*, 94:186–199, 2017.

Clement De Chaisemartin and Xavier d'Haultfœuille. Two-way fixed effects estimators with heterogeneous treatment effects. *American Economic Review*, 110(9):2964–96, 2020.

Clément de Chaisemartin and Xavier D'Haultfœuille. Difference-in-differences estimators of intertemporal treatment effects. *Review of Economics and Statistics*, forthcoming, 2024.

Rodolphe Desbordes, Markus Eberhardt, and Mario Larch. A democratic dividend in trade? evidence from a flexible empirical implementation. Working Paper 11735, CESifo, 2025.

Markus Eberhardt. Democracy, growth, heterogeneity, and robustness. *European Economic Review*, 147(104173), 2022.

Amanda Edgell, Seraphine Maerz, Laura Maxwell, Rick Morgan, Juraj Medzihorsky, Matthew Wilson, Vanessa Alexandra Boese, Sebastian Hellmeier, Jean Lachapelle, Patrik Lindenfors, Anna Lührmann, and Staffan Ingemar Lindberg. Episodes of regime transformation dataset, version 1.0. *Available at https://github.com/vdeminstitute/ERT*, 2020.

Amanda B Edgell, Vanessa A Boese, Seraphine F Maerz, Patrik Lindenfors, and Staffan I Lindberg. The institutional order of liberalization. *British Journal of Political Science*, 52 (3):1465–1471, 2022.

Manuel Funke, Moritz Schularick, and Christoph Trebesch. Populist leaders and the economy. *American Economic Review*, 113(12):3249–3288, 2023.

Scott Gehlbach and Alberto Simpser. Electoral manipulation as bureaucratic control. *American Journal of Political Science*, 59(1):212–224, 2015.

Laurent Gobillon and Thierry Magnac. Regional policy evaluation: Interactive fixed effects & synthetic controls. *Review of Economics and Statistics*, 98(3):535–51, 2016.

Edoardo Grillo and Carlo Prato. Reference points and democratic backsliding. *American Journal of Political Science*, 67(1):71–88, 2023.

Edoardo Grillo, Zhaotian Luo, Monika Nalepa, and Carlo Prato. Theories of democratic backsliding. *Annual Review of Political Science*, 27:381–400, 2024.

Jacob M Grumbach. Laboratories of democratic backsliding. *American Political Science Review*, 117(3):967–984, 2023.

Sergei Guriev and Elias Papaioannou. The political economy of populism. *Journal of Economic Literature*, 60(3):753–832, 2022.

Yohan Iddawela, Neil Lee, and Andrés Rodríguez-Pose. Quality of sub-national government and regional development in africa. *The Journal of Development Studies*, 57(8):1282–1302, 2021.

Carl Henrik Knutsen. A business case for democracy: regime type, growth, and growth volatility. *Democratization*, 28(8):1505–1524, 2021.

Carl Henrik Knutsen, Kyle L Marquardt, Brigitte Seim, Michael Coppedge, Amanda B Edgell, Juraj Medzihorsky, Daniel Pemstein, Jan Teorell, John Gerring, and Staffan I Lindberg. Conceptual and measurement issues in assessing democratic backsliding. *PS: Political Science & Politics*, 57(2):162–177, 2024.

Andrei A Levchenko. Institutional quality and international trade. *The Review of Economic Studies*, 74(3):791–819, 2007.

James Levinsohn and Amil Petrin. Estimating production functions using inputs to control for unobservables. *Review of Economic Studies*, 70(2):317–341, 2003.

Steven Levitsky and Daniel Ziblatt. *How democracies die*. Crown, 2019.

Andrew T Little and Anne Meng. Measuring democratic backsliding. *PS: Political Science & Politics*, 57(2):149–161, 2024a.

Andrew T Little and Anne Meng. What we do and do not know about democratic backsliding. *PS: Political Science & Politics*, 57(2):224–229, 2024b.

Licheng Liu, Ye Wang, and Yiqing Xu. A practical guide to counterfactual estimators for causal inference with time-series cross-sectional data. *American Journal of Political Science*, 68 (1):160–176, 2024.

Anna Lührmann and Staffan I Lindberg. A third wave of autocratization is here: what is new about it? *Democratization*, 26(7):1095–1113, 2019.

Anna Lührmann, Marcus Tannenberg, and Staffan I Lindberg. Regimes of the World (RoW): Opening New Avenues for the Comparative Study of Political Regimes. *Politics & Governance*, 6(1):60–77, 2018.

Seraphine F Maerz, Amanda B Edgell, Matthew C Wilson, Sebastian Hellmeier, and Staffan I Lindberg. Episodes of regime transformation. *Journal of Peace Research*, 61(6):967–984, 2024.

Maximiliano Marzetti and Rok Spruk. Long-term economic effects of populist legal reforms: Evidence from argentina. *Comparative Economic Studies*, 65(1):60, 2022.

Julian Michel. The subnational roots of democratic stability. Ucla phd dissertation in political science, 2024.

Hyungsik Roger Moon and Martin Weidner. Linear regression for panel with unknown number of factors as interactive fixed effects. *Econometrica*, 83(4):1543–1579, 2015.

Petra Moser, Alessandra Voena, and Fabian Waldinger. German jewish émigrés and us invention. *American Economic Review*, 104(10):3222–3255, 2014.

Petra Moser, Sahar Parsa, and Shmuel San. Immigration and innovation: Lessons from the quota acts. Technical report, unpublished mimeo, 2025.

Mathias Wullum Nielsen, Sharla Alegria, Love Börjeson, Henry Etzkowitz, Holly J. Falk-Krzesinski, Aparna Joshi, Erin Leahey, Laurel Smith-Doerr, Anita Williams Woolley, and Londa Schiebinger. Gender Diversity Leads to Better Science. *Proceedings of the National Academy of Sciences*, 114(8):1740–1742, 2017. doi: 10.1073/pnas.1700616114.

Marina Nord, Martin Lundstedt, David Altman, Fabio Angiolillo, Cecilia Borella, Tiago Fernandes, Lisa Gastaldi, Ana Good God, Natalia Natsika, and Staffan I Lindberg. Democracy Report 2024: Democracy Winning and Losing at the Ballot. University of Gothenburg, V-Dem Institute, 2024.

Nathan Nunn. Relationship-specificity, incomplete contracts, and the pattern of trade. *The Quarterly Journal of Economics*, 122(2):569–600, 2007.

G Steven Olley and Ariel Pakes. The dynamics of productivity in the telecommunications equipment industry. *Econometrica*, 64(6):1263–1297, 1996.

Elias Papaioannou and Gregorios Siourounis. Democratisation and growth. *Economic Journal*, 118(532):1520–1551, 2008.

Torsten Persson and Guido Tabellini. *Political economics: Explaining economic policy*. MIT Press, 2002.

M Hashem Pesaran. Estimation and inference in large heterogeneous panels with a multifactor error structure. *Econometrica*, 74(4):967–1012, 2006.

Rachel Beatty Riedl, Paul Friesen, Jennifer McCoy, and Kenneth Roberts. Democratic backsliding, resilience, and resistance. *World Politics*, 77:151–151, 2024.

Peter Rousseeuw and Annick Leroy. *Robust Regression and Outlier Detection*. John Wiley & Sons, 2005.

Patrick Royston and Nicholas J Cox. A multivariable scatterplot smoother. *Stata Journal*, 5 (3):405–412, 2005.

Xavier Sala-i Martin. I just ran two million regressions. *American Economic Review Papers & Proceedings*, 87(2):178–183, 1997.

Xavier Sala-i Martin, Gernot Doppelhofer, and Ronald I Miller. Determinants of long-term growth: A Bayesian averaging of classical estimates (BACE) approach. *American Economic Review*, 94(4):813–835, 2004.

Yuko Sato, Martin Lundstedt, Kelly Morrison, Vanessa A Boese, and Staffan I Lindberg. Institutional order in episodes of autocratization. *V-Dem Working Paper*, 133, 2022.

Andreas Schedler. The menu of manipulation. *Journal of Democracy*, 13(2):36–50, 2002.

Joseph A Schumpeter. *Capitalism, Socialism and Democracy*. London and New York: Harper & Brothers, 1942.

Liugang Sheng and Dennis Tao Yang. Expanding export variety: The role of institutional reforms in developing countries. *Journal of Development Economics*, 118:45–58, 2016.

Liyang Sun and Sarah Abraham. Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics*, 225(2):175–199, 2021.

David Waldner and Ellen Lust. Unwelcome change: Coming to terms with democratic backsliding. *Annual Review of Political Science*, 21(1):93–113, 2018.

Matthew C Wilson, Juraj Medzihorsky, Seraphine F Maerz, Patrik Lindenfors, Amanda B Edgell, Vanessa A Boese, and Staffan I Lindberg. Episodes of liberalization in autocracies: a new approach to quantitatively studying democratization. *Political Science Research and Methods*, 11(3):501–520, 2023.

L Guillermo Woo-Mora. Populism's original sin: Short-term populist penalties and uncertainty traps. *European Economic Review*, 172:104917, 2025.

Yiqing Xu. Generalized synthetic control method: Causal inference with interactive fixed effects models. *Political Analysis*, 25(1):57–76, 2017.

Miaojie Yu. Trade, democracy, and the gravity equation. *Journal of Development Economics*, 91(2):289–300, 2010.

Ben Zissimos. A theory of trade policy under dictatorship and democratization. *Journal of International Economics*, 109:85–101, 2017.

Online Appendix
— not for publication —

Contents

A State Democracy Index

To identify episodes of autocratization in U.S. states, we rely on the State Democracy Index (SDI) developed by Grumbach (2023), which tracks changes in the quality of democratic institutions at the subnational level between 2000 and 2023. This index is constructed from 49 indicators across multiple domains of electoral democracy and is publicly available. For a full breakdown of all individual indicators and their sources, see Grumbach (2023), Supplementary Information, Table A1.

The SDI is based on indicators capturing changes in election laws and practices, voting access, and electoral integrity such as: voter registration, voter identification laws, early voting and absentee voting access, vote-by-mail policies, polling place availability, pre-registration and automatic registration, felony disenfranchisement, gerrymandering (partisan bias in districting), voter roll purging, access to registration (e.g., online registration), registration deadlines, state election administration partisanship, mail ballot verification rules, ballot collection restrictions, signature matching policies.

Interpretation and Scope

The SDI is specifically designed for use within the United States between 2000 and 2023, a context in which other baseline features of electoral democracy — such as civil liberties, separation of powers, and judicial independence — were established. As such, it is not a generalizable measure of electoral democracy suitable for cross-national or historical comparisons. Instead, it should be understood as a tool to capture subnational variation in the quality of democratic institutions within an already democratic federal system, rather than a measure to assess whether electoral democracy exists in the first place.

To illustrate this distinction, consider how other prominent democracy indices, such as the V-Dem measure of electoral democracy, include additional components that are absent in the SDI — such as freedom of expression, independent media, and the ability of civil society to operate freely. These elements are central to the functioning of electoral democracy in a cross-national context, but are implicitly assumed to be present in the U.S. federal system. This highlights that Grumbach's index does not operationalize electoral democracy per se, but instead reflects fluctuations in its quality across U.S. states. Recognizing this distinction is conceptually important for understanding what the index captures — and what it does not.

One key advantage of the SDI is that it captures both *de jure* aspects of electoral democracy (e.g., automatic voter registration, early voting laws) and *de facto* outcomes (e.g., gerrymandering bias measures, felony disenfranchisement rates, ballot rejection rates), allowing for a comprehensive assessment of democratic quality at the state level.

For our purposes, episodes of democratization and autocratization refer to substantive changes in the composite index derived from these indicators (as discussed in the main section of the paper).

In conclusion, while the measure is not suited for cross-national comparisons, it provides a valid and well-grounded instrument for studying subnational variation in democratic quality across U.S. states, i.e. within a federal democracy and during the sample period.

B Data, Sample Makeup, and Descriptives

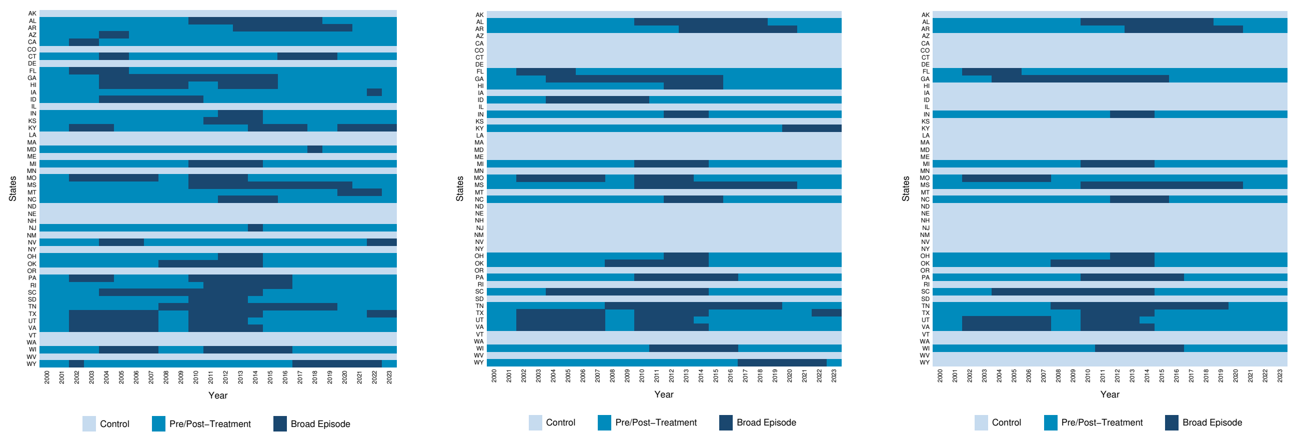
Table B-1: Distribution and Length of Autocratisation Episodes

Definition	**'Broad'** # Episodes				**'Intermediate'** # Episodes				**'Narrow'** # Episodes		
Years	1	2	3	Total	1	2	3	Total	1	2	Total
1	3			3				0			0
2	2			2				0			0
3	3			3	2			2	2		2
4/**LQ**	4			4	4			4	2		2
5	1	2		3	1			1	2		2
6	1	2		3	2			2	2		2
7/**MD**	2	2		4	3			3	2		2
8	1			1	1			1	1		1
9	1			1	1			1	1		1
10		10		10		4		4		2	2
11/**UQ**	2	2	3	7	2	2		4	2	2	4
12	2			2	2			2	2		2
13			3	3			3	3			
Total episodes	22	18	6	46	18	6	3	27	16	4	20
Total states	22	9	2	33	18	3	1	22	16	2	18

Notes: We present details of the distribution and length of episodes based on our three definitions (broad, intermediate, narrow) in the respective three blocks of columns. The numbers in these columns represent frequencies (number of episodes), while the rows represent the number of years states spend in these episodes. Starting in the top-left corner of the table, the entry '3' indicates that three states each experienced single-year episodes exactly once using the 'Broad' definition of autocratisation. The eye-catching entry of '10' in the row labelled '10' and the column labelled '2' indicates that there were five states, each experienced two episodes that, in each state, lasted a combined total of ten years (the distribution across episodes is not reported here). While any entry into a column labelled '1' can be read as the number of states (as well as the number of episodes), entries in columns labelled '2' and '3' are episode counts and hence need to be divided by two and three, respectively, to arrive at the number of states. We report in bold the lower quartile (LQ), median (MD), and upper quartile (UQ) years spent in an episode (treated samples only).

Figure B-1: Treatment and Control Observations by Episode Variant

(a) Broad (b) Intermediate (c) Narrow



Notes: We plot the treated and control observations by state and year for our three variants of autocratisation episodes.

Table B-2: Sample Makeup — Democratic Quality

		Rescaled Grumbach SDI				'Broad' Episode			'Intermed.' Episode			'Narrow' Episode			Always
ID	State	2000	2023	Δ	%Δ	Ep	Years	Sample	Ep	Years	Sample	Ep	Years	Sample	T/C
AK	Alaska	0.72	0.74	0.016	2.2	0	0	control	0	0	control	0	0	control	control
AL	Alabama	0.43	0.22	-0.211	-48.6	1	9	treated	1	9	treated	1	9	treated	treated
AR	Arkansas	0.69	0.37	-0.319	-46.0	1	8	treated	1	8	treated	1	8	treated	treated
AZ	Arizona	0.69	0.78	0.095	13.9	1	2	treated	0	0	control	0	0	control	
CA	California	0.85	0.90	0.050	5.8	1	2	treated	0	0	control	0	0	control	
CO	Colorado	0.90	0.95	0.046	5.1	0	0	control	0	0	control	0	0	control	control
CT	Connecticut	0.72	0.80	0.082	11.3	2	6	treated	0	0	control	0	0	control	
DE	Delaware	0.60	0.70	0.106	17.7	0	0	control	0	0	control	0	0	control	control
FL	Florida	0.52	0.61	0.091	17.6	1	4	treated	1	4	treated	1	4	treated	treated
GA	Georgia	0.56	0.43	-0.127	-22.8	1	12	treated	1	12	treated	1	12	treated	treated
HI	Hawaii	0.79	0.75	-0.043	-5.4	2	10	treated	1	4	treated	0	0	control	
IA	Iowa	0.83	0.75	-0.084	-10.1	1	1	treated	0	0	control	0	0	control	
ID	Idaho	0.62	0.52	-0.101	-16.3	1	7	treated	1	7	treated	0	0	control	
IL	Illinois	0.61	0.93	0.319	52.1	0	0	control	0	0	control	0	0	control	control
IN	Indiana	0.71	0.38	-0.325	-45.8	1	3	treated	1	3	treated	1	3	treated	treated
KS	Kansas	0.64	0.72	0.072	11.2	1	4	treated	0	0	control	0	0	control	
KY	Kentucky	0.73	0.44	-0.290	-39.7	3	11	treated	1	4	treated	0	0	control	
LA	Louisiana	0.55	0.58	0.028	5.1	0	0	control	0	0	control	0	0	control	control
MA	Massachusetts	0.67	0.76	0.088	13.1	0	0	control	0	0	control	0	0	control	control
MD	Maryland	0.75	0.62	-0.129	-17.2	1	1	treated	0	0	control	0	0	control	
ME	Maine	0.90	0.91	0.010	1.2	0	0	control	0	0	control	0	0	control	control
MI	Michigan	0.50	0.78	0.273	54.1	1	5	treated	1	5	treated	1	5	treated	treated
MN	Minnesota	0.79	0.90	0.108	13.6	0	0	control	0	0	control	0	0	control	control
MO	Missouri	0.68	0.38	-0.295	-43.6	2	10	treated	2	10	treated	1	6	treated	treated
MS	Mississippi	0.66	0.36	-0.307	-46.3	1	11	treated	1	11	treated	1	11	treated	treated
MT	Montana	0.83	0.85	0.022	2.6	1	3	treated	0	0	control	0	0	control	
NC	North Carolina	0.75	0.61	-0.138	-18.5	1	4	treated	1	4	treated	1	4	treated	treated
ND	North Dakota	0.88	0.79	-0.093	-10.6	0	0	control	0	0	control	0	0	control	control
NE	Nebraska	0.70	0.70	0.004	0.6	0	0	control	0	0	control	0	0	control	control
NH	New Hampshire	0.80	0.65	-0.148	-18.6	0	0	control	0	0	control	0	0	control	control
NJ	New Jersey	0.67	0.92	0.248	37.1	1	1	treated	0	0	control	0	0	control	
NM	New Mexico	0.82	0.84	0.022	2.7	0	0	control	0	0	control	0	0	control	control
NV	Nevada	0.77	0.76	-0.017	-2.2	2	5	treated	0	0	control	0	0	control	
NY	New York	0.56	0.80	0.245	44.0	0	0	control	0	0	control	0	0	control	control
OH	Ohio	0.58	0.49	-0.093	-15.9	1	3	treated	1	3	treated	1	3	treated	treated
OK	Oklahoma	0.75	0.45	-0.301	-40.1	1	7	treated	1	7	treated	1	7	treated	treated
OR	Oregon	0.87	0.97	0.101	11.7	0	0	control	0	0	control	0	0	control	control
PA	Pennsylvania	0.70	0.81	0.112	16.1	2	10	treated	1	7	treated	1	7	treated	treated
RI	Rhode Island	0.62	0.71	0.093	15.1	1	6	treated	0	0	control	0	0	control	
SC	South Carolina	0.52	0.42	-0.102	-19.5	1	11	treated	1	11	treated	1	11	treated	treated
SD	South Dakota	0.77	0.68	-0.084	-10.9	1	4	treated	0	0	control	0	0	control	
TN	Tennessee	0.57	0.10	-0.471	-82.6	1	12	treated	1	12	treated	1	12	treated	treated
TX	Texas	0.71	0.57	-0.139	-19.6	3	13	treated	3	13	treated	1	5	treated	treated
UT	Utah	0.88	0.58	-0.303	-34.3	2	10	treated	2	10	treated	2	10	treated	treated
VA	Virginia	0.71	0.87	0.152	21.3	2	11	treated	2	11	treated	2	11	treated	treated
VT	Vermont	0.77	0.78	0.009	1.2	0	0	control	0	0	control	0	0	control	control
WA	Washington	0.89	1.00	0.105	11.7	0	0	control	0	0	control	0	0	control	control
WI	Wisconsin	0.81	0.26	-0.551	-68.2	2	10	treated	1	6	treated	1	6	treated	treated
WV	West Virginia	0.70	0.59	-0.105	-15.1	0	0	control	0	0	control	0	0	control	control
WY	Wyoming	0.85	0.55	-0.296	-35.0	2	7	treated	1	6	treated	0	0	control	
	Average	0.71	0.66	-0.05	-6.89	0.92	4.46		0.54	3.34		0.40	2.68		

Notes: We present sample statistics related to state democracy. A first block of columns after the state identifier and name covers the Grumbach (2023) State Democracy Index (SDI), which we rescale to [0,1]. We report the rescaled index values for 2000 and 2023 as well as the total change over these 24 years and the percentage change (%Δ). The next three blocks of columns present details of our three alternative definitions of autocratisation episodes: a liberal, intermediate, and conservative version. In each case, we report the episode count (Ep), the total years spent in episode(s) (Years), the sample affiliation of the state (treated or control). For the intermediate and conservative definitions, we highlight which states switch from treated to control samples relative to the liberal and intermediate definitions, respectively. The final column indicates states that are either always in the treated or always in the control sample.

Table B-3: Sample Makeup — Macroeconomic Variables

		Real GDP pc (US$)		**MD HH Inc** (US$)		**Poverty Rate** (%)		**ΔEmploym** (%)		**ΔPopulation** (%)	
ID	State	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
AK	Alaska	51,109	5,683	86,697	5,859	10.46	1.89	0.85	1.74	0.57	5.30
AL	Alabama	38,738	3,011	58,579	3,612	15.43	1.30	0.88	1.71	0.50	2.27
AR	Arkansas	38,336	4,652	55,241	4,036	16.43	1.92	0.84	1.30	0.03	2.41
AZ	Arizona	41,634	4,639	69,279	6,751	15.19	3.24	2.03	2.37	1.21	2.12
CA	California	53,226	8,472	80,384	6,391	13.37	1.78	1.24	2.15	0.69	2.65
CO	Colorado	50,971	7,462	83,650	7,023	10.04	1.54	1.72	1.83	1.01	2.16
CT	Connecticut	67,772	4,501	89,197	4,379	9.13	1.17	0.65	1.60	0.02	2.58
DE	Delaware	50,540	1,509	76,319	6,031	9.99	2.18	1.25	1.76	0.52	3.29
FL	Florida	46,607	4,411	63,587	3,612	13.35	1.66	2.26	2.33	1.48	3.03
GA	Georgia	42,705	3,648	66,273	4,049	14.57	2.44	1.71	1.96	1.13	2.53
HI	Hawaii	48,310	3,309	86,421	7,983	10.18	1.60	0.92	3.13	0.71	3.92
IA	Iowa	44,522	4,200	72,256	5,713	9.65	0.97	0.49	1.29	0.72	2.52
ID	Idaho	38,907	4,872	68,534	6,281	11.22	2.08	2.03	2.14	1.40	3.14
IL	Illinois	51,146	4,149	75,745	6,984	11.48	1.66	0.52	1.78	0.13	2.96
IN	Indiana	42,574	4,193	67,618	6,094	11.91	2.43	0.65	1.83	0.48	3.10
KS	Kansas	45,743	4,825	70,310	7,974	11.45	2.30	0.58	1.33	0.55	2.26
KY	Kentucky	38,949	3,293	58,312	4,296	15.98	2.17	0.75	1.64	0.08	3.30
LA	Louisiana	42,324	3,055	55,876	3,882	18.50	2.24	0.77	1.73	-0.13	5.62
MA	Massachusetts	61,659	7,995	87,408	9,108	10.19	1.45	1.01	2.10	0.34	1.96
MD	Maryland	56,223	3,501	94,748	9,095	8.75	1.11	1.16	1.55	0.97	2.26
ME	Maine	44,548	4,008	66,783	6,139	11.37	1.73	0.64	1.64	0.27	1.98
MI	Michigan	43,307	4,065	69,499	4,437	12.37	1.70	0.34	2.42	-0.28	3.47
MN	Minnesota	50,857	4,990	84,601	6,368	8.38	1.64	0.75	1.69	0.55	2.50
MO	Missouri	43,573	3,669	68,579	4,774	11.91	1.99	0.63	1.39	0.06	2.29
MS	Mississippi	35,455	2,776	51,283	2,643	19.38	2.30	0.62	1.42	-0.08	2.74
MT	Montana	42,022	5,403	62,112	8,255	12.55	2.02	1.39	1.45	0.67	2.82
NC	North Carolina	42,880	4,109	62,250	5,186	14.63	1.79	1.43	1.88	0.98	2.56
ND	North Dakota	48,063	8,541	69,982	7,525	10.54	1.27	1.32	2.37	1.61	5.08
NE	Nebraska	47,965	4,754	74,148	6,892	9.85	1.07	0.79	1.07	1.11	2.27
NH	New Hampshire	54,768	5,900	91,755	5,617	6.49	1.20	0.91	1.70	0.35	2.53
NJ	New Jersey	59,755	4,386	89,223	6,453	8.92	1.43	1.16	2.03	0.25	2.49
NM	New Mexico	37,305	4,277	58,434	2,483	18.25	1.89	0.89	1.81	-0.01	3.84
NV	Nevada	46,908	3,465	71,298	6,206	12.26	2.79	2.41	3.61	1.90	3.63
NY	New York	58,469	5,306	72,272	6,192	13.88	1.57	1.08	2.19	0.50	2.65
OH	Ohio	43,755	3,698	67,292	5,056	12.58	1.75	0.42	1.64	0.17	2.68
OK	Oklahoma	40,530	5,528	61,473	4,321	14.33	1.81	0.95	1.50	0.48	2.99
OR	Oregon	43,519	6,778	74,471	9,132	11.70	1.86	1.13	2.02	0.63	2.32
PA	Pennsylvania	48,706	4,953	72,292	5,633	11.08	1.24	0.75	1.66	0.14	2.79
RI	Rhode Island	49,155	3,678	75,542	6,441	11.09	1.76	0.77	2.05	0.07	2.86
SC	South Carolina	39,767	3,922	61,988	5,665	14.36	1.87	1.42	1.97	0.86	2.50
SD	South Dakota	47,624	6,117	68,299	6,650	11.52	2.06	1.12	1.09	0.82	3.45
TN	Tennessee	42,579	4,372	60,462	5,666	14.44	2.27	1.27	1.84	0.86	2.26
TX	Texas	44,283	5,752	68,765	5,929	15.33	1.82	2.25	1.59	1.75	3.37
UT	Utah	40,326	5,832	84,229	8,991	8.81	1.43	2.46	1.98	1.81	2.41
VA	Virginia	52,469	4,639	85,064	5,710	9.70	1.06	1.23	1.45	0.74	2.05
VT	Vermont	47,467	4,709	74,534	6,254	9.24	1.23	0.53	1.78	0.07	2.53
WA	Washington	52,420	7,868	82,565	9,189	10.34	1.61	1.45	2.00	1.24	2.52
WI	Wisconsin	46,110	4,020	73,306	3,870	9.93	1.49	0.58	1.58	0.27	2.40
WV	West Virginia	37,000	3,012	54,674	4,276	15.99	1.65	0.17	1.64	-0.13	3.43
WY	Wyoming	52,871	5,824	71,634	4,236	9.91	1.06	1.42	1.96	0.43	5.59
	Average	46,729	8,311	71,905	12,194	12.17	3.33	1.09	1.93	0.61	3.04

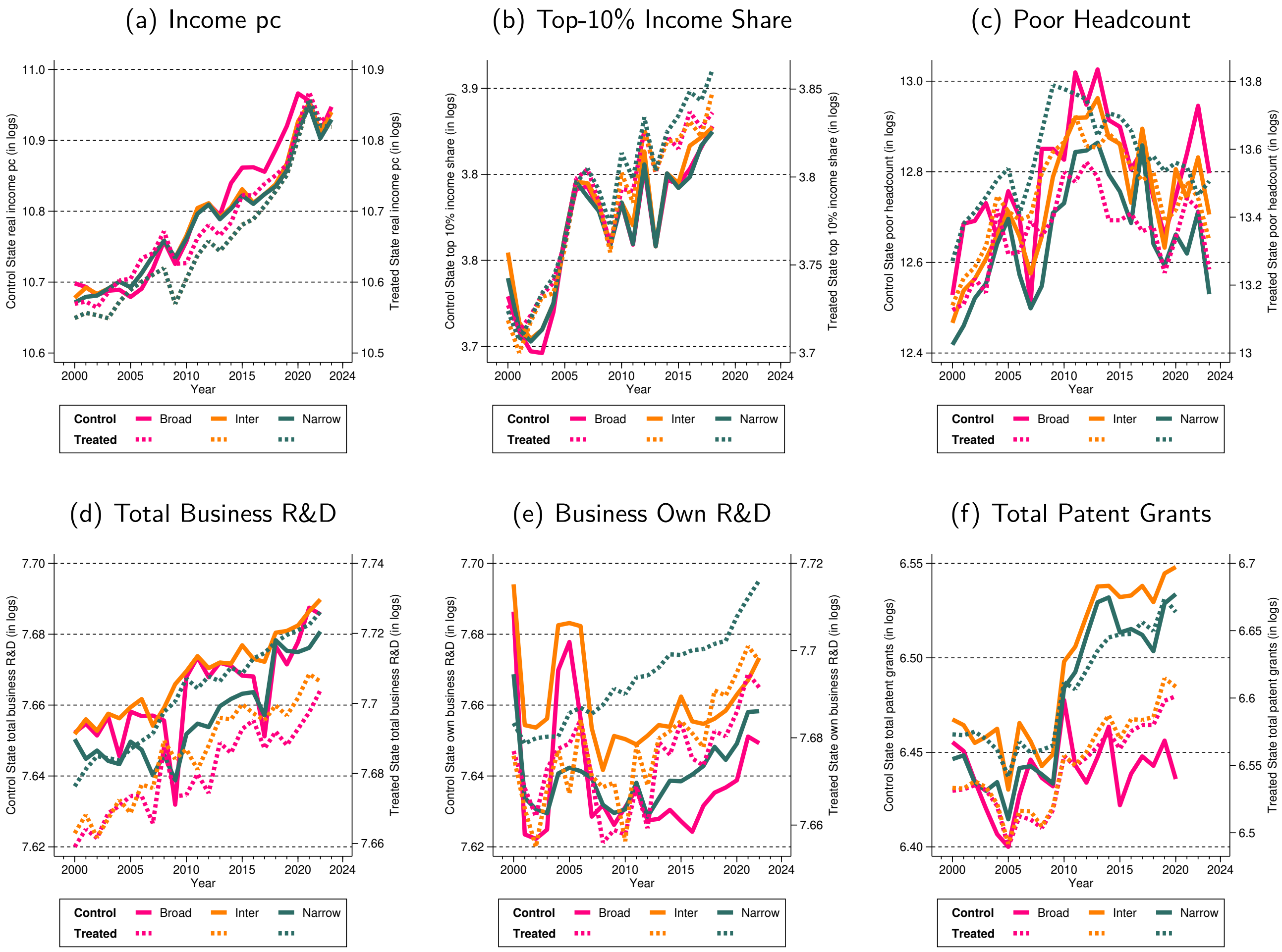
Notes: We present sample statistics (mean, standard deviation) related to state economic outcomes. Per GDP and Median (MD) household (HH) income are both in real values, poverty rate is the share of the state population living in poverty (in %), and the remaining columns report the annual growth rates of employment and population (in %).

Table B-4: Sample Makeup — Other Variables

		Inequality and Poverty Measures								**Business R&D**				**Patent counts**					
		Gini		Top 10% (%)		Top 1% (%)		Uninsured (%)		Total (exp/GDP)		Own (exp/GDP)		Applications '000		Granted '000		Utility only '000	
ID	State	M	SD	M	SD	M	SD	M	SD	M	SD	M	SD	M	SD	M	SD	M	SD
AL	Alabama	0.472	0.009	45.4	2.1	16.7	1.8	12.1	2.2	0.98	0.31	0.49	0.10	0.96	0.09	0.48	0.09	0.41	0.08
AK	Alaska	0.415	0.017	33.9	0.9	11.5	0.9	16.3	3.3	0.20	0.35	0.19	0.37	0.09	0.01	0.05	0.01	0.04	0.01
AZ	Arizona	0.457	0.010	48.1	2.4	18.5	2.2	14.8	3.6	1.77	0.25	1.38	0.16	4.52	0.78	2.26	0.56	2.06	0.52
AR	Arkansas	0.462	0.014	45.1	3.2	17.5	2.4	13.5	4.0	0.35	0.10	0.29	0.05	0.56	0.23	0.26	0.12	0.20	0.11
CA	California	0.477	0.012	50.1	2.6	22.4	2.6	13.9	5.4	4.13	1.10	3.69	1.07	70.87	17.15	32.21	11.37	29.20	10.67
CO	Colorado	0.451	0.011	45.4	2.4	18.6	2.0	12.5	4.1	1.69	0.26	1.48	0.31	5.48	0.81	2.69	0.68	2.40	0.60
CT	Connecticut	0.487	0.013	54.1	2.9	27.9	2.9	7.9	2.2	3.44	0.66	2.82	0.59	4.45	0.64	2.24	0.57	2.04	0.56
DE	Delaware	0.446	0.027	42.3	3.1	15.7	2.1	8.7	2.5	3.21	0.71	2.53	0.49	0.81	0.21	0.38	0.06	0.36	0.05
FL	Florida	0.475	0.011	55.9	4.9	27.1	4.1	16.8	3.7	0.67	0.07	0.47	0.07	8.58	1.54	3.95	1.03	3.25	0.92
GA	Georgia	0.470	0.013	46.4	2.7	17.9	1.7	15.8	2.6	0.83	0.16	0.71	0.10	4.97	0.93	2.24	0.72	1.94	0.63
HI	Hawaii	0.436	0.010	37.2	1.4	13.0	1.4	6.5	2.5	0.28	0.10	0.20	0.09	0.27	0.06	0.12	0.03	0.10	0.03
ID	Idaho	0.433	0.014	41.3	2.5	15.8	2.3	13.8	3.4	2.21	0.60	1.93	0.57	2.13	0.73	1.27	0.37	1.22	0.38
IL	Illinois	0.471	0.011	48.0	2.2	20.7	2.0	10.9	3.3	1.71	0.18	1.60	0.21	9.50	1.37	4.76	1.02	4.02	0.95
IN	Indiana	0.439	0.015	40.6	1.5	14.6	1.1	11.3	2.8	1.94	0.22	1.72	0.18	3.54	0.56	1.86	0.47	1.62	0.44
IA	Iowa	0.431	0.013	37.8	1.6	13.1	1.1	7.3	2.2	1.24	0.35	1.02	0.20	1.65	0.27	0.87	0.20	0.81	0.20
KS	Kansas	0.447	0.012	41.1	1.2	15.9	1.4	10.7	1.6	1.46	0.22	0.97	0.09	1.44	0.28	0.73	0.24	0.64	0.24
KY	Kentucky	0.468	0.011	43.5	2.0	15.3	1.2	10.6	4.1	0.58	0.09	0.47	0.08	1.22	0.20	0.60	0.15	0.54	0.13
LA	Louisiana	0.484	0.011	43.8	3.1	16.8	2.6	14.5	4.9	0.18	0.04	0.15	0.04	0.84	0.08	0.43	0.09	0.38	0.08
ME	Maine	0.443	0.013	41.2	1.8	14.1	1.1	9.3	1.7	0.60	0.13	0.53	0.13	0.38	0.05	0.19	0.04	0.17	0.04
MD	Maryland	0.446	0.010	41.7	1.3	15.7	1.4	9.6	3.1	1.51	0.21	0.96	0.17	3.87	0.39	1.76	0.34	1.61	0.33
MA	Massachusetts	0.475	0.013	51.7	3.2	23.6	3.0	5.4	3.2	4.48	0.90	3.66	0.81	12.95	3.02	5.62	1.88	5.24	1.78
MI	Michigan	0.453	0.014	45.8	3.4	16.9	2.0	8.8	2.9	4.00	0.46	3.73	0.58	8.91	1.55	5.13	1.49	4.57	1.31
MN	Minnesota	0.439	0.015	43.4	1.6	17.1	1.2	6.9	2.0	2.22	0.24	2.08	0.23	7.45	1.03	3.85	0.89	3.48	0.81
MS	Mississippi	0.474	0.012	43.5	2.1	15.0	1.3	15.1	2.9	0.31	0.25	0.21	0.05	0.33	0.03	0.18	0.03	0.15	0.02
MO	Missouri	0.453	0.014	43.8	2.1	17.0	1.5	11.1	1.8	1.69	0.66	1.17	0.16	2.38	0.41	1.13	0.29	0.98	0.28
MT	Montana	0.447	0.015	42.5	1.4	15.8	1.3	13.9	4.4	0.35	0.10	0.32	0.09	0.32	0.10	0.14	0.03	0.12	0.02
NE	Nebraska	0.438	0.012	37.9	1.6	14.9	1.6	9.7	1.8	0.58	0.10	0.55	0.09	0.60	0.10	0.30	0.07	0.25	0.07
NV	Nevada	0.448	0.014	52.8	3.9	27.0	3.8	16.3	4.3	0.46	0.11	0.40	0.08	1.68	0.34	0.66	0.24	0.56	0.22
NH	New Hampshire	0.428	0.014	42.4	2.2	16.7	1.9	8.4	2.4	2.74	0.58	1.25	0.17	1.57	0.21	0.81	0.19	0.73	0.18
NJ	New Jersey	0.468	0.013	49.0	1.3	20.3	1.4	11.2	3.0	3.13	0.45	2.77	0.47	8.94	1.06	4.29	0.79	3.82	0.75
NM	New Mexico	0.469	0.012	43.4	3.2	14.6	1.9	16.3	5.7	0.82	0.40	0.45	0.27	0.86	0.11	0.42	0.09	0.40	0.08
NY	New York	0.503	0.011	57.3	3.0	29.9	3.0	10.1	4.0	1.09	0.18	0.94	0.20	15.75	2.27	8.05	1.76	7.11	1.62
NC	North Carolina	0.466	0.011	44.2	2.7	16.1	1.5	13.7	2.8	1.67	0.36	1.35	0.16	6.32	1.44	2.90	0.82	2.61	0.80
ND	North Dakota	0.446	0.014	38.7	2.7	14.7	2.6	9.1	1.9	0.68	0.38	0.68	0.39	0.21	0.04	0.10	0.02	0.09	0.02
OH	Ohio	0.454	0.014	42.0	1.8	15.4	1.2			1.51	0.11	1.18	0.18	7.95	0.99	3.98	0.70	3.28	0.68
OK	Oklahoma	0.460	0.010	42.0	1.6	16.8	2.0	16.6	2.7	0.42	0.11	0.39	0.11	1.09	0.11	0.57	0.07	0.51	0.06
OR	Oregon	0.452	0.010	44.9	2.2	16.1	1.5	12.0	4.6	2.85	0.78	2.77	0.76	5.03	1.01	2.57	0.82	2.11	0.59
PA	Pennsylvania	0.462	0.012	44.9	1.6	17.6	1.4	8.3	2.3	1.78	0.23	1.63	0.22	7.92	0.86	3.83	0.67	3.42	0.63
RI	Rhode Island	0.461	0.014	44.6	1.4	16.8	1.3			1.70	0.93	0.99	0.17	0.73	0.09	0.38	0.06	0.31	0.05
SC	South Carolina	0.465	0.009	45.3	2.6	16.1	1.7	13.3	2.9	0.76	0.13	0.67	0.14	1.72	0.37	0.85	0.28	0.73	0.27
SD	South Dakota	0.440	0.012	39.7	2.3	16.5	1.8	10.7	1.4	0.32	0.06	0.28	0.06	0.24	0.06	0.11	0.04	0.10	0.03
TN	Tennessee	0.469	0.012	45.2	2.2	18.0	1.8	11.8	2.0	0.58	0.12	0.48	0.11	2.23	0.32	1.04	0.21	0.88	0.20
TX	Texas	0.473	0.008	47.7	2.7	20.8	2.5	21.1	3.2	1.23	0.12	1.08	0.14	17.72	3.70	8.57	2.52	8.00	2.31
UT	Utah	0.416	0.014	42.0	1.8	16.9	2.0	12.4	2.8	1.67	0.27	1.36	0.24	2.76	0.73	1.20	0.48	1.04	0.40
VT	Vermont	0.436	0.014	41.3	1.8	14.7	1.6	7.1	2.9	1.39	0.41	1.23	0.38	0.73	0.19	0.48	0.08	0.45	0.08
VA	Virginia	0.461	0.012	42.6	1.6	15.6	1.2	10.8	2.6	1.16	0.20	0.72	0.15	4.03	1.12	1.76	0.62	1.61	0.61
WA	Washington	0.448	0.014	45.8	2.4	19.0	2.2	10.6	3.6	4.51	1.24	4.37	1.25	12.05	3.71	5.22	2.38	4.72	2.23
WV	West Virginia	0.458	0.013	43.5	1.6	13.8	1.0	11.3	4.3	0.42	0.11	0.36	0.10	0.26	0.05	0.13	0.02	0.12	0.02
WI	Wisconsin	0.432	0.015	41.4	1.7	16.0	1.2	7.6	1.9	1.54	0.21	1.39	0.14	4.33	0.41	2.31	0.37	1.87	0.32
WY	Wyoming	0.429	0.015	49.6	10.2	28.5	10.5	13.6	1.9	0.34	0.65	0.32	0.65	0.20	0.06	0.09	0.03	0.08	0.03
	Average	0.455	0.022	44.4	5.3	17.7	4.8	11.7	4.5	1.51	1.24	1.26	1.12	5.27	10.61	2.52	5.02	2.25	4.57

Notes: We present sample statistics (means, standard deviations) related to state economic outcomes. Inequality Measures — Gini coefficient, income share of the top-10% (top-1%) earners (in percent), share of population without insurance coverage (in percent). Business R&D expenditure expressed as a share of state GDP (in percent) — 'Total' includes funding from the federal government, businesses and organisations from outside the state (or abroad), among other sources; 'Own' covers only the expenditure made by the firm from its own sources. Patent counts (in thousands) — patent applications (all types); patents granted (dto); utility patents granted. All statistics refer to the annual averages by state. Data for the share of population without insurance are missing for Ohio and Rhode Island.

Figure B-2: Average Evolution of Main Outcomes by Treated vs Control Sample



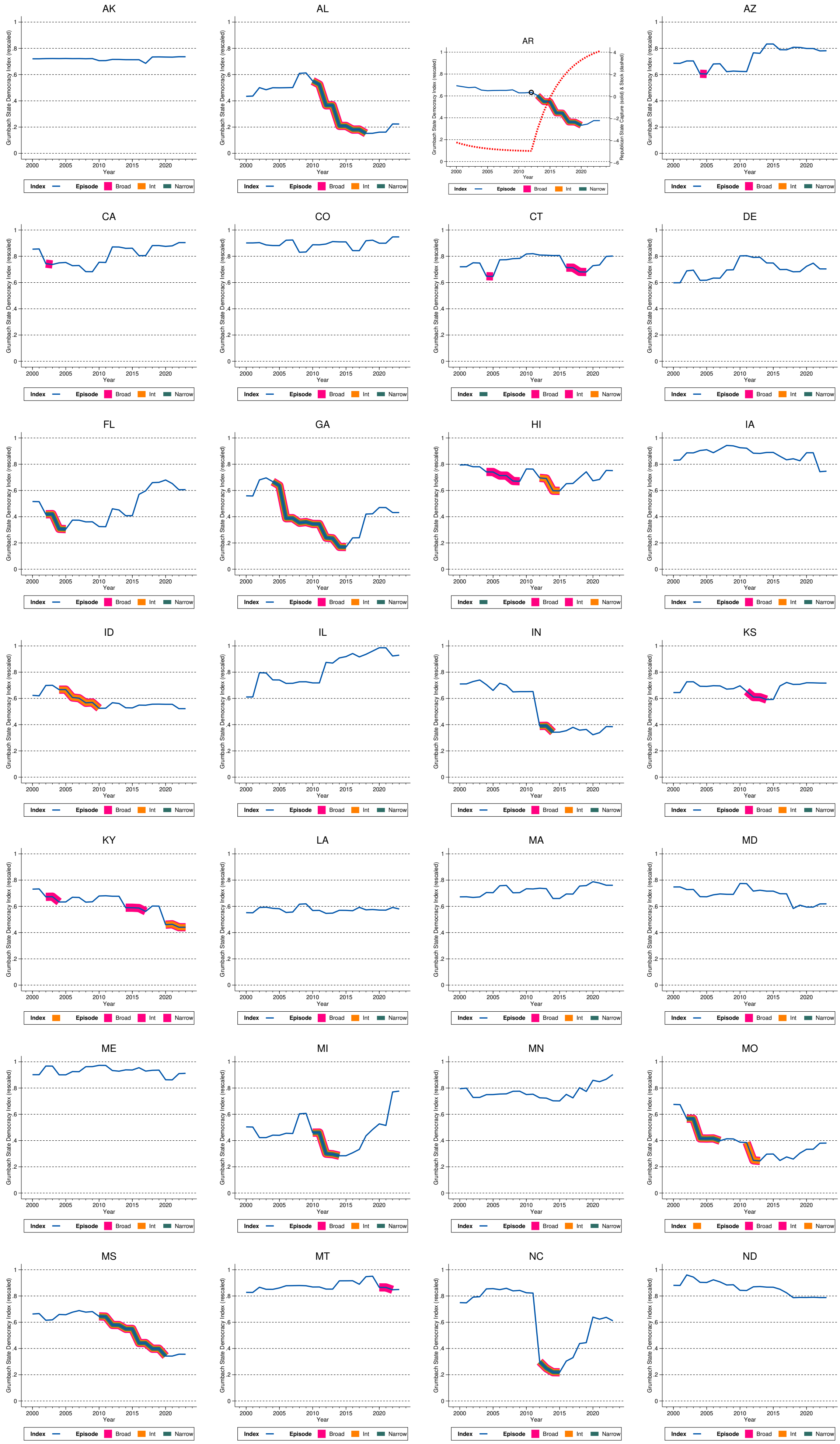
Notes: We plot the median evolution of our primary economic outcomes of interest for treated (dashed lines, right-hand scales) and control (solid lines, left-hand scales) states over the sample period.

Table B-5: Sample/Treatment and the de Chaisemartin and D'Haultfœuille (2024) estimator

	Episode Definition		
	Broad	Inter	Narrow
Treated States	33	22	18
(i) Distribution of Switchers			
2000	0	0	0
2001	0	0	0
2002	9	5	4
2003	1	0	0
2004	9	3	2
2005	2	0	0
2006	3	1	1
2007	1	0	0
2008	7	6	5
2009	0	0	0
2010	10	8	7
2011	4	2	1
2012	4	4	3
2013	1	1	1
2014	5	2	1
2015	9	7	7
2016	4	3	2
2017	4	3	2
2018	2	0	0
2019	2	1	1
2020	4	2	1
2021	2	2	2
2022	3	1	0
2023	3	1	0
(ii) Availability of 4+ pre-treatment obs			
States w/ fewer obs	9	5	4
(iii) Dynamics of Switching			
Year 25% treated	2002	2002	2002/4
Year 50% treated	2004	2008/10	2008/10
Year 75% treated	2011	2011/12	2011
(iv) Pathway Heterogeneity			
2+ Switches	65%	41%	35%

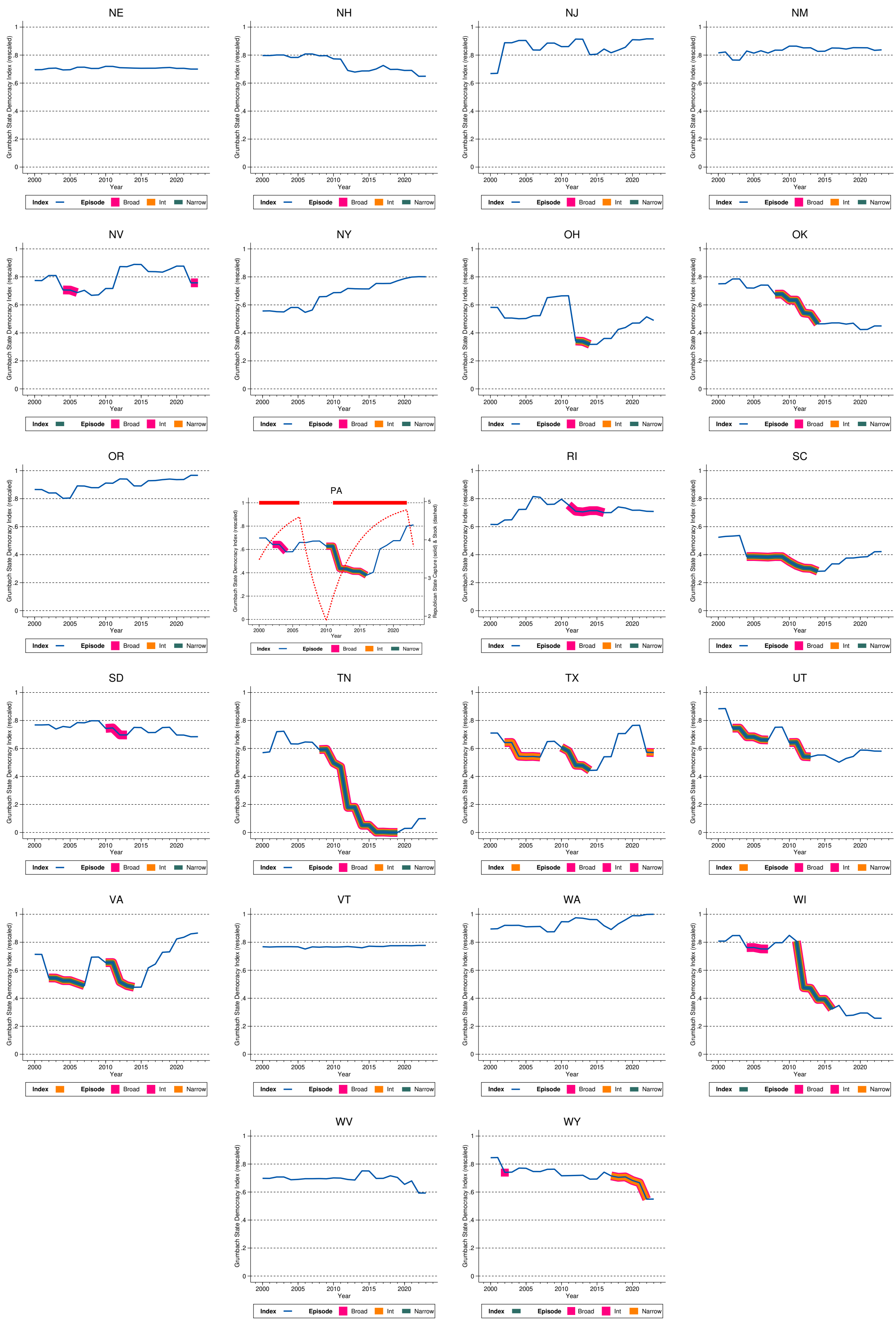
Notes: We present some basic statistics for our sample/treatment characteristics informing the potential applicability of de Chaisemartin and D'Haultfœuille's (2024) `did_multiplegt_dyn` estimator. The major concerns are that (i) very few treatment years have at least 5 'switchers' (shaded in green), suggesting a weak identification problem; (ii) a range of treated states have fewer than 4 pre-treatment observations/years; (iii) there are comparatively few treatments in the second half of the sample (constraining lag dynamics — a minor concern here); and (iv) the proliferation of many different pathways due to many states with two or more staggered switches (especially for the broad definition of autocratisation episodes), implying fragmented treatment histories, undermining credible counterfactual comparison.

Figure B-3: Episodes of Democratic Erosion — Illustration by State



(continued overleaf)

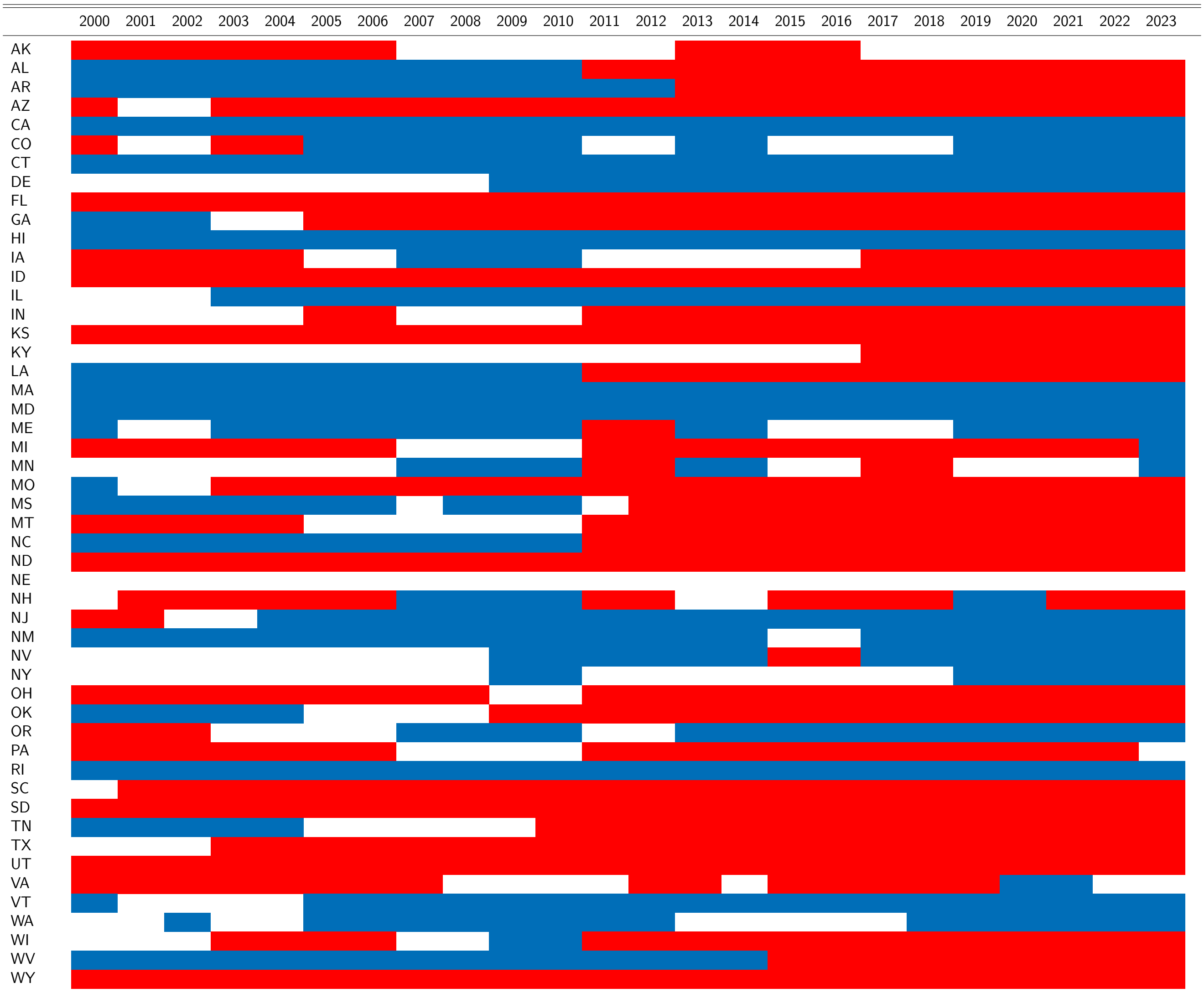
Figure B-3: Episodes of Democratic Erosion — Illustration by State (cont'd)



Notes: These plots present the time series evolution of each (rescaled) Grumbach (2023) democracy index (in blue) by state. They further highlight the autocratisation episodes as determined by our three alternative definitions: pink bands for 'liberal', orange for 'intermediate', and emerald for 'conservative'. For instance, Alaska (AK) experienced no episodes under any definition, Alabama (AL) one lengthy episode in any of the definitions, and Arizona (AZ) one single-year episode in the 'liberal' definition but not in either of the other two.

C Episodes and State Party Partisanship

Table C-1: Partisan states



	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Red	18	18	17	21	21	20	20	14	13	13	14	26	28	27	26	30	30	31	31	29	28	29	29	27
Blue	19	16	17	17	18	19	19	22	23	27	27	15	15	17	17	11	11	13	14	18	19	18	17	19
Mixed	13	16	16	12	11	11	11	14	14	10	9	9	7	6	7	9	9	6	5	3	3	3	4	4
Net Red	-1	2	0	4	3	1	1	-8	-10	-14	-13	11	13	10	9	19	19	18	17	11	9	11	12	8

Notes: We present the dynamics of state partisan ‘capture’ (Republicans or Democrates have majority in both state house <u>and</u> senate vs. mixed). Red is for Republican, Navy for Democrat capture, white for mixed. Note that mixed can mean a powersharing arrangement <u>or</u> that one party has the majority in the state house and the other in the state senate.

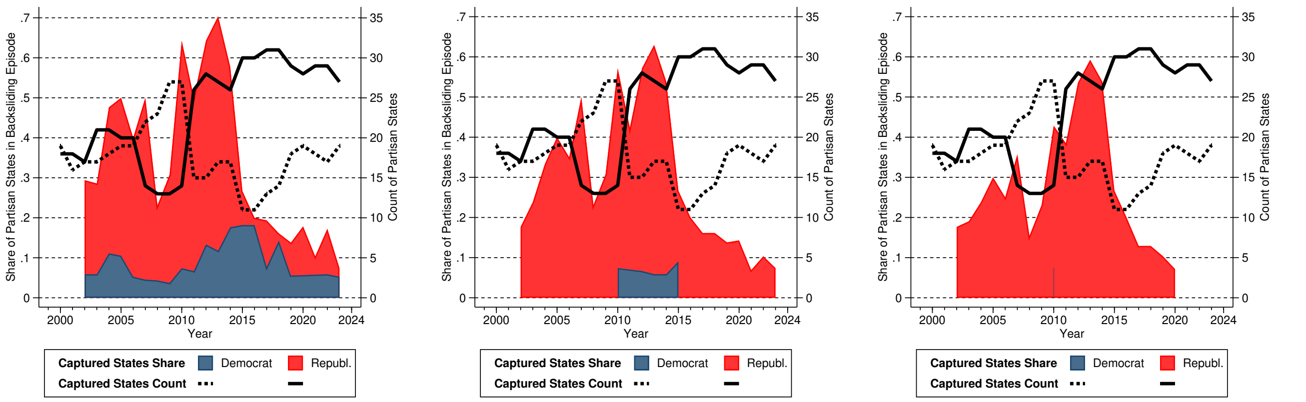
Table C-2: Episodes and Political Parties

	Episode Definition		
Party in Power during Episode	**Broad**	**Intermed.**	**Narrow**
Pure Republican	**24**	**16**	**11**
	52%	59%	55%
One year Democrat then Republican	2	2	2
One year Mixed then Repub.	5	5	4
One Year Dem and Mixed, resp., then Repub.	1	1	1
Two or more years Mixed then Republican	3	2	2
	24%	37%	45%
All Republican	**35**	**26**	**20**
	76%	96%	100%
Pure Mixed	**2**	**0**	**0**
	4%	0%	0%
Pure Democrat	**9**	**1**	**0**
	20%	4%	0%
Total Episodes	46	27	20

Notes: We present the distribution of autocratisation episodes across parties dominating state politics. We assign a state-year as (a) Democrat or Republican if the respective party has majorities in both the state house and senate, and as (b) Mixed if the house and senate are dominated by different parties or there is power sharing. We distinguish a number of typologies for the case of the Republican party, when episodes occur during the party's dominance but start off either in Mixed regimes or under a Democratic majority. Data Source: https://ballotpedia.org, Historical partisan composition of state legislatures (accessed June 10, 2025).

Figure C-1: Captured States and Autocratisation Episodes

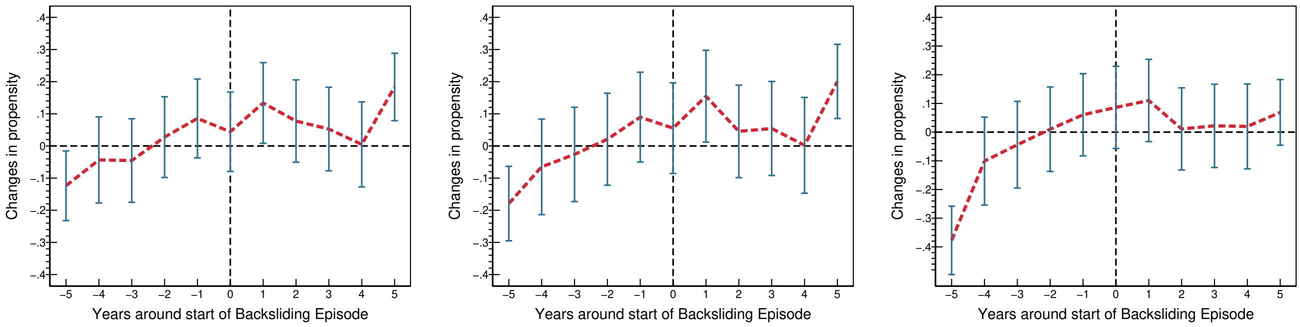
(a) Broad definition (b) Intermediate definition (c) Narrow definition



Notes: The plots chart the number of states captured by the Republicans (solid black line) and the Democrats (dashed black line), using the right scales. The red (blue) areas highlight, using the left scales, the share of states captured by the Republicans (Democrats) in year t which were experiencing an autocratisation episode. The three plots are for the broad, intermediate, and narrow definition, respectively. For instance, in the left-most plot, 50% of the 20 states captured by the Republicans in 2005 experienced an autocratisation episode in that year.

Figure C-2: Republican State Capture and Autocratisation Episodes: event analysis

(a) Broad definition (b) Intermediate definition (c) Narrow definition



Notes: The plots present univariate event analyses for the autocratisation episode and the dummy for Republican state capture. The red line shows the propensity of a state being captured by the Republicans (estimated using robust regression). Blue bars are 90% CI. Regressions only include states that experienced at least one episode.

Table C-3: Autocratisation Episodes and State Party Partisanship

	(1)	(2)	(3)	(4)	(5)	(6)
Episode Definition	Broad		Intermediate		Narrow	
Republican Capture	20.33***	10.79**	20.18***	11.82**	16.70***	9.88**
(47% of obs)	(4.41)	(4.95)	(4.68)	(4.75)	(3.53)	(3.72)
Democrat Capture	-8.86**	-5.03	-8.66**	-5.30	-9.84**	-7.10*
(36% of obs)	(4.04)	(4.18)	(3.85)	(3.93)	(3.77)	(3.81)
State FE	×	×	×	×	×	×
Grumbach Index		×		×		×

Notes: We present the results from simple linear probability models of autocratisation — all coefficients represent marginal effects and are multiplied by 100. Standard errors (in parentheses) are clustered at the state level. In addition to state and year fixed effects, the regression includes two dummies for state-years when Republicans, respectively Democrats, enjoy the majority in both the state house and senate — no majority is the omitted category. The models in even columns include the rescaled Grumbach Index as additional control. Each regression includes 1,200 observations for 50 states.

Table C-4: Autocratisation Episodes and SDI Components

Episode definition	(1) Broad	(2) Inter	(3) Narrow
Panel A: No State FE			
Electoral Performance (21)			
Registration/absentee ballot problems (off-year)	1.7		
Percent of eligible voters who register	-0.7	-2.6	
Registrations rejected		-1.4	-0.7
Website with polling place		0.6	0.6
Registration/absentee ballot problems (on-year)		9.8	2.4
Voting wait times		0.6	2.8
Data completeness		1.6	
Voters deterred: disability/illness (off-year)		-1.7	
Voters deterred: disability or illness (on-year)		0.2	
Online registration		-3.5	
Provisional ballots rejected		-0.6	
Under- and over-votes cast in an election		0.4	
Website for absentee status		-0.2	
Voter Rights (11)			
Voter ID (any)	1.1	0.2	0.4
Voter ID (strict)		2.9	3.0
Absentee voting		-0.1	
Number of ineligible felons (share of pop)		5.5	
Same day registration		1.5	
Gerrymandering (17)			
Mean-median difference (state leg.)	4.0	4.2	2.4
Mean-median difference (state leg.-pres.)	5.2	5.9	6.0
Partisan symmetry (Cong.-Pres.)	6.8	9.4	9.0
Efficiency gap (state leg.-pres.)		-3.3	-0.6
District compactness		3.9	1.7
Declination (state leg.-pres.)		4.2	
Declination (state leg.)		-0.1	
Efficiency gap (Cong.)		1.5	
Partisan symmetry (state leg.-pres.)		-0.9	
Panel B: State FE			
Electoral Performance (21)			
Restrictions on voter registration drives	-1.2	-0.8	
Data completeness	3.5		
Online registration		-0.7	
Voter Rights (11)			
Number of ineligible felons (share of pop)	2.0	14.6	
Voter ID (strict)	1.2		1.2
Gerrymandering (17)			
Mean-median difference (state leg.)	0.4		
Partisan symmetry (Cong.-Pres.)	10.2	12.0	10.0
Partisan symmetry (state leg.)		2.8	1.9
Declination (state leg.-pres.)		0.6	

Notes: We present predictions from elastic net LASSO regressions for 49 indicators related to electoral performance (21), voter rights (11), and gerrymandering (17), used in the construction of the Grumbach (2023) SDI. The dependent variable is a dummy for an episode of the respective definition of autocratisation. In Panel A we absorb year fixed effects, in Panel B we additionally absorb state fixed effects. We present marginal effects (in percent) from the linear probability model of retained covariates for a 1SD increase in the respective variable. Each model includes 950 observations (2000-2018) for 50 states.

D Diagnostic Testing

Table D-1: Alpha and Wald Test Results

			Democratic Erosion: Definition						
			Broad		Intermed.		Narrow		
Dependent Variable	**Specification**	**Statistic**	f=4	f=5	f=4	f=5	f=4	f=5	**Figure**
Panel A: Average incomes									
GDP per capita	No controls	Alpha t ratio	-1.673		-0.633		-0.169		E-2 (c)
	Control: Employment Growth Rate	Alpha t ratio	-1.702		-1.073		-0.604		3 (a)
		Wald p value	0.90	0.91	0.31	0.23	0.76	0.32	
GDP per worker	No controls	Alpha t ratio	-3.423		-2.872		-1.969		E-3 (a)
	Control: Employment Growth Rate	Alpha t ratio	-3.389		-2.962		-2.068		E-3 (a)
		Wald p value	0.01	0.00	0.00	0.00	0.00	0.00	
Panel B: Inequality									
Median HH income	No controls	Alpha t ratio	-1.816		-3.392		-2.881		E-2 (c)
	Control: Employment Growth Rate	Alpha t ratio	-1.893		-3.409		-2.833		E-2 (b)
		Wald p value	0.39	0.82	0.71	0.79	0.86	0.47	
Gini Coefficient	Control: Employment Growth Rate	Alpha t ratio	-1.304		-0.608		0.212		E-3 (b)
		Wald p value	0.52	0.12	0.97	0.49	0.98	0.95	
Top-10% Income Share	No controls	Alpha t ratio	1.409		1.558		0.324		E-4 (a)
	Control: Employment Growth Rate	Alpha t ratio	2.046		2.315		1.329		AOR
		Wald p value	0.00	0.00	0.00	0.00	0.00	0.00	
Top-1% Income Share	No controls	Alpha t ratio	1.354		0.845		-1.497		3 (b)
	Control: Employment Growth Rate	Alpha t ratio	1.764		1.350		-0.705		AOR
		Wald p value	0.00	0.00	0.00	0.00	0.00	0.00	
Panel C: Poverty									
Poverty headcount	No controls	Alpha t ratio	1.074		-0.632		-0.566		E-2 (c)
	Control: Employment Growth Rate	Alpha t ratio	0.986		-0.306		-0.059		3 (c)
		Wald p value	0.09	0.04	0.85	0.84	0.99	0.99	
Poverty Rate	No controls	Alpha t ratio	1.392		-0.161		0.184		AOR
	Control: Employment Growth Rate	Alpha t ratio	0.900		-0.400		0.112		E-3 (c)
		Wald p value	0.27	0.42	0.67	0.62	0.37	0.25	
Uninsured headcount	No controls	Alpha t ratio	-2.938		-3.203		-3.421		AOR
	Control: Employment Growth Rate	Alpha t ratio	-2.905		-3.530		-3.734		E-3 (d)
		Wald p value	0.08	0.02	0.49	0.09	0.25	0.00	
Panel D: Innovation									
Total Business R&D real expend	No controls	Alpha t ratio	-0.068		0.512		-1.497		4 (a)
	Control: Employment Growth Rate	Alpha t ratio	-0.079		-0.337		-1.683		AOR
		Wald p value	0.00	0.00	0.01	0.02	0.02	0.00	
Business own R&D real expend	No controls	Alpha t ratio	-1.246		-0.688		-2.914		4 (b)
	Control: Employment Growth Rate	Alpha t ratio	-0.982		-0.778		-3.524		AOR
		Wald p value	0.00	0.00	0.00	0.00	0.01	0.01	
All Patent Applications	No controls	Alpha t ratio	0.382		0.238		1.708		AOR
	Control: Employment Growth Rate	Alpha t ratio	0.530		0.308		1.916		AOR
		Wald p value	0.00	0.00	0.11	0.03	0.07	0.93	
All Patents Granted	No controls	Alpha t ratio	1.722		0.700		2.560		4 (c)
	Control: Employment Growth Rate	Alpha t ratio	2.034		0.868		3.306		AOR
		Wald p value	0.00	0.00	0.18	0.21	0.04	0.81	
Utility Patent Granted	No controls	Alpha t ratio	1.708		0.681		2.777		4 (d)
	Control: Employment Growth Rate	Alpha t ratio	2.053		0.850		3.585		AOR
		Wald p value	0.00	0.00	0.05	0.06	0.01	0.47	

Notes: We present the diagnostic test results for the PCDID regressions underlying the running line plots we present in this paper. The Alpha test investigates average factor loading equality between treated and control samples (null: model not misspecified) and produces a t-ratio with the standard interpretation. The Wald test asks whether the additional control in the PCDID regression is a 'bad control' (in the sense of Angrist and Pischke, 2008), with the null that the control is fine — we report the p-value from this test. AOR — plots available on request.

Table D-2: Alpha and Wald Test Results (post-treatment effect models)

Dependent Variable	Specification	Statistic	Democratic Erosion: Definition: Broad f=4	Broad f=5	Intermed. f=4	Intermed. f=5	Narrow f=4	Narrow f=5	Figure
State GDP per capita	Control: Employment Growth Rate	Alpha t ratio	-1.418		-1.397		-0.855		?? (a)
		Wald p value	0.90	0.91	0.31	0.23	0.76	0.32	
Top-10% Income Share	No controls	Alpha t ratio	0.510		0.707		-0.940		?? (b)
Poverty headcount	Control: Employment Growth Rate	Alpha t ratio	1.498		0.498		0.620		?? (c)
		Wald p value	0.09	0.04	0.85	0.84	0.99	0.99	
Total Business R&D real exp	No controls	Alpha t ratio	0.282		0.829		-0.784		?? (d)
Business own R&D real exp	No controls	Alpha t ratio	-0.879		-0.373		-2.504		?? (e)
All Patent Granted	No controls	Alpha t ratio	1.756		0.952		3.342		?? (f)
Utility Patent Granted	No controls	Alpha t ratio	1.726		1.044		4.099		?? (g)

Notes: We present the diagnostic test results for the PCDID regressions underlying the running line plots we present in this paper. This table focuses on the robustness analysis, including four post-treatment dummies in the treatment regressions. We use the same setup with regards to employment growth as additional control (or not) as in the main results in Figures 3 and 4 in the main text. For all other details, see notes to Appendix Table D-1.

Table D-3: Alpha Test Results (export flow analysis)

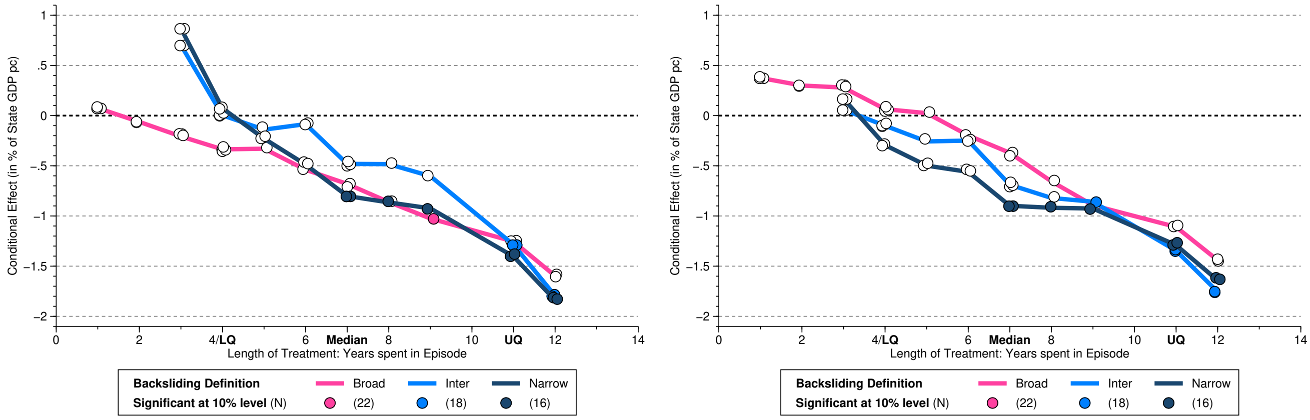
Dep. Var.	Destinations	US States	Destination Regime	Democratic Erosion: Definition: Broad	Intermed.	Narrow	Figure
Exports	All	All	Any	0.635	0.036	0.043	6 (a) left
			Democracies	0.837	0.007	0.030	6 (b) left
			Autocracies	0.481	0.374	0.670	6 (c) left
Exports	All	Single episode	Any	0.556	0.039	0.314	6 (a) right
			Democracies	0.791	0.008	0.049	6 (b) right
			Autocracies	0.215	0.366	0.996	6 (c) right
Exports	Top-100	All	Any	0.243	0.728	0.089	E-5 (a) left
			Democracies	0.062	0.152	0.535	E-5 (b) left
			Autocracies	0.171	0.008	0.014	E-5 (c) left
Exports	Top-100	Single episode	Any	0.371	0.739	0.068	E-5 (a) right
			Democracies	0.060	0.150	0.486	E-5 (b) right
			Autocracies	0.607	0.004	0.005	E-5 (c) right
Exports	Top-50	All	Any	0.877	0.967	0.041	AOR
			Democracies	0.766	0.831	0.285	AOR
			Autocracies	0.299	0.604	0.026	AOR
Exports	Top-50	Single episode	Any	0.246	0.984	0.081	AOR
			Democracies	0.123	0.856	0.374	AOR
			Autocracies	0.435	0.631	0.055	AOR
Exports‡	All	All	Any	0.063	0.293	0.757	E-6 (a) left
			Democracies	0.166	0.732	0.116	E-6 (b) left
			Autocracies	0.163	0.442	0.563	E-6 (c) left
Exports‡	All	Single episode states	Any	0.075	0.258	0.929	E-6 (a) right
			Democracies	0.138	0.919	0.041	E-6 (b) right
			Autocracies	0.446	0.424	0.477	E-6 (c) right

Notes: We present the diagnostic test results for the PCDID regressions underlying the running line plots for state export flows we present in this paper. The Alpha test investigates average factor loading equality between treated and control samples (null: model not misspecified) and, in our implementation, produces a χ^2-statistic with two degrees of freedom — we report the associated p-value for this test. Results are available for all export destinations, the top-100 and the top-50 (based on ranks for 2002-2023); for all U.S. states which experienced one or more episodes and for those which experienced just a single episode; for any destination (country), and for those destinations that are democratic, respectively autocratic, throughout the sample period. AOR — plots available on request. ‡ indicates the diagnostics for specifications where we include year dummies (2020-22) for the Covid period.

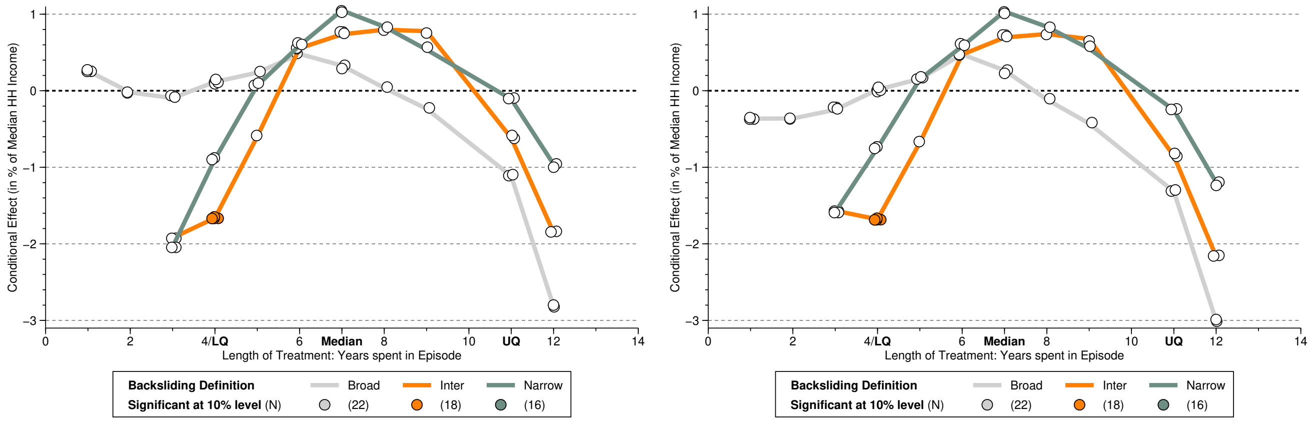
E Additional Results

Figure E-1: Episodes of Democratic Erosion and Economic Outcomes — Single Episode States

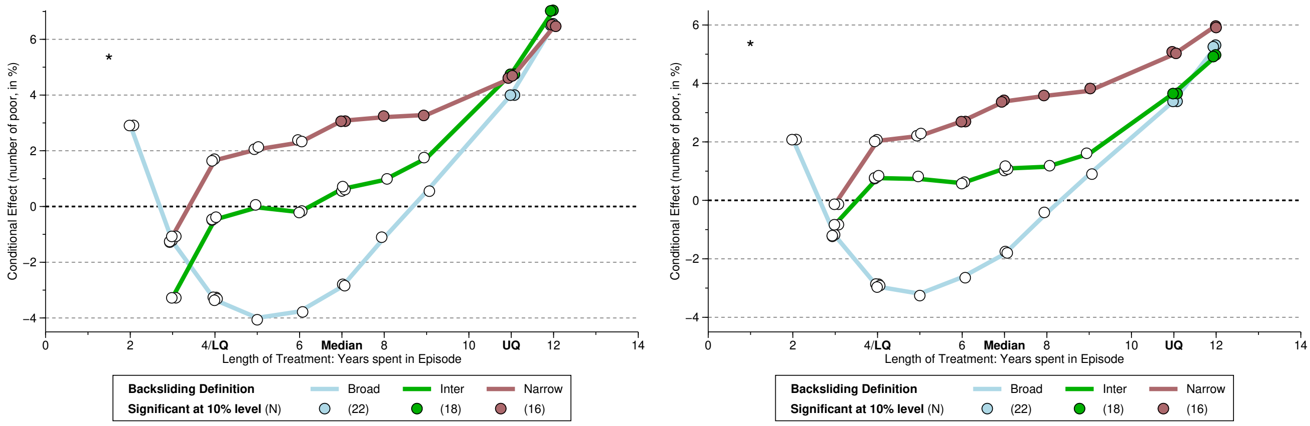
(a) GDP per capita,••• specifications with 4 (left) and 5 factors (right)



(b) Median HH Income,•◦◦ specifications with 4 (left) and 5 factors (right)



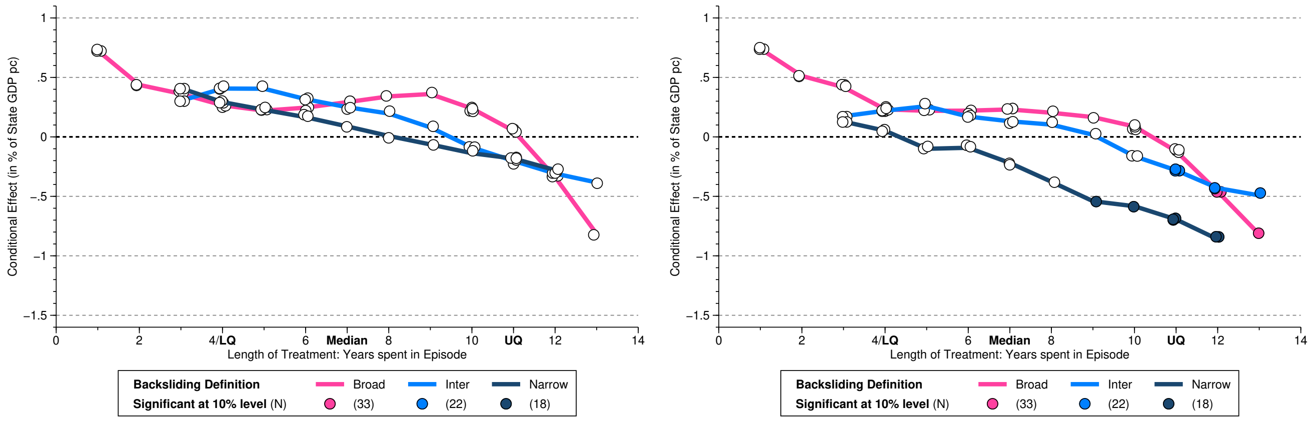
(c) Number of People Living in Poverty,••• specifications with 4 (left) and 5 factors (right)



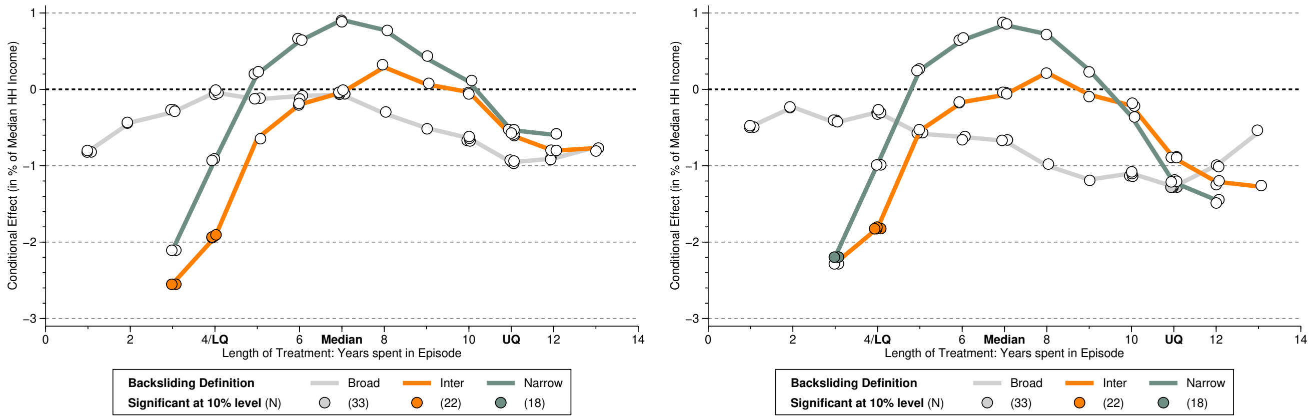
Notes: We present results from the PCDID specifications with the employment growth rate as an additional control variable. Here, we only plot treatment effects if states experienced a single episode of autocratisation. Models using the intermediate and conservative definitions in panel (b) fail the Alpha test. For all other details, see notes to Figure 3 in the main text.

Figure E-2: Episodes of Democratic Erosion and Economic Outcomes — No Controls

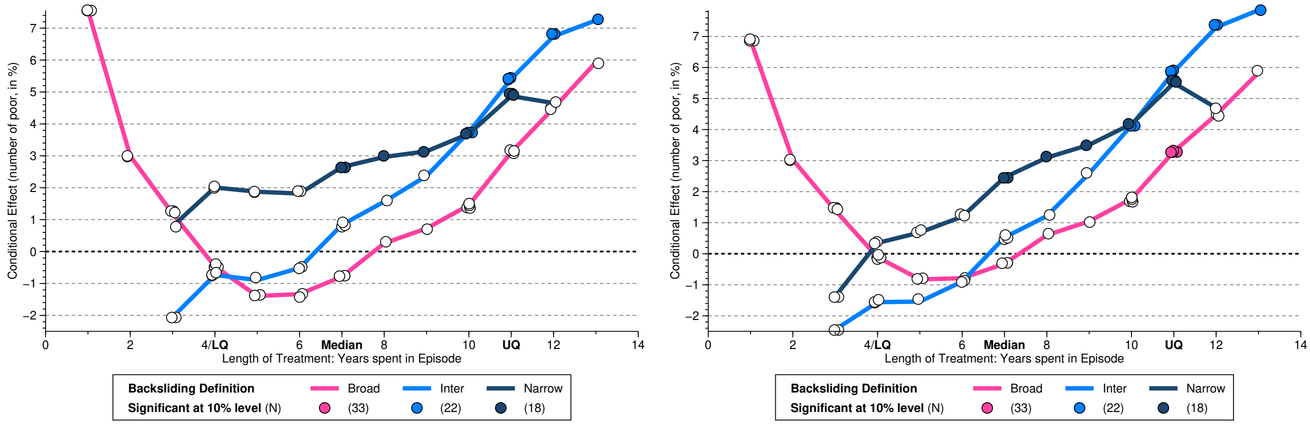
(a) State GDP per capita,••• specifications with 4 (left) and 5 factors (right)



(b) State Median HH Income,•◦◦ specifications with 4 (left) and 5 factors (right)



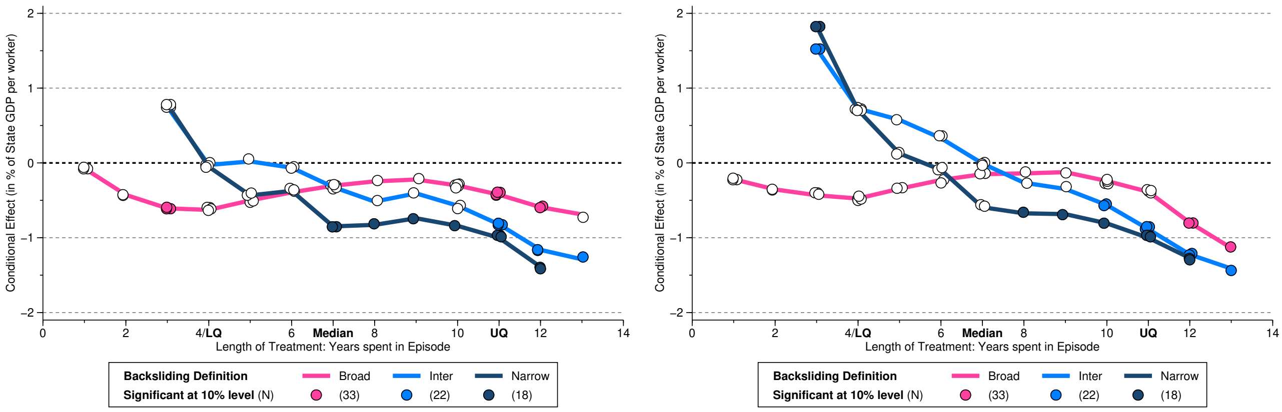
(c) Number of People Living in Poverty,••• specifications with 4 (left) and 5 factors (right)



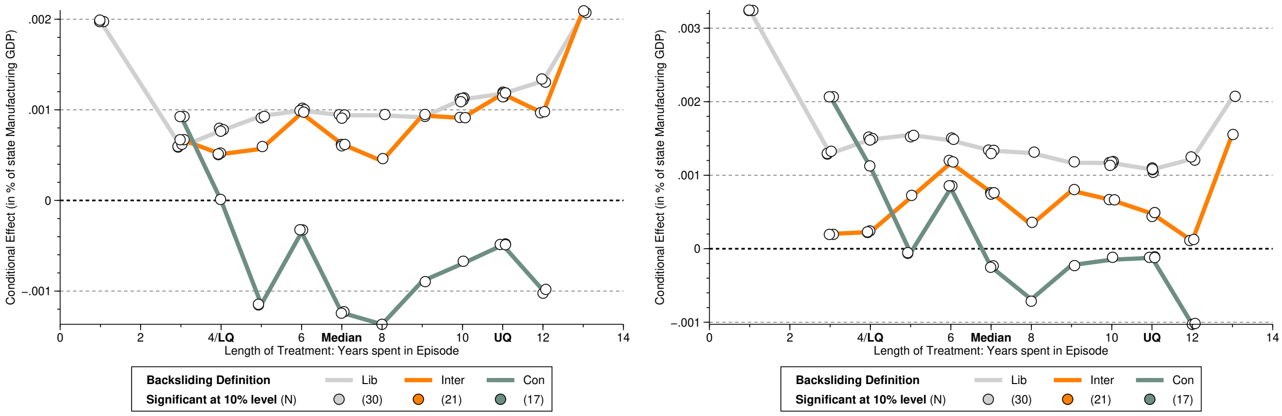
Notes: We present results from the PCDID specifications without any additional control variables. Models using the intermediate and conservative definitions of autocratisation in panel (b) fail the Alpha test. For all other details, see notes to Figure 3 in the main text.

Figure E-3: Episodes of Democratic Erosion and Economic Outcomes — Alternative Outcomes

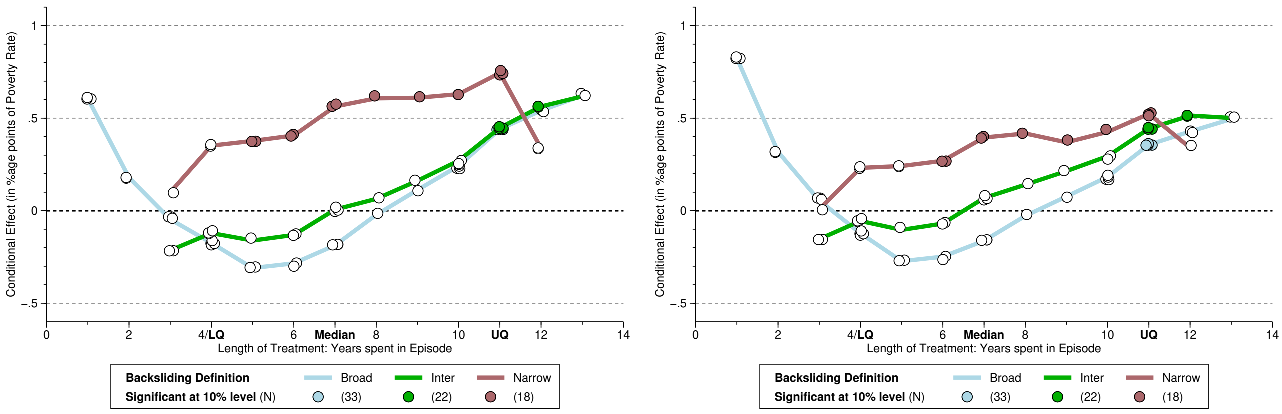
(a) State GDP per worker,°°° specifications with 4 (left) and 5 factors (right)



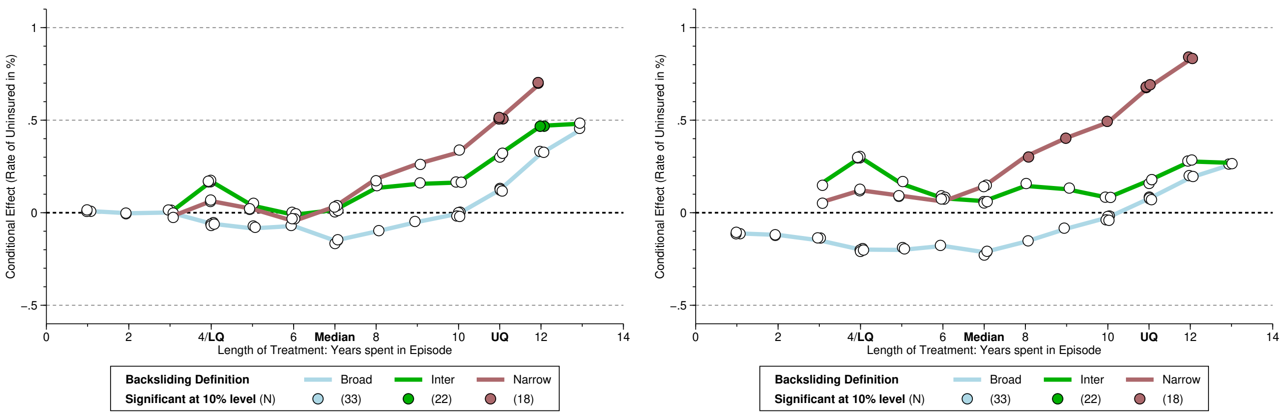
(b) State Gini coefficient,•°° specifications with 4 (left) and 5 factors (right)



(c) State Poverty Rate,••• specifications with 4 (left) and 5 factors (right)



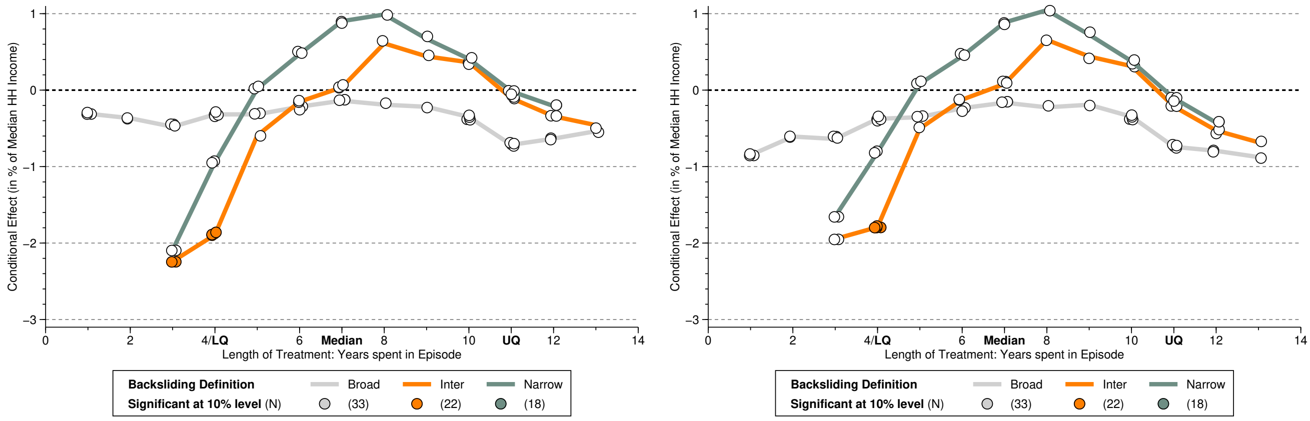
(d) Number of Uninsured Individuals,°°° specifications with 4 (left) and 5 factors (right)



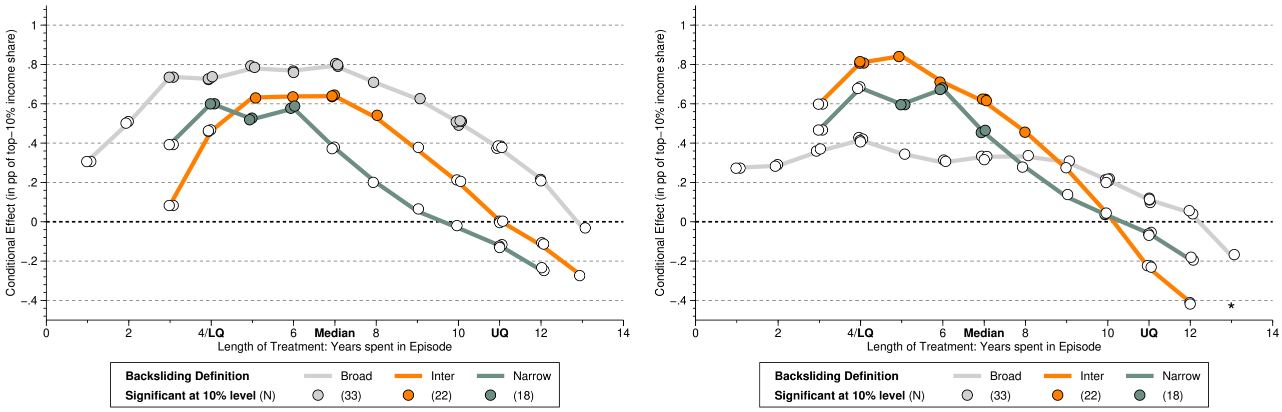
Notes: We present results from the PCDID specifications with the employment growth rate as an additional control variable. For all other details, see notes to Figure 3 in the main text.

Figure E-4: Episodes of Democratic Erosion and Economic Outcomes — More Alternative Outcomes

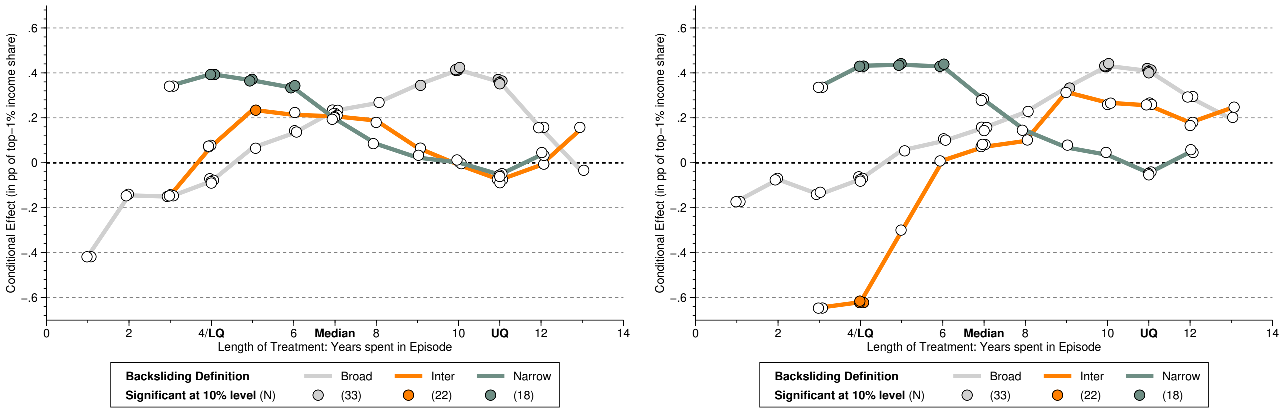
(a) Median HH Income,•○○ specifications with 4 (left) and 5 factors (right)



(b) Top 10% Income Share,••• specifications with 4 (left) and 5 factors (right)



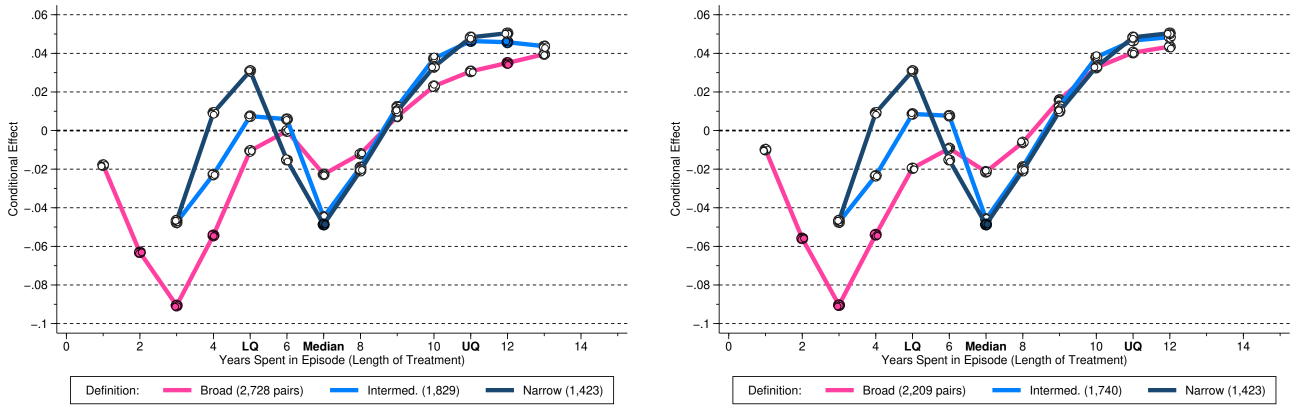
(c) Top 1% Income Share,••• specifications with 4 (left) and 5 factors (right)



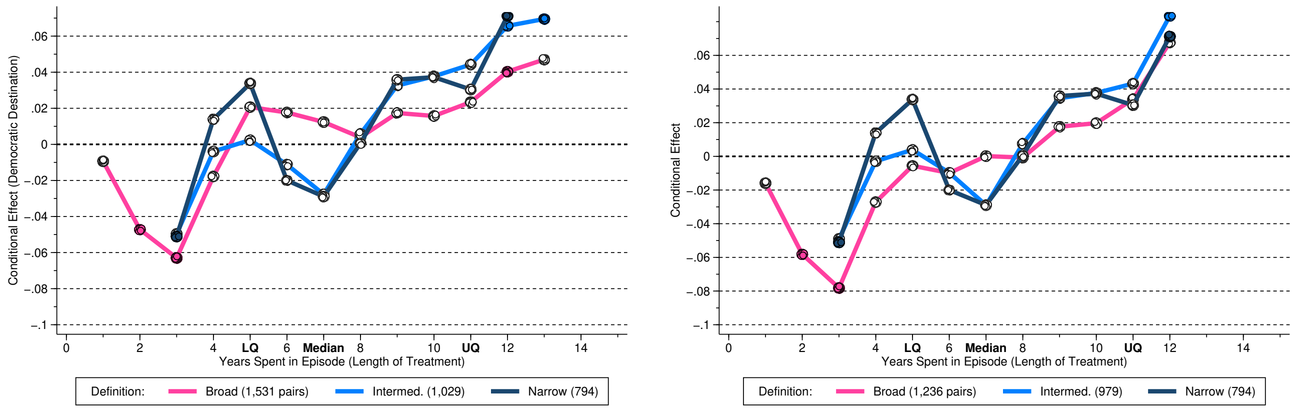
Notes: We present results from the PCDID specifications without any additional control variable — specifications with employment growth as control always fail the Wald test for 'bad controls' and some further fail the Alpha test. The sample for these results is for 2000-2018 only. For all other details, see notes to Figure 3 in the main text.

Figure E-5: Episodes of Democratic Erosion and Exports (Top-100 Destinations)

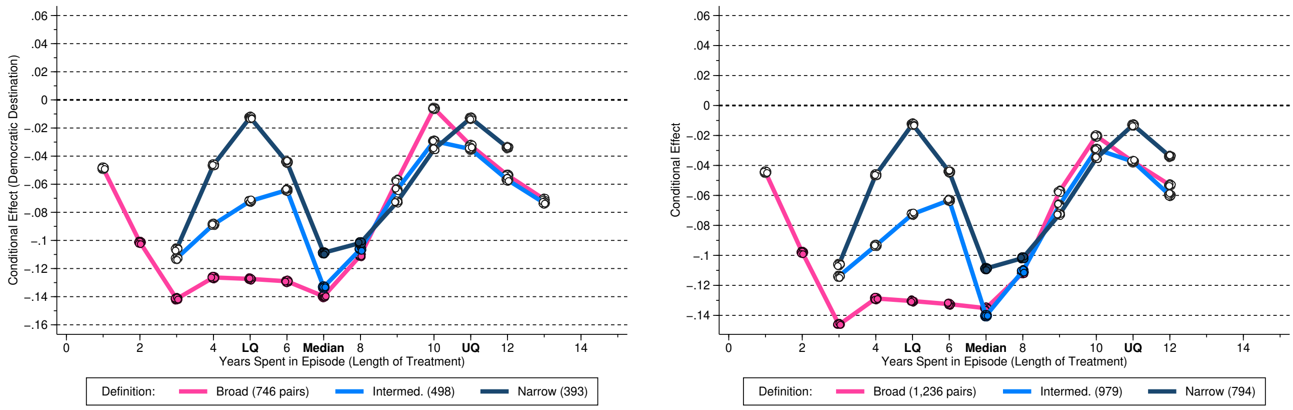
(a) Any Destination, all states●●○ (left) and states with single episode●●○ (right)



(b) Democratic Destinations, all states○●● (left) and states with single episode○●● (right)



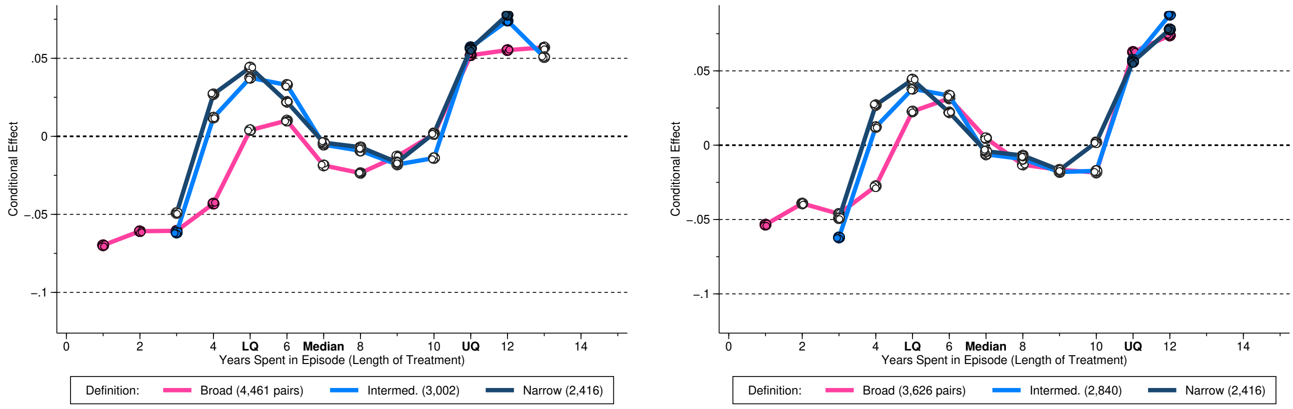
(c) Autocratic Destinations, all states●○○ (left) and states with single episode●○○ (right)



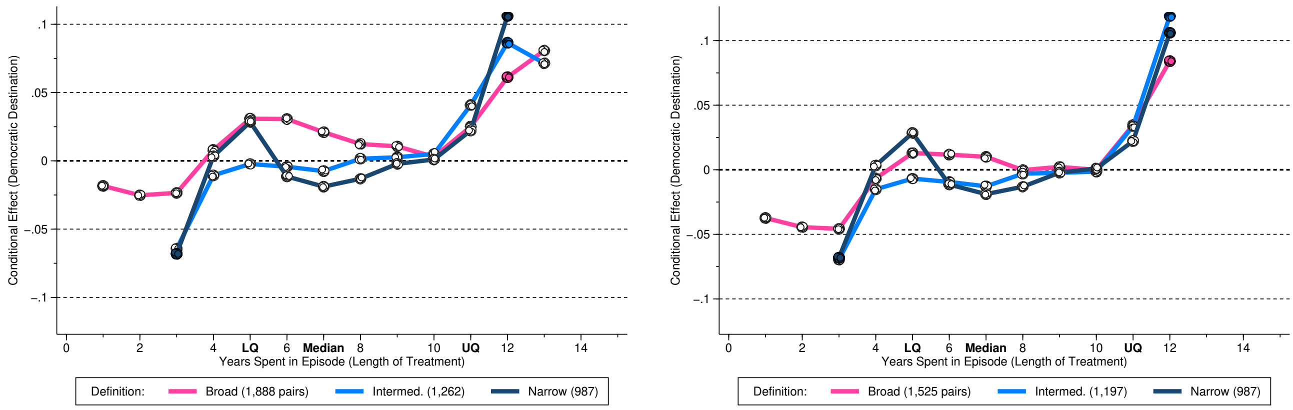
Notes: We present treated-state predictions from multivariate running line regressions (Royston and Cox, 2005) illustrating the economic effects of autocratisation episodes for state exports to the top-100 U.S. export destinations. The (total) number of years a state has spent in one or more episodes is shown on the x-axis of each plot, and the economic effect on the y-axis. The running line regressions control for the number of times a state experienced an episode (dummies for two or three) and use weights derived from the robust ATET estimate using an M-estimator (Rousseeuw and Leroy, 2005). An estimate of -1 indicates a 1% reduction in export values. Each plot presents results for our three definitions of autocratisation: a broad one (33 treated states), an intermediate one (22), and a narrow one (18) — see main text for definitions. A filled (white) marker indicates statistical (in)significance at the 10% level. We use ● and ○ in the subfigure title to indicate whether a specification passes or fails the Alpha diagnostic test at the 10% level: ●●○ indicates that the liberal and intermediate definitions pass the test, whereas the conservative definition does not.

Figure E-6: Episodes of Democratic Erosion and Exports (including Covid dummies)

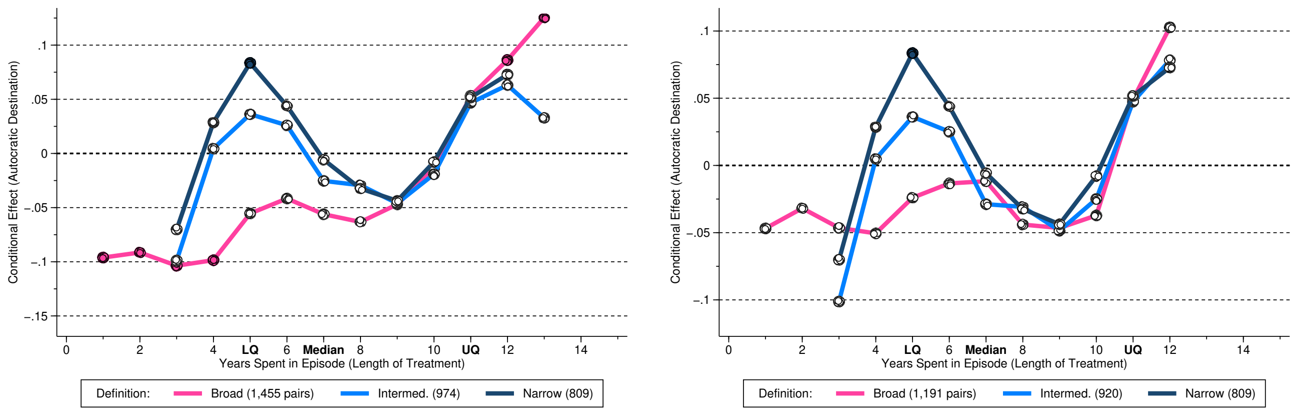
(a) Any Destination, all states○●● (left) and states with single episode○●● (right)



(b) Democratic Destinations, all states●●● (left) and states with single episode●●○ (right)



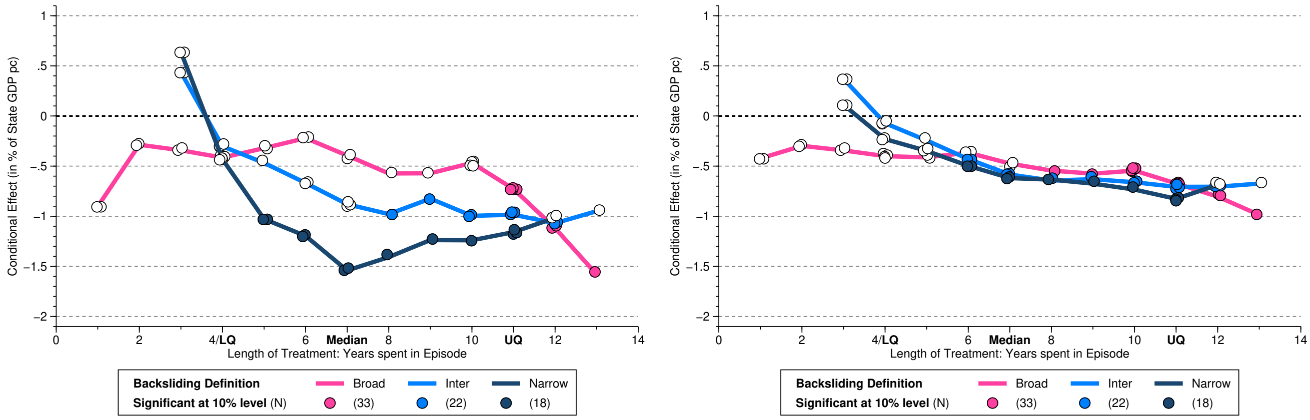
(c) Autocratic Destinations, all states●●● (left) and states with single episode●●● (right)



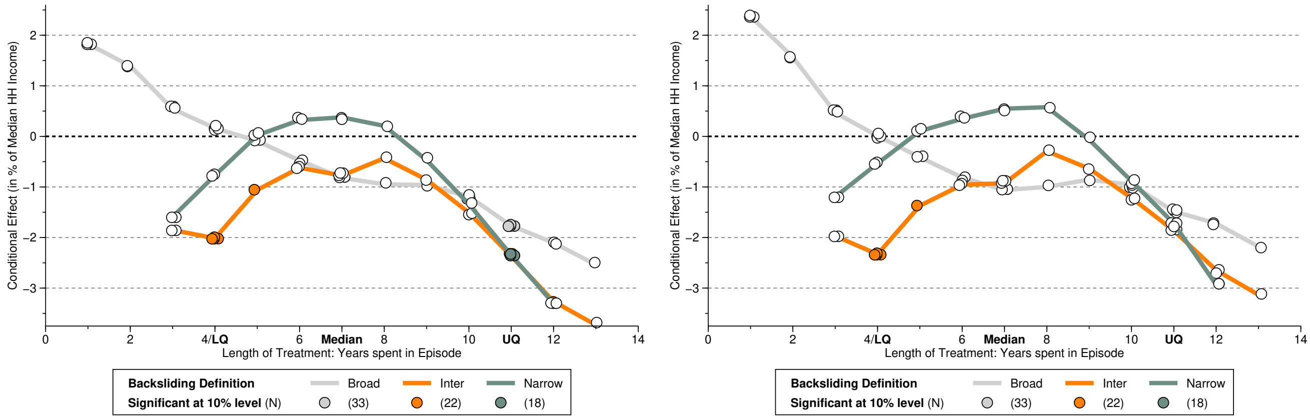
Notes: We present treated-state predictions from multivariate running line regressions (Royston and Cox, 2005) illustrating the economic effects of autocratisation episodes for state exports to the top-100 U.S. export destinations. The (total) number of years a state has spent in one or more episodes is shown on the x-axis of each plot, and the economic effect on the y-axis. The running line regressions control for the number of times a state experienced an episode (dummies for two or three) and use weights derived from the robust ATET estimate using an M-estimator (Rousseeuw and Leroy, 2005). An estimate of -1 indicates a 1% reduction in export values. Each plot presents results for our three definitions of autocratisation: a broad one (33 treated states), an intermediate one (22), and a narrow one (18) — see main text for definitions. A filled (white) marker indicates statistical (in)significance at the 10% level. We use ● and ○ in the subfigure title to indicate whether a specification passes or fails the Alpha diagnostic test at the 10% level: ●●○ indicates that the liberal and intermediate definitions pass the test, whereas the conservative definition does not.

Figure E-7: Episodes of Democratic Erosion — Main results (with Covid dummies)

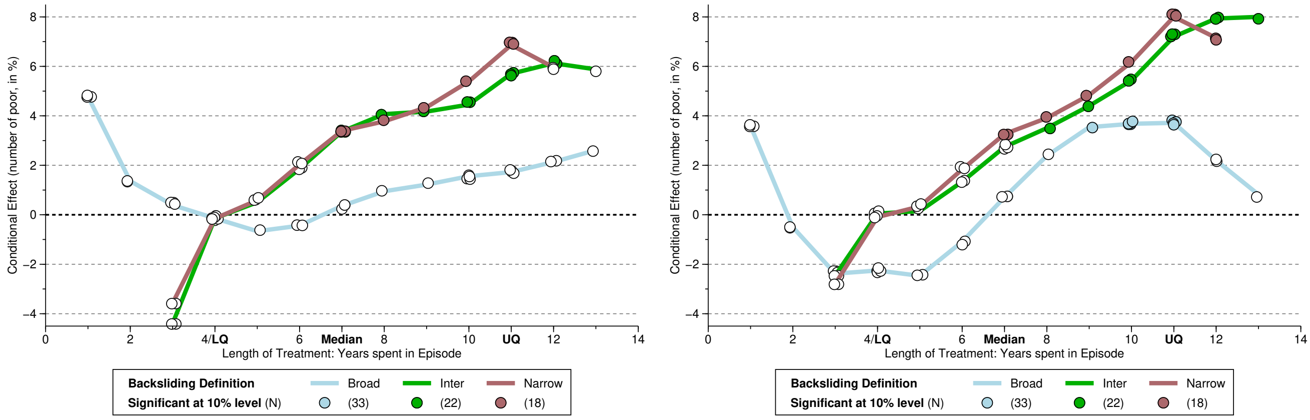
(a) GDP per capita,••• specifications with 4 (left) and 5 factors (right)



(b) Top 10% Income Share,°°° specifications with 4 (left) and 5 factors (right)



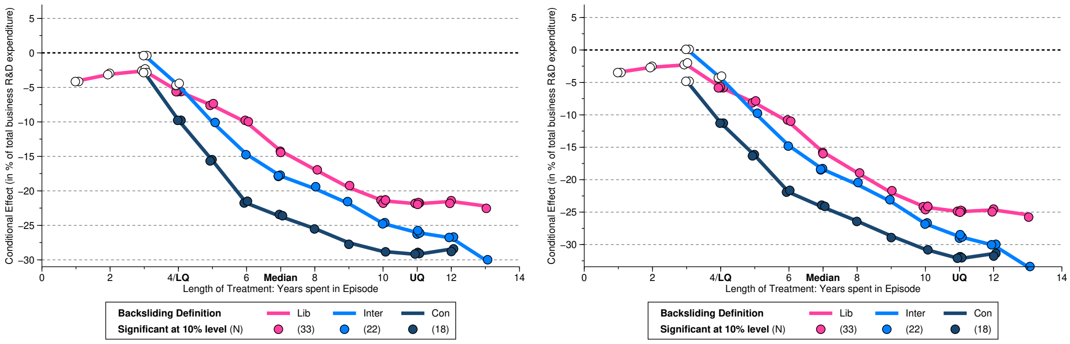
(c) Poverty headcount,••• specifications with 4 (left) and 5 factors (right)



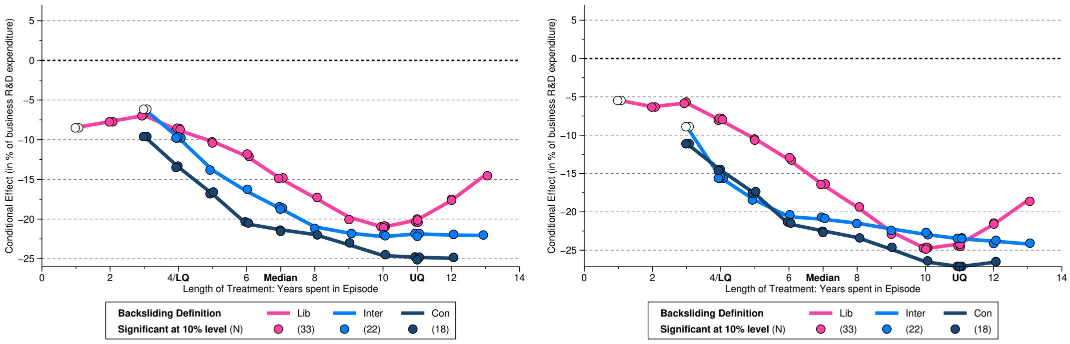
Notes: We present treated-state predictions from multivariate running line regressions (Royston and Cox, 2005), illustrating the economic effects of democratic erosion. The underlying PCDID specifications have the employment growth rate as additional control and augment the treatment regression with four or five estimated factors, as indicated (models with three or six estimated factors yield qualitatively very similar results, available on request). The (total) number of years a state has spent in one or more autocratisation episodes is shown on the x-axis of each plot, and the economic effect on the y-axis. The running line regressions control for the number of times a state experienced an episode and use weights derived from the robust ATET estimate using an M-estimator (Rousseeuw and Leroy, 2005). A filled (white) marker indicates statistical (in)significance at the 10% level. An estimate of -1 (+1) indicates a 1% reduction (increase) in income, median household income, and the number of poor people, in panels (a) to (c), respectively. Each plot presents results for our three definitions of autocratisation: a broad one (33 treated states), an intermediate one (22), and a narrow one (18) — see text for definitions. We use ••• to signal which specifications using these three different definitions (Broad, Inter, Narrow) pass (•) or fail (°) the Alpha test. For other details, see notes to Figure 3 in the main text.

Figure E-8: Episodes of Democratic Erosion — Innovation (with Covid dummies)

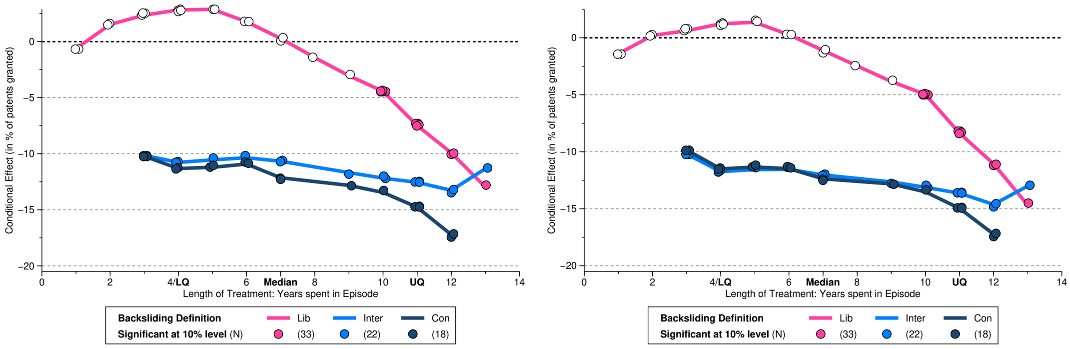
(a) Total Business R&D Expenditure,$^{\bullet\bullet\bullet}$ specifications with 4 (left) and 5 factors (right)



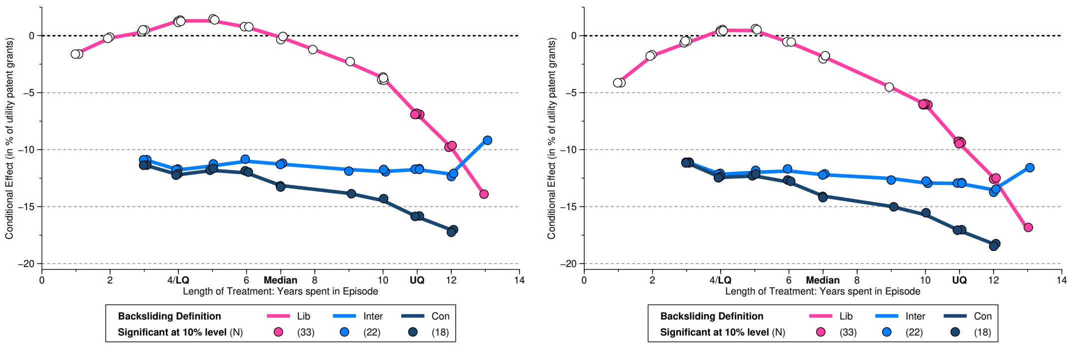
(b) R&D Expenditure by Businesses,$^{\bullet\bullet\bullet}$ specifications with 4 (left) and 5 factors (right)



(c) Total Patents Granted,$^{\bullet\bullet\circ}$ specifications with 4 (left) and 5 factors (right)



(d) Utility Patents Granted,$^{\bullet\bullet\circ}$ specifications with 4 (left) and 5 factors (right)



Notes: We present results from the PCDID specifications without additional control variable but including up to three Covid dummies (for 2020, 2021, and 2022, if applicable). The R&D (patent) data are for 2000-2022 (2000-2020). 'Total business R&D' includes expenditure by businesses sourced from the federal government (e.g. grants) and from 'others' (i.e. other firms located inside or outside the US, state or foreign government agencies, and other organizations inside or outside the US). 'Business R&D' refers to the expenditure paid for by the firms themselves. 'Total Patents' includes utility, plant and design patents. We use $^{\bullet\bullet\circ}$ to indicate whether specifications using the liberal, intermediate, and conservative definitions of autocratisation pass ($^\bullet$) or fail ($^\circ$) the Alpha test at the 10% significance level. For other details, see notes to Figure 3 in the main text.

Table E-1: Visualisation — Average Treatment Effects — bilateral exports

	Benchmark			With Covid dummies		
	(1)	(2)	(3)	(4)	(5)	(6)
(i) Robust Mean ATET						
Episode	Broad	Inter	Narrow	Broad	Inter	Narrow
Exports	-0.011	0.009	0.011	0.017	0.027*	0.023
	[0.011]	[0.014]	[0.014]	[0.012]	[0.015]	[0.015]
Pairs	5,163	3,463	2,929	5,144	3,454	2,925
(ii) Marginal Effect at Median						
	(1)	(2)	(3)	(4)	(5)	(6)
Episode	Broad	Inter	Narrow	Broad	Inter	Narrow
Exports	-0.010	0.009	0.011	0.019	0.022	0.021
	[0.011]	[0.014]	[0.014]	[0.012]	[0.016]	[0.015]
Pairs	5,163	3,463	2,929	5,144	3,454	2,925

Notes: We report various average treatment effects on the treated for the preferred specifications of our CCE-DID estimator for dyadic state-level exports. In Panel (i) we compute the robust mean across all pair estimates, in (ii) we regress the pair-wise treatment estimate on time spent in democracy and compute the marginal effect at the median treatment length (7 years). Standard errors in parentheses, shading indicates positive (insignificant or significant) and negative (only insignificant) average effects. We distinguish results from the benchmark specification and the specification augmented with Covid year dummies.

Table E-2: Controlling for the 'Stock' of Republican State Capture — ATETs

Episode	Broad	Inter	Narrow	Broad	Inter	Narrow
Factors	4	4	4	5	5	5
Panel A: Baseline Model						
(i) Robust Mean ATET						
State GDP pc	0.210	0.670*	0.901**	0.266	0.384	0.557*
Poverty headcount	1.372	2.749	4.571*	2.295	3.565*	4.870**
Top-10% income share †	0.556***	0.614***	0.739***	0.304	0.900***	0.845***
Total Business R&D	-4.076*	-0.849	-1.406	-1.239	0.323	-1.703
Business Own R&D	-3.462	-3.580	-2.736	-3.192	-4.809	-2.364
All granted patents	-2.450**	-2.286*	-6.217***	-2.573**	-3.403**	-3.851*
Granted utility patents	-2.009**	-2.840***	-2.265***	-1.256	-2.599**	-2.270**
(ii) Robust Mean ATET evaluated at the Median						
State GDP pc	0.216	0.740*	1.069***	0.243	0.432	0.650**
Poverty headcount	1.309	1.914	4.377*	2.252	2.103	4.754**
Top-10% income share †	0.645**	0.687**	0.756**	0.407	1.259***	0.921**
Total Business R&D	-4.172*	-0.023	-0.895	-1.601	1.328	-0.794
Business Own R&D	-2.510	-5.854*	-2.959	-2.182	-6.351*	-2.744
All granted patents	-2.346**	-2.325	-6.704***	-2.420**	-3.102*	-4.118**
Granted utility patents	-1.678*	-2.982***	-2.526***	-1.104	-2.764*	-3.557***
(iii) Robust Mean ATET evaluated at the Upper Quartile						
State GDP pc	0.153	0.381	0.105	0.092	0.115	-0.059
Poverty headcount	1.695	4.858**	9.548**	3.742*	5.984**	10.606**
Top-10% income share †	0.569***	0.513***	0.609*	0.328	0.431	0.691**
Total Business R&D	-5.335*	-4.857	-9.221	-4.517	-4.668	-6.970*
Business Own R&D	-5.255**	-0.851	-1.950	-6.004***	-3.797	0.262
All granted patents	-3.415***	-3.252**	-9.141*	-3.483***	-3.695**	-6.513**
Granted utility patents	-2.027*	-2.006	-2.204	-1.752	-2.774**	-2.990
Panel B: Model including Covid dummies						
(i) Robust Mean ATET						
State GDP pc	0.026	0.621**	0.912***	-0.111	0.243	0.575*
Poverty headcount	1.723	4.690***	5.217**	2.188	5.421***	5.139**
Total Business R&D	-1.202	-1.809	-5.322*	-0.514	-1.054	-4.747
Business Own R&D	-2.181	-3.170	-5.295**	-1.754	-1.099	-4.322*
All granted patents	-0.660	-1.923	-2.943	-1.089	-2.125	-3.073
Granted utility patents	-0.755	-2.808**	-1.801*	0.863	-1.801	-2.152
(iii) Robust Mean ATET evaluated at the Median						
State GDP pc	0.147	0.552	1.013***	-0.040	0.181	0.634*
Poverty headcount	1.886	4.021***	5.051***	2.366	4.366**	5.365***
Total Business R&D	-0.381	-2.322	-3.918	-0.730	-1.460	-3.619
Business Own R&D	-2.176	-4.682*	-5.854***	-1.774	-3.657	-4.492**
All granted patents	-0.956	-1.723	-3.487	-0.739	-1.648	-3.496
Granted utility patents	-0.620	-2.527**	-2.102*	0.957	-1.706	-2.833**
(iii) Robust Mean ATET evaluated at the Upper Quartile						
State GDP pc	0.135	0.531*	0.321	-0.015	0.326	0.336
Poverty headcount	3.705	7.715***	11.466***	4.516**	7.888***	11.426***
Total Business R&D	-3.221	-5.771**	-10.297***	-2.860	-4.570	-9.506***
Business Own R&D	-3.844**	-1.581	-3.490	-3.955*	-1.436	-3.149
All granted patents	-2.318	-2.432	-4.661	-1.893	-1.236	-4.835
Granted utility patents	-1.117	-1.453	-1.739	-0.081	-1.674	-1.720

Notes: We report average treatment effects on the treated for the preferred specifications of our PCDID estimator, constructed as robust estimates from all 18, 22, or 33 state-specific treatment estimates using an M-estimator (Rousseeuw and Leroy, 2005). In contrast to the main results in Table 1, we include a measure for the stock of Republican 'capture' of a state to determine whether our results are driven by democratic erosion or partisanship. Here, in this table, the results are based on the Republican power variable with a depreciation rate of 10% rather than 20% as in Table 3.